

Fiscal Beliefs and the Effectiveness of Fiscal Consolidation

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Abstract

This paper studies how the distribution of household fiscal beliefs shapes the effectiveness of tax-based fiscal consolidation. Motivated by empirical evidence of heterogeneous tax-hike expectations in the FRBNY Survey of Consumer Expectations, I incorporate fiscal beliefs into a tractable model in which expectations drive aggregate demand, the endogenous tax base, and government debt. The model shows that, holding average beliefs fixed, the distribution of fiscal beliefs across households with different marginal propensities to consume (MPCs) and wealth substantially alters debt dynamics. I then quantify this mechanism in a two-group HANK model calibrated to cross-group differences in fiscal beliefs, MPCs, income shares, and wealth. Following a persistent one-percentage-point labor-income-tax increase, average fiscal beliefs raise the fiscal sacrifice ratio—cumulative output loss per unit of debt reduction—over the medium run relative to the full-information rational-expectations benchmark. However, belief heterogeneity offsets about one third of this increase because beliefs closer to full-information rational expectations are held by households that account for a disproportionate share of aggregate demand.

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Keywords: fiscal beliefs; heterogeneous expectations; fiscal policy; marginal propensities to consume; government debt.

1 Introduction

Fiscal consolidation is not a purely accounting exercise. A tax increase or spending reduction improves the primary balance mechanically, but its ultimate effect on government debt also depends on how households adjust consumption, how aggregate output responds, and how these changes feed back into the tax base. The strength of these endogenous fiscal feedbacks depends critically on what households understand the fiscal intervention.

Standard macroeconomic models typically assume that households observe fiscal shocks and correctly infer their implications for current disposable income, future taxes, and equilibrium activity. This full-information rational-expectations (FIRE) benchmark is difficult to reconcile with survey evidence documenting substantial heterogeneity in households' perceptions of public debt and future fiscal policy. Households differ not only in what they expect the government to do, but also in how strongly those expectations are reflected in their income and consumption

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plans. These differences raise a question that is largely absent from standard analyses of debt sustainability: does it matter which households understand fiscal policy correctly?

This paper studies how the distribution of fiscal beliefs across households shapes the effectiveness of fiscal consolidation. The central argument is that average fiscal beliefs are not sufficient for evaluating fiscal policy. The same belief can have different aggregate consequences depending on the MPC, wealth, and tax exposure of the household that holds it. As a result, two economies with identical average fiscal beliefs can exhibit different paths for aggregate demand, tax revenue, and government debt when those beliefs are distributed differently across households.

I begin by presenting descriptive evidence from the Federal Reserve Bank of New York Survey of Consumer Expectations (SCE), combining the Public Policy Survey with variables from the core questionnaire. Tax-hike expectations vary systematically with respondents' perceived exposure to tax changes and with household income. Households that expect tax changes to affect them negatively assign higher probabilities to future income-tax increases, and low-income households report higher tax-hike expectations than middle- and high-income households across perceived-exposure groups.

These patterns point to a feature that standard analyses of debt sustainability often abstract from: fiscal beliefs vary systematically with household characteristics that shape fiscal transmission. I use this survey evidence to discipline heterogeneity in fiscal beliefs. Then the models clarify how that heterogeneity interacts with MPCs, income, and wealth to affect aggregate outcomes and the effectiveness of fiscal consolidation.

I first study the mechanism in an overlapping-generations model with government debt and deficit shocks, building on Angeletos et al. [2024]. To depart from full-information rational expectations, I adopt the generalized non-FIRE framework of Angeletos et al. [2025]. Household beliefs are summarized by two parameters: λ^{PE} , which disciplines how correctly households perceive the fiscal innovation, and λ^{GE} , which captures how households believe others perceive the shock, thereby governing general equilibrium effects. This flexible formulation nests a wide range of behavioral models, making it particularly suitable for my application. While my empirical findings document that belief heterogeneity exists, they do not uniquely identify the underlying behavioral mechanism. Therefore, I discipline beliefs using the SCE data without committing to a specific cognitive friction. Firms and the fiscal authority remain fully informed.

The one-type model isolates two channels through which household characteristics shape the effectiveness of fiscal consolidation following a tax shock. First, comparing a FIRE economy to a non-FIRE economy, debt stabilizes more effectively under non-FIRE beliefs. Intuitively, when households perceive the tax shock to be smaller than it actually is, aggregate demand adjusts less, mitigating tax base erosion and allowing debt to stabilize more rapidly. This constitutes the **belief channel**. Second, comparing a high-MPC economy to a low-MPC economy, a higher aggregate MPC yields higher remaining debt. Because high-MPC households respond more strongly to tax shocks, the tax base erodes more severely, dampening debt reduction. This constitutes the **MPC channel**.

I then extend the framework to a model with two household types featuring heterogeneous MPCs and government debt holdings. Key to this extension is that fiscal belief heterogeneity enters

the model jointly with MPC and wealth heterogeneity. Holding average beliefs fixed, I compare a benchmark common-belief scenario against two heterogeneous-belief scenarios: one where low-MPC households hold beliefs closer to FIRE, and another where high-MPC households hold beliefs closer to FIRE. Relative to the common-belief baseline, belief heterogeneity deteriorates fiscal consolidation effectiveness in both cases.

However, the mechanism through which the distribution of fiscal beliefs impairs policy outcomes varies depending on which channel dominates. When low-MPC households hold beliefs closer to FIRE, weaker demand and higher debt arise through their larger wealth and population shares, which carry greater weight in the aggregate economy. In contrast, when high-MPC households hold beliefs closer to FIRE, higher debt is driven by their strong responsiveness to tax shocks (their high MPC), despite their small population and wealth shares. The two-type model thus demonstrates that average fiscal beliefs are insufficient for evaluating fiscal consolidation effectiveness. It is the distribution of fiscal beliefs, rather than average beliefs alone, that shapes the effectiveness of fiscal consolidation. Next, I quantify these mechanisms in a two-group HANK model with richer household heterogeneity.

The quantitative model is a two-group HANK model featuring heterogeneity in MPCs, income shares, and wealth holdings. I model fiscal consolidation as one-percentage-point increase in the common labor income tax rate, which provides a tractable mapping to the SCE measure of expected tax rate changes. I discipline the first-order belief parameters using the ratio of the subjective tax expectation response (from the SCE) to the realized tax rate response following a high-frequency fiscal shock. For second-order beliefs, I adopt a conservative approach by specifying them as a weighted average of first-order beliefs. I subsequently evaluate the robustness of this assumption using prior predictive analysis. The remaining parameters are calibrated to target the liquid wealth distribution in the Survey of Consumer Finances (SCF) and income shares from the Current Population Survey (CPS).

I compare three belief specifications: FIRE, a common-belief counterfactual that preserves the population-average belief while removing cross-group dispersion, and calibrated heterogeneous beliefs. Comparing FIRE to the common-belief scenario isolates the **level effect** of fiscal beliefs, whereas comparing the common-belief scenario to the heterogeneous-belief scenario isolates the **dispersion effect** of fiscal beliefs.

To evaluate fiscal consolidation, I define the fiscal sacrifice ratio as the cumulative output loss per unit of debt reduction. This measure recognizes that a larger debt reduction does not necessarily represent a more effective consolidation if it is achieved through a disproportionately large and persistent contraction in economic activity. Following a persistent one-percentage-point labor income tax increase, average belief distortions raise the sacrifice ratio from 2.33 under FIRE to 3.25 under common beliefs over a six-year horizon. Belief heterogeneity lowers it to 2.95, offsetting approximately one-third of this increase. This offset arises because beliefs closer to full information are held by households that account for most of aggregate demand. The sign and magnitude of the offset are robust to an alternative timing of tax perception and to prior-predictive exercises that vary belief and structural parameters.

This paper contributes to three strands of literature. First, it contributes to the literature on fiscal consolidation. Alesina et al. [2015] use narratively identified multi-year fiscal plans to study

the output effects of consolidation across OECD countries, while Erceg and Lindé [2013] compare tax- and spending-based consolidations in a currency union. I complement this literature by studying how household beliefs shape the output–debt trade-off of tax-based consolidation. The same statutory tax increase can generate different output and debt paths depending on how the fiscal consolidation and its future consequences are perceived across households.

Second, the paper contributes to the literature on fiscal policy under departures from full-information rational expectations. Existing work shows that cognitive discounting, sparse information, and higher-order belief distortions can affect fiscal multipliers and generate departures from Ricardian equivalence [Bianchi-Vimercati et al., 2024, Eichenbaum et al., 2026, Gabaix, 2020]. My contribution is threefold. First, I use the generalized non-FIRE framework to discipline beliefs with survey evidence without committing to a particular cognitive friction. Second, I shift the focus from average beliefs to their distribution across households. Third, I study government-debt dynamics and the output cost of debt reduction rather than fiscal multipliers alone.

Third, the paper connects behavioral belief heterogeneity to the heterogeneous-agent literature, which emphasizes the role of MPCs and liquid wealth in aggregate-policy transmission [Angeletos et al., 2021]. The closest related paper is Guerreiro [2023], who shows that the covariance between attention and exposure to aggregate fluctuations can amplify business cycles and alter government-spending multipliers. Guerreiro [2023] endogenizes attention as an optimal response to income cyclicalities and studies how inattention shapes business-cycle amplification and spending multipliers. I take a complementary approach along two dimensions. First, rather than deriving attention from a model of information choice, I discipline perception of tax policy directly with SCE evidence on how fiscal beliefs vary across households. Second, rather than studying spending shocks, I study tax-based consolidation in an economy with government debt and a fiscal rule, where beliefs about future taxes and debt govern the response. This shifts the question from how much beliefs amplify fluctuations to how they change the output cost of reducing debt.

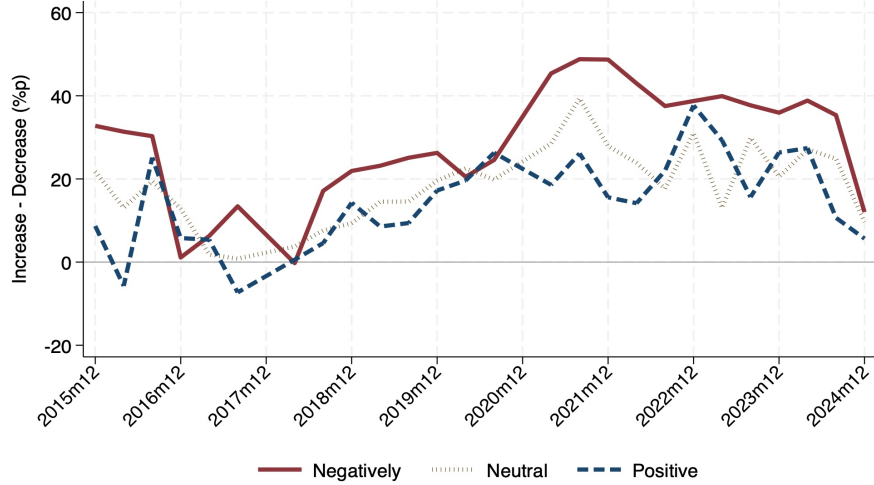
The remainder of the paper is organized as follows. Section 2 presents descriptive evidence on heterogeneity in household fiscal expectations and their association with household characteristics. Section 3 develops the generalized non-FIRE OLG framework, first with a representative household type and then with two types that differ in their MPCs, government debt holdings, and fiscal beliefs. Section 4 quantifies this mechanism using the HANK model. Section 5 concludes.

2 Empirical Evidence

This section documents heterogeneity in household fiscal beliefs using Federal Reserve Bank of New York Survey of Consumer Expectations Public Policy Survey. The survey elicits households’ subjective expectations about future changes in government policies, including whether taxes or benefits are likely to increase, decrease, or remain unchanged. I focus on expectations about income tax rates, since income taxes constitute a central component of federal revenues. I measure net tax-increase expectations as the reported probability of an increase minus the

reported probability of a decrease, expressed in percentage points. The survey also asks respondents whether they expect a given policy change to affect them personally in a positive, negative, or neutral way. I use this response to construct a measure of perceived exposure to fiscal policy. Further details on the questionnaire are provided in the Appendix A.

Figure 1. Tax-Hike Expectations by Perceived Exposure



Source: FRBNY SCE Public Policy Survey. After winsorizing the data at the 1st and 99th percentiles, approximately 74%, 17%, and 9% of households in the full sample reported being negatively, neutrally, and positively affected, respectively.

Figure 1 plots average net tax-hike expectations by perceived exposure from 2015 to 2024. Households that expect tax changes to affect them negatively assign systematically higher probabilities to future income-tax increases than households expecting neutral or positive effects. This pattern indicates that fiscal expectations are heterogeneous across households and vary systematically with respondents' perceived exposure to fiscal policy.

Figure 2. Tax-Hike Expectations by Income Group and Perceived Exposure

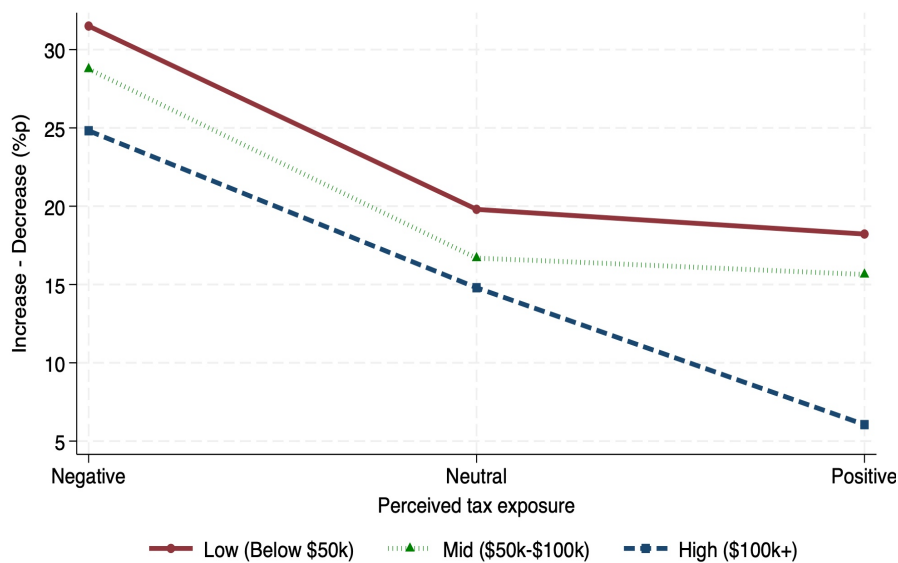


Figure 2 further disaggregates net tax-hike expectations by income group. Within each income group, negative perceived exposure is associated with higher tax-hike expectations. In addi-

tion, low-income households report higher tax-hike expectations than middle- and high-income households across perceived-exposure categories. These patterns suggest that fiscal beliefs are systematically related to household income, a key dimension of heterogeneity in the marginal propensity to consume.

Taken together, the evidence highlights the following empirical pattern. Tax-hike expectations vary systematically with both perceived exposure to fiscal policy and household income. This heterogeneity suggests that the aggregate consumption implications of fiscal beliefs may depend not only on average tax expectations but also on how those expectations are distributed across income groups.

The empirical results motivate a departure from the benchmark full-information rational expectations environment in which households share common fiscal beliefs. If households differ both in their fiscal beliefs and in the extent to which those beliefs are reflected in their income and consumption expectations, the effectiveness of fiscal consolidation may depend on the distribution of beliefs across households. I therefore use the survey evidence to motivate two qualitative features of the model: fiscal beliefs vary across households, and their association with spending differs across household groups. The model developed below therefore introduces non-FIRE distortions in fiscal beliefs and examines how their interaction with MPC heterogeneity shapes debt dynamics and the effectiveness of fiscal consolidation.

3 Simple model

I develop a tractable OLG framework to study how fiscal beliefs affect aggregate demand, the endogenous tax base, and government debt. I adopt the generalized non-FIRE approach of Angeletos, Guerreiro, and Zhang (2025), which parameterizes households' first- and higher-order beliefs without committing to a particular cognitive friction. Section 3.1 introduces fiscal-belief distortions into a one-type OLG model. Section 3.2 extends the model to two household types that differ in their MPCs and government debt holdings, allowing the allocation of fiscal beliefs to operate through both response and aggregation-weight margins.

3.1 One-Type OLG

I build on the OLG framework of Angeletos, Lian, and Wolf (2024), which features fiscal innovations, by incorporating a non-FIRE information structure. In this setup, we distinguish between the non-FIRE beliefs, denoted by $\bar{E}$, and the standard rational expectations under FIRE, denoted by $\mathbb{E}$.

3.1.1 Model

Current consumption is determined by the marginal propensity to consume out of total wealth, which comprises financial wealth (d_t) and human wealth (the expected value of $y_{t+k} - t_{t+k}$). The final term captures the intertemporal substitution effect of varying interest rates

$$c_t = (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k}) \right] \right) - \left(\sigma\beta\omega - (1 - \beta\omega) \frac{D}{Y} \right) \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k r_{t+k} \right]$$

New Keynesian Phillips Curve:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa y_t$$

Tax policy and flow budget constraint:

$$\begin{aligned} T_t &= \tau_y y_t + \tau_d(d_t + \varepsilon_t) - \varepsilon_t \\ d_{t+1} &= \frac{1}{\beta}(d_t - t_t) + \frac{D}{Y}r_t - \frac{D}{Y}(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]) \end{aligned}$$

The central bank sets the interest rate according to a simple feedback rule responding to output.

$$r_t = \phi y_t$$

We abstract from active monetary policy ($\phi = 0$)¹, and goods market clearing condition $y_t = c_t$,

$$y_t = (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - T_{t+k}) \right] \right)$$

Following Angeletos, Guerreiro, and Zhang (2025), I make the following assumption about households' first- and second-order beliefs about a fiscal shock.

Assumption 1. Following a fiscal shock, a fraction $\lambda^{\text{PE}} \in [0, 1]$ of the population learns it immediately and perfectly, while the rest never learn it. Furthermore, everyone believes that the fraction of agents that learn the innovation is $\lambda^{\text{GE}} \in (0, 1]$, where λ^{GE} may or may not coincide with λ^{PE} . Then

$$\begin{aligned} \bar{E}_t \varepsilon_{t+k} &= \lambda^{\text{PE}} \varepsilon_{t+k} \\ \bar{E}_t \bar{E}_t \varepsilon_t &= \lambda^{\text{GE}} \varepsilon_t \\ \bar{E}_t \bar{E}_{t+k} \varepsilon_{t+k} &= \lambda^{\text{GE}} \lambda^{\text{PE}} \varepsilon_{t+k} \end{aligned}$$

The final condition implies that the period- t expectation of the average period- $t+h$ forecast can be also interpreted as the fraction of informed agents at t (λ^p) times what these agents believe that the fraction of informed individuals will be at $t+h$ (λ^g). The non-FIRE departure applies only to households; firms and policy authorities retain full information.

Assumption 2. The fiscal shock ε_t follows an AR(1) process.

$$\varepsilon_{t+1} = \rho_\varepsilon \varepsilon_t + u_{t+1}$$

Assume a one-time fiscal innovation u_0 , then

$$\bar{E}_t \bar{E}_{t+k} \varepsilon_{t+k} = \lambda^{\text{GE}} \lambda^{\text{PE}} \rho_\varepsilon^k \varepsilon_t$$

Throughout the quantitative exercises, I consider a one-time negative innovation, $u_t < 0$. Because ε_1 enters the tax rule with a minus sign, this innovation raises current taxes and therefore represents a tax-based fiscal consolidation. Impulse responses are reported for a tax increase equal to 1 percent of GDP in absolute value.

¹Following the original calibration in Angeletos et al. [2024], I set this parameter to exactly zero. This assumption can easily be relaxed.

With assumption 1 and assumption2, the solution to this model becomes simple. The model can be solved using a standard guess-and-verify method. Assume:

$$y_t = \psi_d d_t + \psi_\varepsilon \underbrace{\bar{E}_t \varepsilon_t}_{\hat{\varepsilon}_t}$$

$$d_{t+1} = \phi_d d_t + \phi_\varepsilon \underbrace{\bar{E}_t \varepsilon_t}_{\hat{\varepsilon}_t}$$

where $\bar{E}_t \varepsilon_t$ denotes the average belief of a shock.

Substituting the tax rule into the consumption,

$$y_t = (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k \{ (1 - \tau_y) y_{t+k} - \tau_d d_{t+k} - (\tau_d - 1) \varepsilon_{t+k} \} \right] \right)$$

$$= (1 - \beta\omega) (d_t + (1 - \tau_y) S_y - \tau_d S_d - (\tau_d - 1) S_\varepsilon)$$

where

$$S_\varepsilon = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t \varepsilon_{t+k} = H \varepsilon_t, \quad S_d = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t d_{t+k} = A_d d_t + A_\varepsilon H \varepsilon_t, \quad S_y = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t y_{t+k} = B_d d_t + B_\varepsilon \varepsilon_t$$

and

$$H = \frac{\lambda^{\text{PE}}}{1 - \beta\omega\rho_\varepsilon}, \quad A_d = \frac{1}{1 - \beta\omega\phi_d}, \quad A_\varepsilon = \frac{\beta\omega\phi_\varepsilon\lambda^{\text{GE}}}{1 - \beta\omega\phi_d}, \quad B_d = \psi_d A_d, \quad B_\varepsilon = (\psi_d A_\varepsilon + \psi_\varepsilon \lambda^{\text{GE}} H)$$

Combined,

$$y_t = (1 - \beta\omega) \left(\underbrace{[1 + (1 - \tau_y) B_d - \tau_d A_d]}_{\text{Coefficient on } d_t} d_t + \underbrace{[(1 - \tau_y) B_\varepsilon - \tau_d A_\varepsilon - (\tau_d - 1) H]}_{\text{Coefficient on } \varepsilon_t} \varepsilon_t \right)$$

$$= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t \tag{1}$$

Note substitute in the tax rule,

$$d_{t+1} = \frac{1}{\beta} (d_t - t_t) + \frac{D}{Y} \underbrace{r_t}_{=0} - \frac{D}{Y} \underbrace{(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}])}_{=0}$$

$$= \frac{1}{\beta} \left((1 - \tau_d) d_t - \tau_y (\psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t) - (\tau_d - 1) \frac{\hat{\varepsilon}_t}{\lambda^{\text{PE}}} \right)$$

$$= \phi_d d_t + \phi_\varepsilon \lambda^{\text{PE}} \varepsilon_t \tag{2}$$

where $d_0 = -\frac{D}{Y} \pi_0$.

I consider an impulse response to a one-time fiscal shock (ε_t) at $t = 0$. Since no further innovations occur, the post- $t = 0$ path is deterministic, implying $\pi_{t+1} = \mathbb{E}_t \pi_{t+1}$ for $t \geq 0$. Hence I abstract from surprise inflation in the government budget constraint. Therefore,

$$\phi_d = \frac{1}{\beta} (1 - \tau_d - \tau_y \psi_d)$$

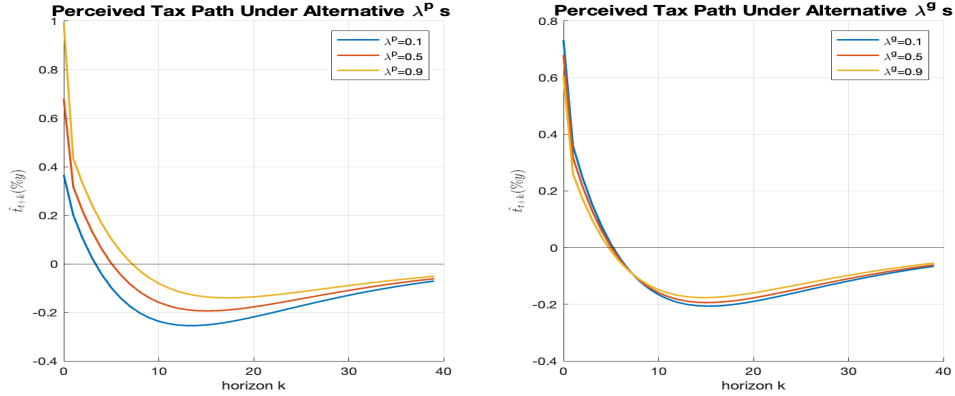
$$\phi_\varepsilon = \frac{1}{\beta} \left(-\tau_y \psi_\varepsilon - \frac{\tau_d - 1}{\lambda^{\text{PE}}} \right)$$

Therefore, $\psi_d, \psi_\varepsilon, \phi_d, \phi_\varepsilon$ are solved as a fixed-point to this problem. Plug these into equations 1 and 2, and I can plot the impulse response functions. See Appendix B.1 for detailed derivations.

3.1.2 Results

Figure ?? plots the perceived tax path under alternative λ values in response to a one-time, unexpected tax increase. A higher λ^{PE} scales up the initial perceived tax shock. Given the underlying AR(1) process, a larger initial perception proportionally shifts the entire expected path of future tax increases upward.

Figure 3. Perceived Tax Path Under Alternative λ values



Notes: Impulse responses to a one-time tax increase of 1% of GDP. t is reported in percent of GDP, so a value of 1 corresponds to 1% of GDP.

A higher λ^{GE} strengthens households' expectations of future average perceptions, thereby increasing the perceived persistence of the aggregate shock. Both a higher λ^{PE} and a higher λ^{GE} amplify the contractionary response through a perceived negative wealth effect: households anticipate a larger decline in lifetime disposable income and reduce current consumption. Quantitatively, however, the impact of λ^{GE} is dampened because households discount future tax liabilities at rate $\beta\omega$. In contrast, λ^{PE} exerts a first-order effect on lifetime wealth by directly scaling the immediate innovation.

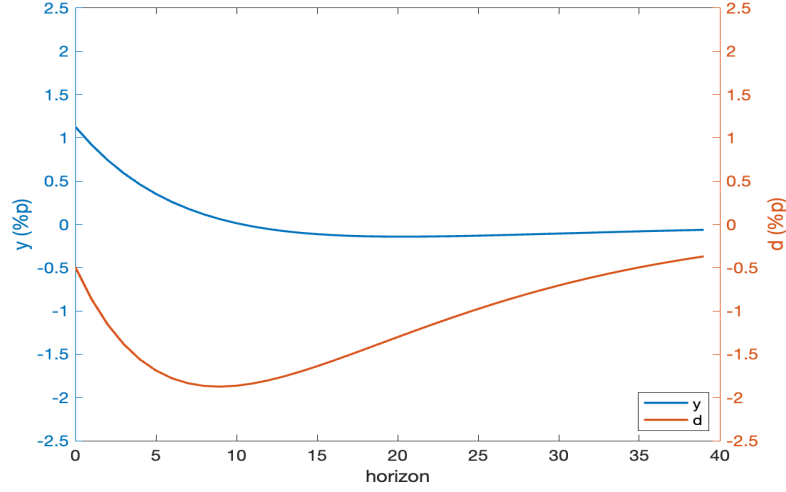
Having established how belief parameters shape tax expectations, I conduct a counterfactual analysis to isolate how specific household characteristics alter the transmission of a fiscal shock. I examine two distinct specifications:

1. **Belief Channel:** Comparing a non-FIRE economy to a FIRE economy by varying λ values while holding ω constant.
2. **MPC Channel:** Comparing a high-MPC economy to a low-MPC economy by varying ω values while holding λ constant.

Figure 4 plots the differences in output and debt impulse responses under alternative values of λ , comparing non-FIRE fiscal beliefs against the FIRE benchmark. Non-FIRE beliefs lead to higher output (a milder contraction) and lower debt relative to the FIRE economy. Intuitively, when households perceive the tax shock as smaller than it actually is, aggregate demand adjusts less. Consequently, the tax base erodes less severely, facilitating more effective debt stabilization.

Figure 5 plots the differences in impulse responses under alternative values of ω , effectively comparing a high-MPC economy to a low-MPC economy. A higher-MPC economy produces a deeper demand contraction and a higher debt path relative to a lower-MPC economy. Because

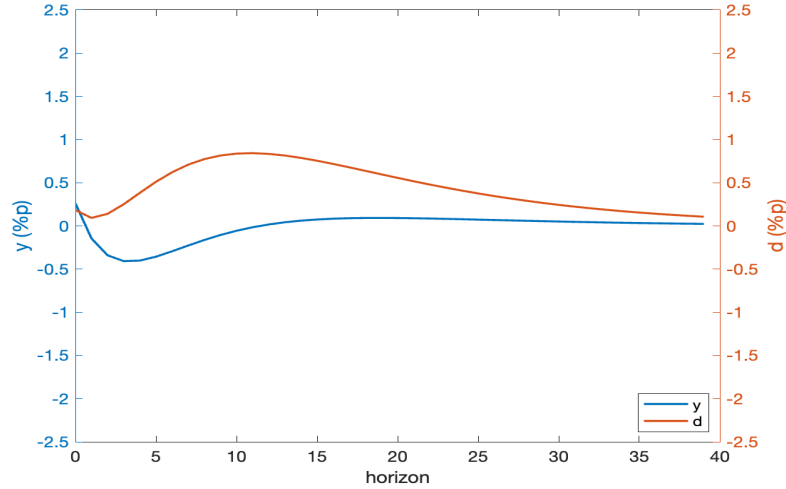
Figure 4. Impulse Responses of Output and Debt Under Alternative λ Values



Notes: Differences in impulse responses relative to the FIRE benchmark following a one-time tax increase of 1% of GDP. Output (y) is reported in percentage points (%), and debt (d) is reported in percent of GDP.

households respond more strongly to tax shocks when MPCs are high, aggregate demand and the endogenous tax base drop significantly, causing government debt to rise relative to the baseline.

Figure 5. Impulse Responses of Output and Debt Under Alternative ω Values



Notes: Differences in impulse responses following a one-time tax increase of 1% of GDP. Output (y) is reported in percentage points (%), and debt (d) is reported in percent of GDP.

This analytical mechanism is corroborated by Table 1, which presents the model solution coefficients from the output and debt equations under alternative λ and ω values. Increasing λ leads to a larger contemporaneous output response (higher $\lambda\psi_\varepsilon$) and a weaker debt-reduction response (lower $\lambda\phi_\varepsilon$) to an actual fiscal shock (ε_t). Intuitively, when agents perceive a larger tax increase on impact (higher λ^{PE} for $\varepsilon < 0$), contemporaneous demand contracts more sharply, eroding the tax base ($\tau_y y$) and resulting in a smaller reduction in government debt.

Conversely, increasing ω (which lowers the MPC) yields a smaller output response (lower $\lambda^{\text{PE}}\psi_\varepsilon$) and a stronger debt-reduction response (higher $\lambda^{\text{PE}}\phi_\varepsilon$). When agents are less responsive to a tax shock (higher ω), aggregate demand contracts less, preserving the tax base and allowing

government debt to decline more effectively.

Table 1. Coefficient Solutions Under Alternative λ s and ω s

Param.	Output Response to Shock (ψ)			Debt Response to Shock (ϕ)		
	ψ_d	ψ_ϵ	$\lambda^{\text{PE}}\psi_\epsilon$	ϕ_d	ϕ_ϵ	$\lambda^{\text{PE}}\phi_\epsilon$
Panel A: Varying λ values						
λ value						
0.10	0.22	0.94	0.09	0.91	9.52	0.95
1.00	0.22	1.33	0.13	0.91	0.54	0.05
Panel B: Varying ω values						
ω value						
0.55	0.86	1.98	0.19	0.70	0.32	0.03
0.865	0.22	0.86	0.08	0.91	0.70	0.07

Note: ψ_ϵ and ϕ_ϵ are the coefficients on the perceived fiscal innovation, $\hat{\epsilon}_t$, whereas $\psi_\epsilon\lambda^{\text{PE}}$ and $\phi_\epsilon\lambda^{\text{PE}}$ are the coefficients on the actual fiscal innovation, ϵ_t .

The counterfactual analysis establishes that belief distortions can materially alter the transmission of a fiscal shock. The one-type model, however, cannot address whether aggregate outcomes depend on the joint distribution of fiscal beliefs, MPCs, and wealth. To study how the distribution of fiscal beliefs across households affects aggregate outcomes, I extend the model to two household types in the following section.

3.2 Two-Type OLG: The Allocation of Fiscal Beliefs

I extend the model to two household types that differ in their survival probabilities, ω_i , and hence in their MPCs, as well as in their holdings of government debt. I allow the belief parameters, $(\lambda_i^{\text{PE}}, \lambda_i^{\text{GE}})$, to vary across types. Key to this extension is that fiscal belief heterogeneity enters the model jointly with MPC and wealth heterogeneity. The results show that evaluating fiscal consolidation requires looking beyond average fiscal beliefs: the distribution of beliefs across households is critical. Furthermore, who holds specific beliefs matters, as aggregate policy effectiveness depends on whether non-FIRE beliefs are concentrated among high-MPC or high-wealth households.

3.2.1 Model

Assume each agent i owns a s_i fraction of government debt.

$$d_t^i = s_i d_t, \quad s_H = s, s_L = 1 - s$$

Note S_y, S_d, S_t in section 3.1 have i superscripts where

$$S_\epsilon^i = \sum_{k=0}^{\infty} (\beta\omega_i)^k E_t \epsilon_{t+k} = H_i \epsilon_t, \quad S_d^i = A_d^i d_t + A_\epsilon^i H \epsilon_t, \quad S_y^i = B_d^i d_t + B_\epsilon^i \epsilon_t \quad \text{where } i \in \{H, L\}$$

where

$$H_i = \frac{\lambda_i^{\text{PE}}}{1 - \beta\omega_i\rho_\varepsilon}, \quad A_d^i = \frac{1}{1 - \beta\omega_i\phi_d}, \quad A_\varepsilon^i = \frac{\beta\omega_i\phi_\varepsilon\lambda_i^{\text{GE}}}{1 - \beta\omega_i\phi_d}, \quad B_d^i = \psi_d A_d^i, \quad B_\varepsilon^i = \psi_d A_\varepsilon^i + \psi_\varepsilon \lambda_i^{\text{GE}}$$

Aggregate output is

$$\begin{aligned} y_t &= \mu_1 c_t^1 + \mu_2 c_t^2 = \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) \left(s_i d_t + (1 - \tau_y) S_y^i - \tau_d S_d^i - (\tau_d - 1) S_\varepsilon^i \right) \\ &= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t \end{aligned} \quad (3)$$

Then

$$\psi_d = \frac{\Xi_s - \tau_d \Xi_d}{1 - (1 - \tau_y) \Xi_d}, \quad \psi_\varepsilon = \frac{((1 - \tau_y)\psi_d - \tau_d)\Xi_A - (\tau_d - 1)\Xi_H}{\sum_i \lambda_i^p - (1 - \tau_y)\Xi_G}.$$

where

$$\begin{aligned} \Xi_s &\equiv \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) s_i, \quad \Xi_d \equiv \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) A_d^i, \quad \Xi_A \equiv \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i A_\varepsilon^i, \\ \Xi_H &\equiv \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i, \quad \Xi_G \equiv \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i \lambda_i^{\text{GE}}. \end{aligned}$$

Finally, from the government budget constraint,

$$\begin{aligned} d_{t+1} &= \frac{1}{\beta} \left((1 - \tau_d) d_t - \tau_y y_t - (\tau_d - 1) \varepsilon_t \right) \\ &= \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t \\ \phi_d &= \frac{1 - \tau_d - \tau_y \psi_d}{\beta}, \quad \phi_\varepsilon = \frac{-(\tau_d - 1) - \tau_y \psi_\varepsilon \sum_i \mu_i \lambda_i^{\text{PE}}}{\beta \sum_i \mu_i \lambda_i^{\text{PE}}} \end{aligned} \quad (4)$$

Then, $\psi_d, \psi_\varepsilon, \phi_d, \phi_\varepsilon$ are solved as a fixed-point to this problem. Plug this into M, and we can plot the impulse response functions. See Appendix B.2 for detailed derivations.

The calibration of the model parameters is summarized in Table 2. For the preference and policy parameters (β, τ_y, τ_d) , I adopt the baseline values from Angeletos et al. [2024]. The persistence of the deficit shock is set to a high value to clearly isolate the transmission mechanism of a one-time deficit shock. The population shares of high- and low-MPC agents are calibrated based on summary statistics from the FRBNY SCE. In the survey, individuals with low-to-middle income and education levels constitute approximately 65% of respondents; accordingly, I set the population shares to $\mu_H = 1/3$ and $\mu_L = 2/3$. Regarding the survival probability parameters, I set ω_L (the low-MPC group) following Angeletos et al. [2024], and calibrate ω_H (the high-MPC group) to match an aggregate average MPC of 0.25. Under the calibrated ω parameters, the implied MPC profile is front-loaded over the first two years, consistent with micro evidence on total spending (including durables) out of income gains. Finally, reflecting the limited asset participation of these households, I set the share of government debt held by the high-MPC group, s , to 0.1.

To study how the allocation of fiscal beliefs interacts with the two margins, I hold population-average beliefs fixed and redistribute beliefs across household types. In Scenario 1, the high-MPC

Table 2. Calibration

Preferences & Policy			Agents		
Param.	Value	Description	Param.	Value	Description
β	0.99		(ω_H, ω_L)	(0.65, 0.865)	Mortality risks
τ_y	0.33	Tax rule	(μ_H, μ_L)	(0.33, 0.67)	Share of High/Low-MPC agents
τ_d	0.033	Tax rule	(s_H, s_L)	(0.1, 0.9)	Share of government debt holdings
ρ_ϵ	0.80	Deficit shock persistence			
$\frac{D}{Y}$	1.04	Steady-state Debt-to-GDP			

type holds more distorted beliefs, while the type with larger population and government-debt shares holds beliefs closer to FIRE. Scenario 2 reverses this allocation, assigning beliefs closer to FIRE to the high-MPC type (Table 3).

Table 3. Counterfactual analysis

Scenarios	λ^H	λ^L
Benchmark	0.5	0.5
Scenario 1: High-MPC strongly Non-FIRE	0.1	0.7
Scenario 2: High-MPC close to FIRE	0.9	0.3

The exercise compares allocations in which beliefs closer to FIRE operate primarily through the aggregation-weight margin or through the response margin.

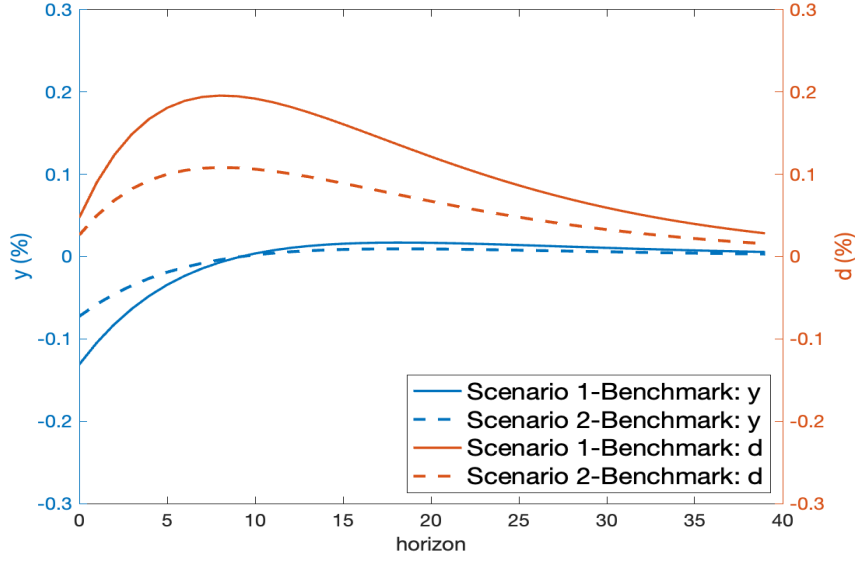
3.2.2 Results

Figure 5 reports the differences in the impulse responses of output and government debt between the common-belief benchmark and two heterogeneous-belief scenarios: Scenario 1 (solid line) and Scenario 2 (dashed line). Under both scenarios, the debt difference rises above zero, indicating that debt stabilizes less relative to the common-belief benchmark. This demonstrates that belief heterogeneity deteriorates the effectiveness of fiscal consolidation.

While both scenarios deteriorate fiscal consolidation effectiveness, they generate this result through distinct margins. In Scenario 1, high-MPC households hold more distorted beliefs ($\lambda_H = 0.1$), whereas low-MPC households hold beliefs closer to FIRE ($\lambda_L = 0.7$). Because low-MPC households account for larger population and government-debt shares, they receive substantial weight in aggregate consumption and hold most of the economy’s financial wealth. Although each low-MPC household adjusts consumption relatively modestly, their more accurate perception of the fiscal shock produces a sizable aggregate wealth and demand contraction.

In Scenario 2, the allocation of beliefs is reversed: high-MPC households hold beliefs closer to FIRE ($\lambda_H = 0.9$), whereas low-MPC households hold more distorted beliefs ($\lambda_L = 0.3$). High-MPC households have smaller population and government-debt shares, meaning their beliefs receive less weight through the population and wealth margins. Nevertheless, these households translate a much larger fraction of changes in perceived disposable income into current consumption. Their high MPC thus generates a strong aggregate-demand response despite their

Figure 6. Differences in Output and Debt Responses Relative to the Benchmark



Notes: Impulse responses to a one-time tax increase of 1% of GDP. y is reported in percentage points (%). d is reported in percent of GDP, so a value of 1 corresponds to 1% of GDP.

relatively small size and limited asset holdings.

Thus, fiscal beliefs affect aggregate output not only through their population-weighted average, $\bar{\lambda}$. Rather, belief distortions interact with population shares μ_i , the distribution of government-debt holdings s_i , and type-specific MPCs (governed by $1 - \beta\omega_i$). Scenario 1 assigns beliefs closer to FIRE to the type with the larger aggregation weight, whereas Scenario 2 assigns them to the type with the stronger response margin. Consequently, both reallocations generate larger contractions in aggregate demand than the common-belief benchmark, even though they operate through different margins.

Table 4 provides a quantitative summary of these differences. The coefficients in equations 3 and 4,

$$\psi_\varepsilon \bar{\lambda}^{\text{PE}}, \quad \phi_\varepsilon \bar{\lambda}^{\text{PE}}$$

describe the responses of output and next-period debt to the actual fiscal innovation, respectively. When high-MPC agents perceive the tax increase as larger and more persistent (Scenario 2), the aggregate output response to the true tax increase becomes significantly stronger (higher $\psi_\varepsilon \bar{\lambda}^{\text{PE}}$). Because the experiment uses a negative innovation, $u_0 < 0$, a higher value of $\psi_\varepsilon \bar{\lambda}^{\text{PE}}$ corresponds to a larger contraction in output. The resulting erosion of the tax base weakens the debt improvement generated by the tax increase. Likewise, the lower value of $\phi_\varepsilon \bar{\lambda}^{\text{PE}}$ implies a smaller decline in next-period debt (lower $\phi_\varepsilon \bar{\lambda}^{\text{PE}}$).

In addition, the discounted average responses, denoted by Avg.y and Avg.d, summarize the entire adjustment path rather than only the impact response (Table 5). These discounted averages differ substantially between Scenarios 1 and 2, reflecting differences in both the magnitude and persistence of the demand and debt responses. The differences arise even though the two scenarios have the same average fiscal belief.

The comparison establishes that population-average fiscal beliefs are not a sufficient statistic for

Table 4. Parameter Combinations and Equilibrium Coefficients ($\bar{\lambda}$ fixed)

	Parameters		Coefficients					
	λ^H	λ^L	ψ_d	ψ_ϵ	$\psi_\epsilon \bar{\lambda}^{\text{PE}}$	ϕ_d	ϕ_ϵ	$\phi_\epsilon \bar{\lambda}^{\text{PE}}$
Benchmark	0.50	0.50	0.2715	1.1790	0.5895	0.8863	1.5605	0.7803
Scenario 1	0.10	0.70	0.2715	1.3160	0.6606	0.8863	1.5071	0.7566
Scenario 2	0.90	0.30	0.2715	1.4718	0.7329	0.8863	1.4708	0.7325

Note: ψ_ϵ and ϕ_ϵ are the coefficients on the perceived fiscal innovation, $\hat{\epsilon}_t$, whereas $\psi_\epsilon \sum_i \mu_i \lambda_i^{\text{PE}}$ and $\phi_\epsilon \sum_i \mu_i \lambda_i^{\text{PE}}$ are the coefficients on the actual fiscal innovation, ϵ_t .

Table 5. Average Output and Debt Response

	Avg. y (%)	Avg. d (%)
Benchmark	-0.0358	0.6455
Scenario 1	-0.0380	0.6865
Scenario 2	-0.0406	0.7327

Note: $\text{Avg.}y = \sum_{t=0}^T \beta^t y_t$, $\text{Avg.}d = \sum_{t=0}^T \beta^t d_t$

fiscal transmission. What matters is how those expectations are distributed across households with different sources of aggregate influence. Beliefs assigned to households with larger population and government-debt shares operate through the aggregation-weight margin, whereas beliefs assigned to high-MPC households operate through the response margin. The joint distribution of fiscal beliefs, MPCs, and government-debt holdings therefore determines the strength of the aggregate-demand and tax-base responses and, ultimately, the dynamics of government debt.

4 Quantitative model

While the simple model isolates how the allocation of fiscal beliefs interacts with heterogeneous household spending responses and wealth, it cannot discipline the quantitative importance of these channels. I therefore embed group-specific fiscal beliefs into a two-group HANK model. The calibration combines cross-group empirical evidence on fiscal beliefs from the Survey of Consumer Expectations with differences in liquid wealth from the Survey of Consumer Finances and labor-income shares from the Current Population Survey. The model generates corresponding differences in household consumption responses endogenously.

I model fiscal consolidation as one-percentage point increase in the common labor income tax rate, which provides a tractable mapping to the SCE measure of expected tax rate changes. I discipline the first-order belief parameters using the ratio of the subjective tax expectation response (from the SCE) to the realized tax rate response following a high-frequency fiscal shock. For second-order beliefs, I adopt a conservative approach by specifying them as a weighted average of first-order beliefs. I subsequently evaluate the robustness of this assumption using prior predictive analysis.

The quantitative exercise compares three belief specifications: full-information rational expectations (FIRE), common mean beliefs, and calibrated heterogeneous beliefs. This comparison separates the level effect and dispersion effect of fiscal beliefs. Moving from FIRE to common beliefs isolates the level effect: relative to FIRE, common imperfect beliefs raise the medium-run output cost of consolidation. Moving from common to heterogeneous beliefs then isolates the dispersion effect: heterogeneous beliefs offset approximately one-third of the increase in the fiscal sacrifice ratio associated with common beliefs. The sign and approximate magnitude of the offset are robust to an alternative timing of tax perception and to prior-predictive exercises that vary belief and model parameters.

The remainder of the section proceeds as follows. Section 4.1 sets out the environment and, in Section 4.1.5, states how the first- and second-order beliefs of Assumption 1 are embedded in a HANK model. Section 4.2 calibrates the model and belief parameters. Section 4.3 reports impulse responses to a persistent labor-income-tax consolidation under FIRE, under calibrated heterogeneous beliefs, and under a common-belief counterfactual that holds average beliefs constant. Section 4.4 evaluates the effectiveness of fiscal consolidation under different belief specifications using the sacrifice ratio. Section 4.5 assesses sensitivity to the tax-timing assumption and the model parameters.

4.1 Environment

Throughout this section, capital letters denote levels and lower-case letters denote deviations from the steady state (e.g., $y_t = \frac{Y_t - Y^{ss}}{Y^{ss}}$ and $c_t = \frac{C_t - C^{ss}}{C^{ss}}$). Fiscal variables, such as debt and taxes, are expressed as deviations relative to steady-state GDP: $b_t = \frac{B_t - B^{ss}}{Y^{ss}}$ and $\Delta T_t = \frac{T_t - T^{ss}}{Y^{ss}}$. Finally, interest rates and inflation rates are defined as absolute deviations from their steady-state values: $i_t = I_t - I^{ss}$, $\pi_t = \Pi_t - \Pi^{ss}$.

4.1.1 Households

There are two household groups, $g \in \mathcal{G} = \{H, L\}$. These correspond to the groups analyzed in the empirical section: Group H denotes high-MPC (low-income) households with annual income below \$50,000, and Group L collects the remainder. Within each group g , households face idiosyncratic labor productivity z and trade a single liquid asset, government debt. Preferences are given by

$$\bar{E}_{i,0} \sum_{t=0}^{\infty} \beta_g^t [u(C_{i,g,t}) - v(N_{i,g,t})], \quad (5)$$

where $u(c) = \frac{c^{1-\sigma-1}}{1-\sigma-1}$ and $v(n) = \frac{\xi n^{1+\psi-1}}{1+\psi-1}$. The flow budget constraint for an optimizing household is

$$C_{i,g,t} + A_{i,g,t+1} = (1 - \tau_{g,t}) \varsigma_g W_t Z_{i,g,t} N_{i,g,t} + (1 + R_t) A_{i,g,t}, \quad A_{i,g,t+1} \geq \underline{A}, \quad (6)$$

where $1 + R_t = \frac{(1+I_{t-1})P_{t-1}}{P_t}$ is the gross ex-post real return on government debt held from t to $t-1$, I_{t-1} is the net nominal interest rate set in period $t-1$, and $\Pi_t = P_t/P_{t-1} - 1$. To first order, $r_t = i_{t-1} - \pi_t$. $\varsigma_g = s_Y^g/\mu_g$ denotes group g 's per-capita labor-income exposure, and $z_{i,g,t}$ follows a group-specific finite-state Markov process. The group-specific discount factors β_H and β_L govern asset accumulation and cross-group MPC differences. $\tau_{g,t}$ is the group-average

labor-income tax rate defined in equation (10), w_t is the real wage, and r_t is the realized real return on liquid government debt.

Each group also contains a share π_g of hand-to-mouth (HtM) households. They hold no assets and consume their current after-tax labor income each period:

$$C_{i,g,t}^{\text{HtM}} = (1 - \tau_{g,t})\varsigma_g W_t Z_{i,g,t} N_{i,g,t}, \quad A_{i,g,t}^{\text{HtM}} = A_{i,g,t+1}^{\text{HtM}} = 0.$$

Group-level aggregate choices combine optimizer (O) and hand-to-mouth (HtM) households:

$$X_{g,t} = (1 - \pi_g)X_{g,t}^{\text{O}} + \pi_g X_{g,t}^{\text{HtM}}, \quad X \in \{C, A\}. \quad (7)$$

4.1.2 Firms and labor unions

Output is linear in effective labor, $Y_t = \Theta_t N_t$. Following Guerreiro [2023], differentiated labor unions set wages subject to quadratic adjustment costs and employment is equally rationed across households. This ensures that a perceived tax change affects output solely through consumption, the channel through which the belief-wealth and belief-MPC mechanisms operates. Thus, households differ in assets, idiosyncratic productivity, tax exposure, and beliefs, but do not separately bargain over nominal wages.

With flexible goods prices and constant productivity, price inflation equals wage inflation. The linearized wage Phillips curve² can be written as

$$\pi_t = \beta_{nk}\pi_{t+1} + \kappa_w \left[\frac{1}{\sigma} \frac{C_t - C^{ss}}{C^{ss}} + \frac{1}{\psi} \frac{Y_t - Y^{ss}}{Y^{ss}} + \tilde{\tau}_t^{\text{eff}} \right] \quad (8)$$

where $\tilde{\tau}_t^{\text{eff}} = \sum_g s_g^Y d\tau_{g,t}/(1 - \bar{\tau}_g)$ is the labor-tax wedge.

Firms and unions have full information.

4.1.3 Monetary policy

Policy authorities have full information. Only household beliefs depart from FIRE. The monetary authority follows a Taylor rule that responds to both inflation and the output gap,

$$\dot{i}_t = \rho_i \dot{i}_{t-1} + (1 - \rho_i) (\phi_\pi \pi_t + \phi_y y_t) + u_t^i, \quad \phi_\pi > 1, \quad \phi_y \geq 0. \quad (9)$$

where $y_t \equiv \log(Y_t/Y^{ss})$ is the output gap. I use the steady state Y^{ss} instead of the flexible-wage (natural) output level, Y_t^n . The ex ante real interest rate is $r_t = \dot{i}_t - \mathbb{E}_t \pi_{t+1}$

4.1.4 Fiscal policy

I model fiscal consolidation as a uniform innovation to each group's labor-income tax rate. While the SCE measures expectations of overall effective tax burdens, labor earnings represent the primary income source and tax base for most households, rendering this specification a tractable and empirically grounded proxy. In this baseline setup, equal percentage-point changes in statutory rates generate differential tax incidence across groups through heterogeneity in effective tax rates and labor-income shares.

²See Guerreiro (2023) for derivations. This is based on the optimization problem of labor union.

The group-specific effective labor-income tax rate evolves according to

$$\tau_{g,t} = \bar{\tau}_g + x_t + \phi_b \frac{B_{t-1} - B^{ss}}{Y^{ss}}, \quad g \in \mathcal{G}, \quad (10)$$

where x_t is a common percentage-point innovation to each group's labor-income tax rate. At impact, an innovation of $x_0 = 0.01$ generates direct revenue equal to one percent of quarterly steady-state output, with group g contributing $Y_g^{ss} x_0$ to this direct revenue effect.

The fiscal innovation follows

$$x_t = \rho_x x_{t-1} + \varepsilon_t^\tau, \quad 0 \leq \rho_x < 1. \quad (11)$$

A positive innovation $\varepsilon_t^\tau > 0$ is a tax-based fiscal consolidation. As in Section 3, it generates a persistent increase in the tax burden.

Aggregate tax revenue is

$$T_t = \sum_{g \in \{L,H\}} \tau_{g,t} Y_{g,t} = \sum_{g \in \{L,H\}} \left(\bar{\tau}_g + x_t + \phi_b \frac{B_{t-1} - B^{ss}}{Y^{ss}} \right) Y_{g,t}$$

Let $b_t \equiv \frac{B_t - B^{ss}}{Y^{ss}}$, $\hat{r}_t \equiv r_t - r^{ss}$, and $\Delta T_t \equiv \frac{T_t - T^{ss}}{Y^{ss}}$. A first-order approximation around the steady state yields

$$\Delta T_t = \sum_{g \in \mathcal{G}} \left[\underbrace{Y_g^{ss} x_t}_{\text{direct revenue}} + \underbrace{\bar{\tau}_g y_{g,t}}_{\text{tax-base feedback}} + \underbrace{\phi_b Y_g^{ss} b_{t-1}}_{\text{debt feedback}} \right]. \quad (12)$$

The government budget constraint is

$$B_t = (1 + R_t) B_{t-1} + G_t - T_t. \quad (13)$$

With government purchases fixed at their steady-state level, its first-order approximation is

$$b_t = (1 + R^{ss}) b_{t-1} + \frac{B^{ss}}{Y^{ss}} r_t - \Delta T_t. \quad (14)$$

Therefore, beliefs affect debt through consumption and output, the endogenous tax base, inflation and the real rate, and the induced debt-feedback taxes.

4.1.5 Beliefs

Beliefs are parameterized as in Assumption 1, with one modification. I allow $(\lambda_g^{\text{PE}}, \lambda_g^{\text{GE}})$ to differ across H and L . This generates belief-wealth and belief-MPC correlations studied in Section 3.2. In addition, the baseline timing is important. All households observe the current tax rate, labor income, and asset return before choosing consumption. Limited perception therefore applies to the continuation of the tax innovation and to its future aggregate consequences. This timing assumption follows the convention in the HANK formulations of Guerreiro [2023] and Angeletos et al. [2025]. It also guarantees that the household budget and borrowing constraint are satisfied without an additional after-choice tax-settlement mechanism. I relax this assumption and examine a case in which households do not perceive the contemporaneous shock in Section 4.5.

In the OLG model of Section 3, aggregate demand is a closed-form function of average expectations, so λ^{GE} can be substituted directly into the solution. That shortcut is unavailable here, because household behavior is summarized by Jacobians of a heterogeneous-agent problem rather than by a single Euler equation. Second-order beliefs instead enter through a *subjective equilibrium*: the economy that an informed household believes it inhabits, and against which it forms the forecasts that drive its saving and consumption plan.

Let

$$\hat{z}_g = \left(\hat{y}_g, \hat{\pi}_g, \hat{i}_g, \hat{r}_g, \hat{b}_g \right)^\top$$

denote the *subjective equilibrium* perceived by an informed household in group g : the aggregate path that this household believes the economy will follow. The subjective equilibrium differs from the actual equilibrium whenever households' beliefs diverge from the economy's actual law of motion.

An informed household in group g believes that a fraction λ_g^{GE} of households in each group incorporates the fiscal shock into its forward-looking plan, while the remaining households respond only to realized current conditions. The parameter λ_g^{GE} therefore governs the extent to which the household takes endogenous general-equilibrium adjustment into account. Because an informed household in group g must forecast the behavior of both groups, the subjective equilibrium paths are determined jointly:

$$\hat{z}_H = \Phi_H(\hat{z}_H, \hat{z}_L; \lambda_H^{\text{GE}}), \quad \hat{z}_L = \Phi_L(\hat{z}_H, \hat{z}_L; \lambda_L^{\text{GE}}). \quad (15)$$

where Φ_g collects the subjective household-response equations, market clearing, the wage Phillips curve, monetary policy, and the government budget constraint. Appendix G gives the full matrix representation and solution of this joint fixed-point problem.

Then the group- g 's response aggregates informed and uninformed household's behavior using each group's *first-order* informed share λ_g^{PE} . Responses to realized inputs enter fully, whereas a group acts on its forward-looking plan only to the extent λ_g^{PE} , and it forms that plan at its own perceived path $\hat{z}_g$. To make this precise, write each household FIRE Jacobian as $J_g^X = \mathcal{E}_g^X + K_g^X$, where $\mathcal{E}_g^X$ contains responses to contemporaneously realized inputs and K_g^X contains the still-future component. Informed households respond to both components, evaluating the latter at $\hat{z}_g$, while uninformed households respond only to realized inputs. Taking the λ_g^{PE} -weighted average of the two, group behavior in sequence-space form is

$$X_g = \mathcal{E}_g^X z + \lambda_g^{\text{PE}} K_g^X \hat{z}_g + (\mathcal{E}_g^{X\varepsilon} + \lambda_g^{\text{PE}} K_g^{X\varepsilon}) \varepsilon, \quad X \in \{C, A\}, \quad (16)$$

and aggregates are $X = \sum_g \mu_g X_g$. Thus λ_g^{PE} governs how much a group acts on its forecast, and λ_g^{GE} governs the equilibrium path it forecasts. Setting both weights to one gives FIRE. Appendix G reports the matrix implementation and equilibrium conditions.

4.1.6 General equilibrium

The equilibrium conditions in log-linearized form are:

$$y_t = \sum_g \mu_g c_{g,t} \quad \text{goods market} \quad (17)$$

$$a_t = \sum_g \mu_g a_{g,t} \quad \text{asset market} \quad (18)$$

$$\pi_t = \beta_{nk} \pi_{t+1} + \kappa_w \left[\frac{1}{\sigma} C_t + \frac{1}{\psi} Y_t + \tilde{\tau}_t^{\text{eff}} \right] \quad \text{wage Phillips curve} \quad (19)$$

$$i_t = \rho_i i_{t-1} + (1 - \rho_i) (\phi_\pi \pi_t + \phi_y Y_t) \quad \text{Taylor rule} \quad (20)$$

$$r_t = i_{t-1} - \pi_t \quad \text{Fisher equation} \quad (21)$$

$$b_t = (1 + R^{ss}) b_{t-1} + \frac{B^{ss}}{Y^{ss}} r_t - \Delta T_t. \quad \text{government budget.} \quad (22)$$

The Fisher equation is written without an expectation operator because the model is solved along a perfect-foresight transition, in which the realized and expected inflation paths coincide for the fully informed policy block.

Household responses take the sequence-space form

$$c_g = \mathcal{E}_g^c \hat{z} + \lambda_g^{\text{PE}} K_g^c \hat{z}_g + [\mathcal{E}_g^{c\varepsilon} + \lambda_g^{\text{PE}} K_g^{c\varepsilon}] \varepsilon \quad (23)$$

$$a_g = \mathcal{E}_g^a \hat{z} + \lambda_g^{\text{PE}} K_g^a \hat{z}_g + [\mathcal{E}_g^{a\varepsilon} + \lambda_g^{\text{PE}} K_g^{a\varepsilon}] \varepsilon \quad (24)$$

where $z = (y, \pi, i, r, b)^\top$, J_g^X is the FIRE Jacobian, $\mathcal{E}_g^X$ is the myopic Jacobian that retains only contemporaneous responses, and $K_g^X = J_g^X - \mathcal{E}_g^X$ is the forward-looking kernel, for $X \in \{c, a\}$.

4.2 Calibration

4.2.1 Belief parameters

First-order beliefs. I measure the first-order belief parameter using the relative response of subjective and objective tax changes to the same fiscal-news shock. For income group g , define

$$\lambda_g^p = \frac{\beta_g^{\text{survey}}}{\beta_g^{\text{objective}}}, \quad (25)$$

where β_g^{survey} denotes the response of group g 's expected household tax burden and $\beta_g^{\text{objective}}$ denotes the response of the corresponding group-specific objective tax measure. This equation (25) refers to a Wald IRF ratio: it compares two responses to the same fiscal-news shock.

This relative-response construction follows the measurement logic used by Angeletos et al. [2025]. In their monetary-policy application, the belief parameter is measured by the response of expected interest rates relative to the response of realized interest rates to the same monetary policy shock. I apply the analogous construction to household tax expectations and objective tax changes.

The two responses are estimated separately from

$$\text{objective}_{t+12|t,g} = \alpha_g^{\text{objective}} + \beta_g^{\text{objective}} \varepsilon_t^{\text{FIRE}} + \Gamma_g^{\text{objective}} W_{t-1} + u_{g,t}^{\text{objective}}, \quad (26)$$

$$\text{survey}_{t+12|t,g} = \alpha_g^{\text{survey}} + \beta_g^{\text{survey}} \varepsilon_t^{\text{FIRE}} + \Gamma_g^{\text{survey}} W_{t-1} + u_{g,t}^{\text{survey}}, \quad (27)$$

where ε_t^{FIRE} is the FIRE fiscal-deficit news shock and W_{t-1} contains predetermined controls. The estimated belief parameter is therefore

$$\hat{\lambda}_g^p = \frac{\hat{\beta}_g^{\text{survey}}}{\hat{\beta}_g^{\text{objective}}}. \quad (28)$$

Measurement. The subjective tax-change variable, $\text{survey}_{t+12|t,g}$, is constructed from Q27 of the monthly SCE Core Survey. Q27 asks respondents how much they expect their total household tax payments to change over the next twelve months, holding total household income fixed. The response is therefore interpreted as an expected percentage change in the household’s effective average total-tax burden, rather than as an expectation of a statutory or marginal income-tax rate.

The question covers federal, state, and local income taxes as well as property and sales taxes. I therefore construct an objective measure that aligns as closely as possible with this broad tax concept. Because no single data observes quarterly tax-liability changes separately by income group, I combine aggregate and micro-level data. The Consumer Expenditure Surveys (CEX) provides the cross-sectional distribution of taxable income and tax liabilities at an annual frequency, whereas the National Income and Product Accounts (NIPA) provides the quarterly aggregate effective tax rate. Combining these sources yields the quarterly group-specific objective tax rate

$$\tau_{g,t} = \kappa_g^{\text{CEX}} \tau_t^{\text{NIPA}}.$$

The objective tax-change variable, $\Delta\tau_{t+12|t,g}^{\text{obj}}$, is then constructed by scaling the realized twelve-month change in the aggregate effective personal-tax rate over the corresponding twelve-month horizon by the group-specific tax-incidence loading, κ_g^{CEX} . Appendix G provides more details about estimating κ_g^{CEX} from CEX. AppTable 6 summarizes the data sources, variable definitions, and tax coverage used in this construction.

Table 6. Variables and Data Sources

Object	Data and measure	Coverage
Subjective tax expectation	SCE, Q27	Federal, state, and local income, property, and sales taxes
Objective agg. tax rate	NIPA, W055RC1Q027SBEA/PI	Federal, state, and local personal current taxes relative to personal income
Group tax incidence	CEX, IRS	Total income share relative to broad-base tax share that includes federal and state income tax, property tax, estimated sales tax share.

Notes: Federal and state incidence components are combined using their shares in each group’s federal-plus-state income-tax liability. AGI-based proxy for state income-tax exposure relative to state gross income-tax liability

The fiscal-news shock, ε_t^{FIRE} , is constructed from the daily high-frequency fiscal-news series of Fritsch and Mazlish (2026). Daily innovations are summed within each calendar quarter to match the quarterly aggregation of the SCE responses. The resulting quarterly shock is normalized by its standard deviation over the common estimation sample. One limitation of this shock is that this is a fiscal-deficit news shock rather than a tax-policy shock. It can therefore alter households' expectations of future tax burdens without generating a comparably large change in realized taxes over the same twelve-month horizon. Consistent with this distinction, the estimated objective-tax responses are less precise than the survey responses, as shown in Table 7.

Estimates. Table 7 presents the estimated responses and the implied belief parameters.

$$\hat{\lambda}_{g,2SLS} = \frac{\hat{\beta}_g^{\text{survey}}}{\hat{\beta}_g^{\text{objective}}}.$$

The point estimates lie below one in all specifications, ranging from 0.115 to 0.685. In the benchmark specification—the total-personal-tax measure excluding the Covid period—the estimated belief parameters are 0.310 for the lower-income group and 0.685 for the higher-income group. I use these ratios as the benchmark calibration targets for the quantitative model. The point estimates also display a consistent ordering, with a larger value for the higher-income group, although this ordering should be interpreted cautiously given the limited precision of the denominator.

Table 7. Household Fiscal Beliefs to Tax Changes by Group

Sample	Group	β_{survey} (s.e.)	$\beta_{\text{objective}}$ (s.e.)	First-stage F	λ_{2SLS} (s.e.)	N
<i>Panel A: Total income tax</i>						
Baseline	H ($\leq$ \$50k)	0.188 (0.099)	1.629 (1.168)	1.95	0.115 (0.086)	46
Baseline	L ($>$ \$50k)	0.158 (0.104)	0.835 (0.623)	1.80	0.188 (0.211)	46
Exclude Covid	H ($\leq$ \$50k)	0.351 (0.175)	1.131 (1.319)	0.73	0.310 (0.253)	37
Exclude Covid	L ($>$ \$50k)	0.380 (0.058)	0.555 (0.686)	0.65	0.685 (0.789)	37

Notes: β_{survey} is the response of group g 's SCE expectation of the average income-tax rate to the fiscal-news shock, and $\beta_{\text{objective}}$ the response of the realized CPS-based tax rate, scaled by the group's tax incidence κ_g (1.74 for group H and 0.94 for group L). $\lambda_{2SLS} = \beta_{\text{survey}}/\beta_{\text{objective}}$ is the just-identified IV coefficient. Standard errors are in parentheses.

The precision of $\hat{\lambda}_g^{\text{PE}}$ is limited by the objective-tax response in its denominator. Although the estimated objective-tax responses $\hat{\beta}_{\text{objective}}$ are positive, none is statistically distinguishable from zero. The weak objective response is consistent with fiscal news affecting expectations before announced policies are implemented and reflected in realized effective tax rates. Consequently, the response ratios are sensitive to sampling variation in the denominator and should not be interpreted as precisely estimated structural parameters. I therefore use the estimated ratios

as benchmark calibration targets and report the model’s implications over alternative values of λ_g^{PE} in the economically admissible interval $[0, 1]$. Weak-denominator-robust confidence sets, reported in Appendix E, are unbounded and confirm the substantial uncertainty surrounding the level of the parameter.

Higher-order beliefs. The second-order weight λ_g^{GE} is not separately identified from aggregate dynamics. By equation (15) it multiplies the entire general-equilibrium continuation, so its effect on the impulse responses is observationally close to that of the other equilibrium elasticities, and a fiscal-shock impulse response cannot distinguish it from the slope of the Phillips curve or the monetary rule. I therefore restrict it rather than estimate it, imposing that informed households hold *correct* beliefs about the population-average informed share,

$$\lambda_H^{\text{GE}} = \lambda_L^{\text{GE}} = \mu_H \lambda_H^{\text{PE}} + \mu_L \lambda_L^{\text{PE}} = 0.5498. \quad (29)$$

This is the natural agnostic benchmark. Section 4.5 reports the sacrifice-ratio decomposition over this range and shows that the result is driven by the ordering of the first-order shares $\lambda_H^{\text{PE}} < \lambda_L^{\text{PE}}$, not by the level of λ^g .

4.2.2 Model parameters

Table 8 reports the main quarterly parameters and targets. I keep the group classification from the motivating empirical evidence: low income group and mid/high income group. The population and labor-income shares come from the CPS. Government debt is the only traded asset, with $B_{ss}/Y_{ss} = 1.04$ from Angeletos et al. [2024]. I set the debt-feedback parameter in the tax rule to $\phi_b = 0.12$, the minimum value that ensures equilibrium determinacy at the calibrated steady-state debt level, $B_{ss} = 1.04$. Appendix F provides the corresponding determinacy analysis.

Table 8. Quarterly calibration

Parameter or target	Value	Interpretation
<i>A. Steady state and group composition</i>		
$Y^{ss}, B^{ss}/Y^{ss}$	1, 1.04	Output normalization; liquid assets
$\mu_H, s_Y^H, s_H^{\text{liquid}}$	0.361, 0.0617, 0.0645	H population, income, and asset shares
Aggregate one-quarter MPC	0.25	Targeted rebate MPC
<i>B. Preferences and idiosyncratic risk</i>		
EIS, Frisch	0.5, 0.5	Preference elasticities
$\rho_z, \sigma_z, \underline{a}$	0.966, 0.50, 0	Earnings process; no borrowing
<i>C. Fiscal block</i>		
$r^{ss}, \bar{\tau}_H, \bar{\tau}_L, G^{ss}/Y^{ss}$	0.005, 0.05, 0.18, 0.16	Steady fiscal block
$\rho_\tau, \phi_b, \varepsilon_0^{\tau}$	0.80, 0.12, 0.01	Tax persistence, debt feedback, shock
<i>D. Monetary policy and nominal rigidities</i>		
ρ_i, ϕ_π, ϕ_y	0.70, 1.50, 0.125	Taylor rule
β_{nk}, κ_w	0.99, 0.0062	Wage Phillips curve

Using 2022, I compute ℓ_g^{SCF} , the share of households in group g with non-positive liquid assets, as an empirical proxy for the hand-to-mouth share π_g .

The SCF measure of market-liquid assets includes transaction accounts, certificates of deposit, directly held mutual funds, stocks, and bonds, while excluding retirement accounts, housing, business equity, and other illiquid wealth. It is therefore close to the broad liquid-asset definition used by Kaplan, Violante, and Weidner (2014). This share cannot be equated directly with π_g , however, because some unconstrained households in the model choose to hold zero wealth. Setting $\pi_g = \ell_g^{\text{SCF}}$ would therefore count liquidity-poor households both as rule-of-thumb types and as constrained optimizers. I therefore use the SCF only to discipline the *relative* incidence of liquidity poverty across groups and let the aggregate MPC target determine its overall level:

$$\pi_g = \varphi \ell_g^{\text{SCF}}, \quad g \in \{H, L\}.$$

where the common scaling factor φ preserves the ratio of liquidity-poor shares across groups. The scaling factor φ and the discount factors β_H and β_L are jointly chosen to match three moments: a population-weighted one-quarter transfer MPC of 0.25, conservative relative to the tax-rebate literature [e.g., Parker et al., 2013], asset-market clearing at B^{ss} , and group H's share of aggregate liquid assets in the SCF. The total mass at the borrowing limit, including both rule-of-thumb households and constrained optimizers, is not targeted and is reported below.

Table 9. Stationary household moments

Moment	H	L	Aggregate
Population share	0.3606	0.6394	1.0000
Optimizer discount factor	0.9605	0.9622	
Annualized discount factor, β_g^4	0.8512	0.8573	
Mean liquid assets per capita	0.1861	1.5217	1.0400
Steady consumption per capita	0.1730	1.2324	0.8503
One-quarter transfer MPC	0.2654	0.2413	0.2500
Four-quarter cumulative transfer MPC	0.4408	0.4256	0.4311
Structural HtM share within group	0.0741	0.0441	0.0549
Total zero-asset share within group	0.4645	0.3059	0.3631

Notes: Group-level assets and consumption are per capita within the group; aggregate entries use population weights. The aggregate entry in the structural-HtM row is the population share. Total zero-asset shares include structural HtM households and optimizing households at the borrowing limit.

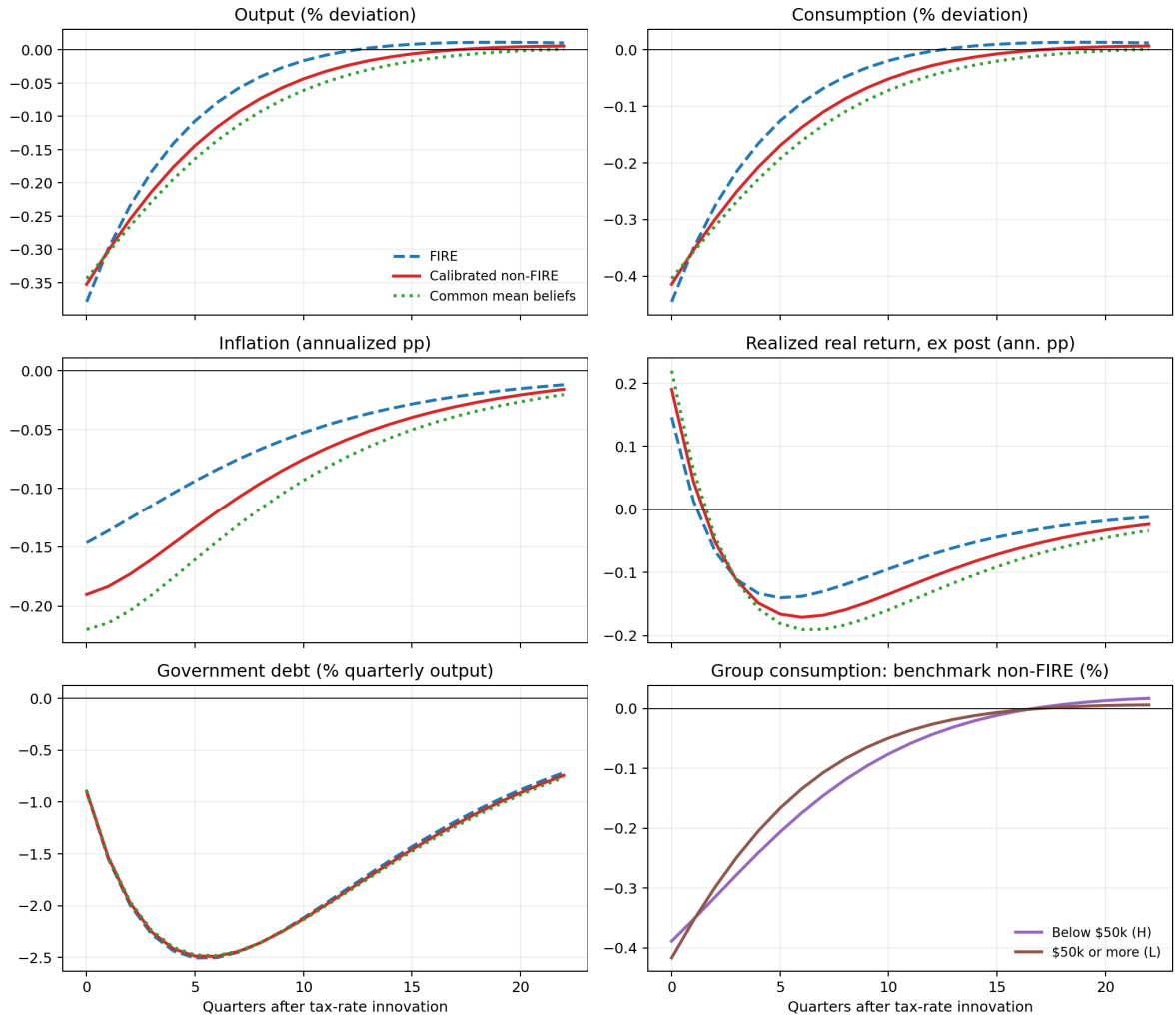
Table 9 reports the resulting stationary household moments. Because group-specific MPCs are not separately targeted, the implied one-quarter MPC difference is modest: the MPC is 0.2654 for group H and 0.2413 for group L. The ranking persists over a four-quarter horizon, with cumulative MPCs of 0.4408 and 0.4256, respectively. This muted MPC heterogeneity reflects the construction of the groups. To align the model with the SCE, households are sorted by income rather than directly by their MPCs. Because income is an imperfect proxy for liquidity and consumption responsiveness, income-defined groups need not exhibit sharply separated MPCs at every horizon. Nevertheless, the below-\$50,000 group has the higher MPC at both horizons. The aggregate zero-asset share is 0.3631. This untargeted moment combines structural hand-to-mouth households with optimizing households at the borrowing limit. It is partly elevated

by the narrow, deposit-based definition of liquid assets used to construct ℓ_g^{SCF} , which raises the measured incidence of liquidity-poor households.

4.3 Quantitative results

Figure 7 compares the responses to a one-percentage-point innovation in the labor-income tax rate under three belief specifications: full-information rational expectations (FIRE), common mean beliefs, and calibrated heterogeneous beliefs. Under the common-belief counterfactual, both groups receive the population-average first-order attention weight, $\lambda_H^{\text{PE}} = \lambda_L^{\text{PE}} = 0.525$, and the higher-order weight remains common across groups, as in the baseline. The counterfactual therefore preserves the population-average belief while removing its cross-group dispersion. I organize the discussion around two steps. Moving from FIRE to common beliefs isolates the *level* effect of fiscal beliefs. Moving from common to heterogeneous beliefs isolates its *dispersion* effect.

Figure 7. Impulse Responses Under Labor-income-tax Consolidation



Level effect of fiscal beliefs: from FIRE to common beliefs. Consumption falls by 0.446 percent on impact under FIRE but by only 0.404 percent under common beliefs, and output falls by 0.379 and 0.344 percent, respectively. Every household observes its current tax payment,

so the attenuation does not come from the current tax. It reflects the underweighting of the expected continuation of the tax increase and of its general-equilibrium consequences.

The same underweighting makes the contraction more persistent. FIRE households anticipate the future relief as well as the future burden: the tax innovation decays, debt reduction lowers future taxes through the debt-feedback term of the fiscal rule, and real rates fall. Non-FIRE households underweight this relief as well, so their spending recovers more slowly. Output returns to steady state in quarter 13 under FIRE but only in quarter 22 under common beliefs.

Because the wage Phillips curve is forward looking and wage setters have full information, the more persistent contraction in demand translates into a deeper decline in wage and price inflation, even though the impact decline in output is smaller. Impact inflation is -0.146 annualized percentage point under FIRE and -0.220 under common beliefs. Because the policy rule is inertial, the nominal rate initially falls by less than expected inflation, and the impact real rate rises by more under common beliefs (0.146 versus 0.220). As the policy rate cuts accumulate, the real rate turns negative from the second quarter onward and falls well below its FIRE counterpart (-0.083 versus -0.145 at $h = 11$).

Dispersion effect of fiscal beliefs: from common to heterogeneous beliefs. The heterogeneous-belief economy lies between FIRE and common beliefs for every variable and at every horizon. Impact inflation is -0.190 , the impact real rate is 0.190 , output is 0.033 percent below steady state at $h = 11$, and output returns to steady state in quarter 19. Belief dispersion therefore moves the economy back toward FIRE.

The group-level responses clarify this mechanism. Group L perceives a tax shock more correctly ($\lambda_L^p = 0.685$) than the common benchmark, leading to a faster consumption recovery than under common beliefs (-0.249 versus -0.269 percent at $h = 3$). Group H , by contrast, perceives the tax shock less precisely ($\lambda_H^{PE} = 0.303$), resulting in a slower recovery (-0.278 versus -0.256 percent). While the two groups feature comparable marginal propensities to consume (0.265 and 0.241), their shares in aggregate spending are highly asymmetric: Group H accounts for just 7.3 percent of aggregate consumption and 6.4 percent of liquid assets. The aggregate response is therefore governed mainly by group L . Weighting these beliefs by consumption shares yields an effective aggregate belief parameter of about $0.073 \times 0.310 + 0.927 \times 0.685 \approx 0.658$, substantially above the population average of 0.545 . Belief heterogeneity thus places beliefs where most of the expectation-sensitive spending occurs.

This is the quantitative counterpart of the mechanism derived in Section 3.2. The aggregate path depends not only on the population mean of the informed share but also on its covariance with each group's weight in aggregate demand, including its marginal propensity to consume, its liquid-asset share, and its labor-income share. In this calibration the MPC gap is small, so the covariance operates mainly through the consumption, asset, and income shares. A representative-agent or common-belief specification sets this covariance to zero by construction and therefore understates effective fiscal beliefs.

Debt dynamics. Surprisingly, debt dynamics are similar across the three specifications, even though the paths of output and real rates differ markedly. At the end of year 5, debt is only 0.032

percent of quarterly output lower under heterogeneous beliefs and 0.049 percent lower under common beliefs than under FIRE, or 3.3 and 5.0 percent of FIRE debt reduction, respectively. To understand why, I decompose the belief-induced difference in debt using the government budget constraint. (With the inclusion of debt maturity, however, debt dynamics differ across specifications; see Appendix H for details.)

Tax revenue consists of three components: the direct revenue from the exogenous tax-rate innovation, $\varepsilon_t Y^{ss}$, the revenue from changes in the tax base, $\bar{\tau} dY_t$, and the debt-feedback revenue implied by the fiscal rule, $\phi_b db_{t-1}$, where d denotes the deviation from steady state. Because the innovation path is exogenous and common to all specifications, the direct revenue is identical across specifications by construction.

Beliefs therefore affect government debt through three channels. First, they change the path of the real interest rate and hence the cost of servicing existing debt. Second, they change the path of output and hence the endogenous labor-income tax base. Third, the fiscal rule responds to the resulting difference in debt through debt-feedback revenue. Let Δ denote the deviation from the FIRE path. Taking the difference of the linearized government budget constraint gives

$$\begin{aligned}\Delta b_t &= (1 + r^{ss}) \Delta b_{t-1} + B^{ss} \Delta r_t - \bar{\tau} \Delta Y_t - \phi_b \Delta b_{t-1} \\ &= \underbrace{B^{ss} \Delta r_t}_{\text{interest}} + \underbrace{-\bar{\tau} \Delta Y_t}_{\text{tax base}} + \underbrace{(1 + r^{ss} - \phi_b) \Delta b_{t-1}}_{\text{inherited gap}},\end{aligned}\tag{30}$$

with $1 + r^{ss} - \phi_b = 0.885$. The last term is the debt gap inherited from the previous quarter, net of the debt-feedback response of the fiscal rule. The first two terms are the new contributions of the interest and tax-base channels in quarter t . Table 10 reports this decomposition.

Table 10. Debt Dynamics Decomposition

Horizon	Interest $B^{ss} \Delta r_t$	Tax base $-\bar{\tau} \Delta Y_t$	Inherited gap $(1 + r^{ss} - \phi_b) \Delta b_{t-1}$	Total Δb_t
<i>Panel A: Heterogeneous beliefs — FIRE</i>				
Impact ($h = 0$)	+0.009	−0.005	0.000	+0.004
End of year 1 ($h = 3$)	−0.004	+0.005	+0.010	+0.011
End of year 3 ($h = 11$)	−0.011	+0.005	−0.017	−0.023
End of year 5 ($h = 19$)	−0.005	+0.002	−0.029	−0.032
<i>Panel B: Common beliefs — FIRE</i>				
Impact ($h = 0$)	+0.014	−0.006	0.000	+0.008
End of year 1 ($h = 3$)	−0.007	+0.008	+0.015	+0.016
End of year 3 ($h = 11$)	−0.016	+0.007	−0.026	−0.035
End of year 5 ($h = 19$)	−0.008	+0.003	−0.044	−0.049

The interest and tax-base channels move in opposite directions at every horizon and therefore largely offset each other. On impact, the higher real rate under non-FIRE beliefs raises debt service, while the smaller output decline sustains the tax base. From the first year onward, the signs reverse: the lower real rate reduces debt service, while the slower recovery weakens the tax base. Because each quarter adds only a small net contribution, the debt gap never

builds up. The fiscal rule also lowers the persistence of the gap from $1 + r^{ss}$ to 0.885, so the inherited gap decays with a half-life of about six quarters. As a result, debt paths remain close across specifications. At the end of year 5, debt is 0.032 percent of quarterly output lower under heterogeneous beliefs and 0.049 percent lower under common beliefs than under FIRE, or 3.3 and 5.0 percent of FIRE debt reduction, respectively.

The offset also explains why the output cost, rather than debt reduction, drives the differences in consolidation effectiveness in Section 4.4. The same slow recovery that raises the cumulative output loss also lowers real rates and therefore debt service, while the fiscal rule gradually eliminates any remaining difference in debt. Belief distortions that substantially change the path of activity therefore leave the path of debt nearly unchanged.

4.4 Fiscal consolidation effectiveness and output costs

A larger decline in debt need not be a more desirable outcome if it is accompanied by a large and persistent contraction in activity. I therefore evaluate the consolidation along two dimensions: its *effectiveness* in reducing government debt and the output cost at which that reduction is achieved. Let $s \in \{F, C, H\}$ index the FIRE, common-belief, and heterogeneous-belief scenarios. I normalize the size of the initial fiscal innovation as

$$\bar{\varepsilon} \equiv \frac{100\varepsilon_0}{Y^{ss}} = 1 \quad (31)$$

so the innovation equals one percent of quarterly steady-state output. I define debt reduction, cumulative output loss, the fiscal sacrifice ratio, and fiscal effectiveness as

$$\mathcal{D}_s(h) = -\frac{100 b_h^s / Y^{ss}}{100 d\mathcal{T}_0^{direct} / Y^{ss}}, \quad (32)$$

$$\mathcal{L}_s(h) = -\frac{\sum_{t=0}^h 100 y_t^s / Y^{ss}}{100 d\mathcal{T}_0^{direct} / Y^{ss}}, \quad (33)$$

$$\mathcal{S}_s(h) = \frac{\mathcal{L}_s(h)}{\mathcal{D}_s(h)}, \quad (34)$$

$$\mathcal{E}_s(h) = \frac{1}{\mathcal{S}_s(h)} = \frac{\mathcal{D}_s(h)}{\mathcal{L}_s(h)}. \quad (35)$$

The sacrifice ratio $\mathcal{S}_s(h)$ measures the cumulative output loss per unit of debt reduction, so a lower $\mathcal{S}_s(h)$ means that a given reduction in the debt stock is obtained at a smaller output cost. Fiscal effectiveness $\mathcal{E}_s(h)$ is its inverse and measures the debt reduction obtained per unit of cumulative output loss. I decompose the difference between the heterogeneous-belief economy and FIRE into a level component, $X_C - X_F$, and a dispersion component, $X_H - X_C$, for $X \in \{\mathcal{S}, \mathcal{E}\}$.

Table 11 presents the main quantitative result. Because debt reduction differs by at most 5 percent across specifications (Panel D), differences in the sacrifice ratio and in fiscal effectiveness are driven almost entirely by differences in cumulative output loss (Panel C). Consequently, the ranking across specifications directly mirrors the cumulative output dynamics shown in Figure 7.

Level effect. On impact, the common-belief distortion lowers the sacrifice ratio from 0.420 to 0.387 and raises fiscal effectiveness from 2.281 to 2.584, because non-FIRE fiscal beliefs

Table 11. Beliefs and the Measures of Fiscal Consolidation

Horizon	Scenario			Decomposition of Δ		
	FIRE	Common	Hetero.	Level (F $\rightarrow$ C)	Dispersion (C $\rightarrow$ H)	Disp./ Level
<i>Panel A: Sacrifice ratio $\mathcal{S}_s(h)$ (cumulative output loss per unit debt reduction)</i>						
Impact ($h = 0$)	0.420	0.387	0.394	-0.033	0.007	-0.21
End of year 1 ($h = 3$)	0.481	0.510	0.498	0.028	-0.012	-0.41
End of year 3 ($h = 11$)	0.794	1.011	0.932	0.217	-0.079	-0.37
End of year 5 ($h = 19$)	1.552	2.111	1.903	0.559	-0.208	-0.37
<i>Panel B: Fiscal effectiveness $\mathcal{E}_s(h)$ (debt reduction per unit of cumulative output loss)</i>						
Impact ($h = 0$)	2.381	2.584	2.538	0.203	-0.046	-0.22
End of year 1 ($h = 3$)	2.079	1.961	2.008	-0.118	0.047	-0.40
End of year 3 ($h = 11$)	1.259	0.989	1.073	-0.270	0.084	-0.31
End of year 5 ($h = 19$)	0.644	0.474	0.525	-0.171	0.052	-0.30
<i>Panel C: Cumulative output loss $\mathcal{L}_s(h)$ (pp-quarters)</i>						
Impact ($h = 0$)	0.379	0.344	0.353			
End of year 1 ($h = 3$)	1.095	1.141	1.122			
End of year 3 ($h = 11$)	1.572	2.028	1.859			
End of year 5 ($h = 19$)	1.514	2.165	1.913			
<i>Panel D: Debt reduction $\mathcal{D}_s(h)$</i>						
Impact ($h = 0$)	0.903	0.890	0.896			
End of year 1 ($h = 3$)	2.276	2.240	2.254			
End of year 3 ($h = 11$)	1.980	2.005	1.995			
End of year 5 ($h = 19$)	0.975	1.026	1.005			

Notes: Because $\mathcal{E}$ is a nonlinear transformation of $\mathcal{S}$, the Disp./Level ratios differ across Panels A and B.

attenuate the immediate contraction in spending. The ranking reverses within the first year, as the contraction becomes more persistent under common beliefs, corresponding to the crossing of the output IRF paths. By the end of year 5, cumulative output loss is 2.165 percentage-point quarters under common beliefs, compared with 1.514 under FIRE, an increase of about 45 percent. Debt reduction rises by only about 5 percent, from 0.975 to 1.026. The sacrifice ratio consequently increases from 1.552 to 2.111, and fiscal effectiveness falls from 0.644 to 0.474, a decline of about 27 percent. The average belief distortion thus shifts the output cost of consolidation from impact toward later quarters and lowers its medium-run effectiveness.

Dispersion effect. Belief dispersion moves both measures in the opposite direction from the level effect at every horizon. On impact it slightly raises the sacrifice ratio (+0.007) and lowers effectiveness (-0.046). From the first year onward, as the output recovery under dispersion accelerates relative to common beliefs (Figure 7), it lowers the sacrifice ratio (-0.079 at $h = 11$ and -0.208 at $h = 19$) and raises effectiveness (+0.084 and +0.052). Over the medium run, the dispersion component offsets about one third of the level effect on the sacrifice ratio ($0.208/0.559 \simeq 0.37$) and about one third of the level effect on effectiveness ($0.052/0.171 \simeq 0.30$).

at $h = 19$). The two offsets differ only because $\mathcal{E}$ is the inverse of $\mathcal{S}$. Both show that the heterogeneous-belief economy lies between FIRE and common beliefs. At the end of year 5, fiscal effectiveness under heterogeneous beliefs is 0.525, about 10 percent higher than under common beliefs. This offset reflects the mechanism in Section 4.3: heterogeneous beliefs place close to full-information rational-expectations on the households that account for most aggregate spending, which brings the economy closer to FIRE.

Implications. The results have two implications for tax-based consolidation under non-FIRE fiscal beliefs. First, misperception postpones the output cost of consolidation. Because households underweight the continuation of the tax increase and its general-equilibrium consequences, the impact contraction is attenuated, but the recovery is delayed. Over the medium horizon, a consolidation that retires essentially the same amount of debt therefore incurs a larger cumulative output cost. Evaluations based on short-run responses thus understate the cost of consolidation when beliefs depart from full information.

Second, this cost depends on how misperception is distributed across households, not only on its average. A common-belief specification with the same population-average fiscal beliefs understates medium-run fiscal effectiveness by about 10 percent (0.474 versus 0.525), because it ignores that beliefs closer to full-information rational-expectation are concentrated among the households that account for most aggregate spending. Belief dispersion offsets roughly one third of the level effect on the sacrifice ratio, so a representative-agent or common-belief specification overstates the cost of non-FIRE beliefs by a quantitatively meaningful margin.

4.5 Robustness

I examine two sources of fragility: the timing of tax perception and parameter uncertainty. Both exercises confirm the two components of the main result: the average level of misperception raises the medium-run output cost of consolidation, and belief dispersion offsets roughly one third of this increase.

4.5.1 An alternative tax timing

The baseline assumes that households observe their current taxes. In the alternative timing, a fraction $1 - \lambda_i^p$ of group i does not perceive the current tax innovation when planning. The unperceived liability is settled after the choice is made: unconstrained optimizers absorb it by reducing end-of-period assets, optimizers at the borrowing limit absorb it by reducing current consumption, and structural hand-to-mouth households reduce current consumption one for one. Because 24.9 percent of optimizers in group H and 25.1 percent in group L are at the borrowing limit, the share of an unperceived liability absorbed through current consumption on impact is 31.1 percent in group H and 28.8 percent in group L once structural hand-to-mouth households are included. The remainder is absorbed through assets.

Table 12 reports the results. The settlement timing deepens the impact contraction in both non-FIRE economies: impact output falls by 0.360 percent under heterogeneous beliefs and 0.359 percent under common beliefs, compared with 0.341 and 0.335 percent in the baseline and 0.370 percent under FIRE. Because unperceived current taxes are still paid, most of the

Table 12. Robustness: Tax Timing

<i>Panel A: Sacrifice ratio $\mathcal{S}_s(h)$</i>					
Horizon	FIRE	Baseline timing		Settlement timing	
		Common	Hetero.	Common	Hetero.
Impact ($h = 0$)	0.397	0.362	0.367	0.392	0.390
End of year 1 ($h = 3$)	0.468	0.492	0.481	0.547	0.521
End of year 3 ($h = 11$)	0.800	1.019	0.946	1.157	1.044
End of year 6 ($h = 23$)	2.333	3.248	2.948	3.720	3.286

<i>Panel B: Decomposition into level ($F \rightarrow C$) and dispersion ($C \rightarrow H$)</i>						
Horizon	Baseline timing			Settlement timing		
	Level	Dispersion	Disp./Level	Level	Dispersion	Disp./Level
Impact ($h = 0$)	−0.034	+0.005	−0.15	−0.004	−0.002	+0.50
End of year 1 ($h = 3$)	+0.024	−0.011	−0.46	+0.079	−0.026	−0.33
End of year 3 ($h = 11$)	+0.219	−0.073	−0.33	+0.357	−0.113	−0.32
End of year 6 ($h = 23$)	+0.915	−0.300	−0.33	+1.387	−0.434	−0.31

Notes: $\mathcal{S}_s(h) = \mathcal{L}_s(h)/\mathcal{D}_s(h)$, normalized by the impact direct-tax revenue. The FIRE path is identical under both timings. Level = $\mathcal{S}_C - \mathcal{S}_F$ and Dispersion = $\mathcal{S}_H - \mathcal{S}_C$; a negative component lowers the sacrifice ratio. At impact under the settlement timing both components are close to zero, so their ratio is not informative.

impact attenuation of the baseline disappears, and the level component on impact is close to zero (−0.004). Beyond impact, settlement raises the sacrifice ratio in both non-FIRE economies at every horizon, and the level component becomes larger: +0.357 at $h = 11$ and +1.387 at $h = 23$, compared with +0.219 and +0.915 in the baseline. The dispersion component grows in proportion, to −0.113 and −0.434. As a result, the ratio of the two components is essentially unchanged: dispersion offsets 32 and 31 percent of the level component at the end of years 3 and 6, compared with 33 percent in the baseline. The tax timing therefore affects the size of the belief distortion but not the share of it that is offset by the cross-group allocation of beliefs.

4.5.2 Prior predictive analysis.

The second exercise addresses the concern that the results reflect weakly identified or conventionally chosen parameters. I place priors over the belief parameters and the aggregate block, solve the model for each draw, and report the induced distribution of the decomposition.

Table 13 lists the priors. The first-order belief parameters are drawn from Beta priors centered on the survey estimates, with standard deviations of about 0.13; λ^g is drawn uniformly on $[0.05, 1]$, essentially its whole admissible range, because it is not separately identified. The debt-feedback coefficient, tax-shock persistence, wage Phillips-curve slope, and Taylor-rule coefficients are drawn over conventional ranges. The prior does not impose $\lambda_H^p < \lambda_L^p$, so draws that reverse the estimated ordering of beliefs remain in the sample and provide a falsification test. Of 600 draws, all solve and 516 yield a determinate equilibrium. I use the determinate draws throughout.

The baseline-prior column of Table 14 reports the results. Three findings stand out. First, both components of the decomposition have robust signs beyond impact. The level component

Table 13. Priors for the Prior Predictive Analysis

Parameter	Description	Prior	Baseline
<i>Belief parameters</i>			
λ_H^p	Informed share, group H	Beta(mean 0.303, $\nu=12$)	0.303
λ_L^p	Informed share, group L	Beta(mean 0.651, $\nu=12$)	0.651
λ^g	Second-order (believed) share	U(0.05, 1.00)	0.525
<i>Fiscal and nominal block</i>			
ϕ_b	Debt feedback in the tax rule	U(0.10, 0.25)	0.12
ρ_τ	Tax-shock persistence	Beta(mean 0.80, $\nu=20$)	0.80
κ_w	Wage Phillips-curve slope	LogNormal(med 0.0062, $\sigma_{\log}=0.5$)	0.0062
ϕ_π	Taylor rule, inflation	U(1.25, 2.50)	1.50
ϕ_y	Taylor rule, output	U(0.00, 0.25)	0.125
ρ_i	Taylor rule, inertia	U(0.50, 0.85)	0.70
<i>Household block (drawn only in the extended exercise)</i>			
$\overline{\text{MPC}}$	One-quarter aggregate rebate MPC	U(0.15, 0.30)	0.25
B^{ss}	Aggregate liquid-asset supply	U(1.04, 2.60)	1.04

Notes: ν is the concentration of the Beta prior; the implied standard deviations are about 0.13 for λ_H^p and λ_L^p and about 0.09 for ρ_τ . With $r^{ss} = 0.005$, the ϕ_b range implies a half-life of the debt gap between about seven quarters at the lower end and two and a half quarters at the upper end, and contains the baseline value of 0.12.

raises the sacrifice ratio in 80 to 95 percent of draws at the end of years 1, 3, and 6, while the dispersion component lowers it in 91 to 96 percent. The two components have opposite signs in 80 to 91 percent of draws, and the median ratio of dispersion to level is -0.26 , -0.31 , and -0.33 at these horizons, close to the one-third offset in the baseline calibration. On impact, the level component instead lowers the sacrifice ratio in most draws, consistent with the attenuation of the impact contraction in Section 4.3.

Second, the sign of the dispersion component is determined by the ordering of beliefs across groups. Among the 495 draws with $\lambda_L^p > \lambda_H^p$, the dispersion component is negative at $h = 23$ in every draw. Among the 21 draws that reverse the ordering, it is negative in none. The magnitude of the offset also increases with the belief gap (correlation -0.38). The mechanism is therefore governed by which group is better informed, as the analytical decomposition in Section 3.2 implies.

Third, λ^g is close to irrelevant. Its correlation with the dispersion component at $h = 23$ is $+0.14$, and the dispersion component is negative in 95 to 97 percent of draws within each tercile of the λ^g prior. This supports the identification argument of Section 4.2 and justifies restricting λ^g rather than estimating it.

One caveat should be stated plainly. The analysis retains only draws that yield a determinate equilibrium, which excludes 14 percent of the baseline draws. In every retained draw, debt reduction is positive at every reported horizon, so the sacrifice ratio is well defined and no further selection is required. The evidence should nonetheless be read as support for the signs of the two components within determinate, debt-reducing calibrations, not as a probability

Table 14. Prior Predictive Analysis: Baseline and Extended Priors

	Baseline priors		Extended priors	
Horizon	Level	Dispersion	Level	Dispersion
<i>Share of draws in which the component lowers $\mathcal{S}$</i>				
Impact ($h = 0$)	0.84	0.46	0.47	0.77
End of year 1 ($h = 3$)	0.20	0.91	0.10	0.95
End of year 3 ($h = 11$)	0.06	0.96	0.06	0.95
End of year 6 ($h = 23$)	0.05	0.96	0.04	0.95
<i>Dispersion component at $h = 23$, by sign of $\lambda_L^p - \lambda_H^p$</i>				
$\lambda_L^p > \lambda_H^p$	1.00		1.00	
$\lambda_L^p \leq \lambda_H^p$	0.00		0.00	
<i>Correlation of the signed dispersion component at $h = 23$ with</i>				
$\lambda_L^p - \lambda_H^p$	−0.38		−0.48	
λ^g	+0.14		+0.10	
MPC target	—		+0.03	
B^{ss}	—		+0.25	
Draws used / solved / requested	516/600/600		133/170/200	

Notes: A valid sacrifice ratio requires positive cumulative debt reduction under all three specifications. Statistics use the determinate draws only. The conditional dispersion entries use $n = 495$ and $n = 21$ in the baseline exercise and $n = 127$ and $n = 6$ in the extended exercise. The 30 failed extended draws are parameter combinations for which the MPC target is unattainable. Correlations are between the drawn parameter and the signed dispersion component at $h = 23$; a negative correlation indicates a larger offsetting effect of dispersion. A dash indicates a parameter that is fixed under that prior.

statement over all admissible economies.

Extending the priors to the household block. The baseline prior predictive exercise holds the aggregate MPC target and the liquid-asset supply fixed, because varying them requires resolving the stationary household problem for each draw. As a more demanding check, I also draw these two parameters, the one-quarter aggregate rebate MPC over $[0.15, 0.30]$ and B^{ss} over $[1.04, 2.60]$, so that the degree of MPC heterogeneity and the level of household wealth are themselves randomized. Of 200 draws, 170 solve and 133 yield a determinate equilibrium. The 30 failures are parameter combinations for which no rule-of-thumb scale reproduces the drawn MPC target.

The extended-prior column of Table 14 shows that the conclusions are unchanged. Beyond impact, the level component raises the sacrifice ratio in 90 to 96 percent of draws, and the dispersion component lowers it in 95 percent. The median ratio of dispersion to level is -0.32 , -0.30 , and -0.31 at the end of years 1, 3, and 6. The sign of the dispersion component again tracks the ordering of attention without exception (127 of 127 draws with $\lambda_L^p > \lambda_H^p$, and none of the 6 draws with the reverse ordering), and it remains nearly uncorrelated with λ^g ($+0.10$). The dispersion component is essentially uncorrelated with the drawn MPC target ($+0.03$); its correlation with B^{ss} ($+0.25$) indicates that greater liquid wealth somewhat reduces the size of

the offset, but does not change its sign. The distributional mechanism therefore does not depend on the particular MPC calibration or asset supply of the baseline economy.

Together, the exercises sharpen the main result. The model does not support a claim that non-FIRE fiscal beliefs make consolidation cheaper: the average level of fiscal beliefs raises the medium-run output cost per unit of debt reduction across tax timings and in the large majority of prior draws. It supports the more specific conclusion that, under the estimated ordering in which the higher-income group is more informed, the cross-group allocation of misperception offsets roughly one third of this increase.

5 Conclusion

This paper studies how the distribution of household fiscal beliefs shapes the effectiveness of tax-based fiscal consolidation. Using the FRBNY Survey of Consumer Expectations, I document that tax-hike expectations vary systematically across income and perceived-exposure groups. These relationships motivate a framework in which fiscal beliefs are jointly distributed with MPCs, income, and wealth.

The generalized non-FIRE OLG model isolates the underlying mechanism. In the one-type model, belief distortions affect consumption, the endogenous tax base, and government debt through households' perceived wealth effects. The two-type extension shows that population-average beliefs are not sufficient to characterize these dynamics. Holding average beliefs fixed, reallocating beliefs across household types changes aggregate demand and debt through two distinct margins: a response margin governed by type-specific MPCs and an aggregation-weight margin governed by population and government-debt shares. Because the OLG model abstracts from heterogeneous tax exposure, this result identifies a general belief-allocation mechanism rather than a mechanical consequence of differential tax incidence

I quantify this mechanism in a two-group HANK model with heterogeneous fiscal beliefs, MPCs, liquid wealth, and labor-income exposure. Following a persistent one-percentage-point labor-income-tax increase, average belief distortions attenuate the initial contraction but slow the subsequent recovery. Output falls by 0.370 percent on impact under FIRE, compared with 0.335 percent under common beliefs and 0.341 percent under heterogeneous beliefs. Despite similar debt reductions across the three specifications, the greater persistence of the contraction under common beliefs produces a larger cumulative output loss.

By the end of year six, the fiscal sacrifice ratio—cumulative output loss per unit of debt reduction—is 2.33 under FIRE and 3.25 under common beliefs. Survey-calibrated belief heterogeneity lowers the ratio to 2.95, offsetting approximately one third of the increase generated by average belief distortions. This offset arises because beliefs closer to full information are held by the households that account for most aggregate spending and tax liabilities. The cross-group MPC difference is modest in the calibration, so the quantitative result should be interpreted as reflecting the joint distribution of fiscal beliefs, MPCs, labor income, and liquid wealth rather than as a pure MPC channel. The sign and approximate magnitude of the offset are robust to an alternative timing of tax perception and to prior-predictive exercises that vary belief and

model parameters.

The main implication is that population-average fiscal beliefs are not sufficient for evaluating fiscal consolidation. A common-belief model can overstate the output cost of debt reduction even when it matches the population-average belief parameter, because it removes the cross-household allocation of belief distortions. The output and debt consequences of consolidation depend on whether beliefs closer to full information are held by households with stronger response margins or larger aggregation weights. Here, effectiveness is defined narrowly as debt reduction relative to cumulative output losses. The analysis does not provide a broader normative assessment of fiscal consolidation.

Several limitations remain. First, the group-specific belief parameters are time-invariant. This restriction reflects the short and relatively infrequent survey sample: the SCE Public Policy Survey has been administered three times per year since November 2015. Identifying time variation in households' responsiveness to fiscal news would require a longer and more frequent series of tax expectations together with well-identified tax-specific fiscal shocks. The current data do not provide sufficient variation to distinguish changes in belief formation from changes in the fiscal environment. Second, income is an imperfect proxy for the MPC, and the calibrated MPC gap across income groups is modest. Third, the quantitative model features a single liquid asset and is solved to first order. Future research could combine direct measures of household MPCs with experimentally induced fiscal-information changes and examine the mechanism in a nonlinear two-asset HANK model. These extensions would permit a sharper separation of the response, labor-income, and balance-sheet components of fiscal transmission under heterogeneous beliefs.

References

- Alesina, A., Favero, C., & Giavazzi, F. (2015). The output effect of fiscal consolidation plans. *Journal of International Economics*, 96(S1), S19–S42. DOI: 10.1016/j.jinteco.2014.11.003.
- Andrade, P., Gautier, E., Mengus, E., & Moench, E. (2025). Household Beliefs about Fiscal Dominance. *Federal Reserve Bank of Boston Research Department Working Paper*, No. 25-2 (March). DOI: 10.29412/res.wp.2025.02.
- Angeletos, G.-M., Guerreiro, J., & Zhang, D. R. (2025). From RANK to HANK, without FIRE. *NBER Working Paper*, No. 34596 (December). DOI: 10.3386/w34596.
- Angeletos, G.-M., & Zhen, H. (2021). Myopia and Anchoring. *American Economic Review*, 111(4), 1166–1200.
- Angeletos, G.-M., Lian, C., & Wolf, C. K. (2024). Can Deficits Finance Themselves? *Econometrica*, 92(5), 1351–1390.
- Auclert, A., Bardóczy, B., Rognlie, M., & Straub, L. (2021). Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models. *Econometrica*, 89(5), 2375–2408.
- Bianchi-Vimercati, R., Eichenbaum, M. S., & Guerreiro, J. (2024). Fiscal Stimulus with Imperfect Expectations: Spending vs. Tax Policy. *Journal of Economic Theory*, 217, 105814. DOI: 10.1016/j.jet.2024.105814.

- Bianchi, F., Dabla-Norris, E., & Khalid, S. (2025). Perceptions of Public Debt and Policy Expectations: Evidence from Cross Country Surveys. *IMF Working Papers*, 2025/197. DOI: 10.5089/9798229025751.001.
- Eichenbaum, M. S., Guerreiro, J., & Obradovic, J. (2026). Ricardian Non-Equivalence. *NBER Working Paper*, No. 34691 (January). DOI: 10.3386/w34691.
- Erceg, C. J., & Lindé, J. (2013). Fiscal consolidation in a currency union: Spending cuts vs. tax hikes. *Journal of Economic Dynamics and Control*, 37(2), 422–445. DOI: 10.1016/j.jedc.2012.09.012.
- Farhi, E., & Werning, I. (2019). Monetary Policy, Bounded Rationality, and Incomplete Markets. *American Economic Review*, 109(11), 3887–3928.
- Kaplan, G., Moll, B., & Violante, G. L. (2018). Monetary Policy According to HANK. *American Economic Review*, 108(3), 697–743.
- Gabaix, X. (2020). A Behavioral New Keynesian Model. *American Economic Review*, 110(8), 2271–2327.
- Guerreiro, J. (2023). Belief Disagreement and Business Cycles. *Working Paper*.
- Iovino, L., & Sergeyev, D. (2023). Central Bank Balance Sheet Policies without Rational Expectations. *The Review of Economic Studies*, 90(6), 2991–3023.
- Fritsch, G., & Mazlish, J. (2026). High-frequency Fiscal Shocks. *Working Paper*.
- Mei, P., & Wu, L. (2025). Thinking about the Economy, Deep or Shallow? *Working Paper*.
- Parker, J. A., Souleles, N. S., Johnson, D. S., & McClelland, R. (2013). Consumer spending and the economic stimulus payments of 2008. *American Economic Review*, 103(6), 2530–2553. DOI: 10.1257/aer.103.6.2530.

Appendix A FRBNY SCE Details

A.1 Survey Sample and Data Treatment

The empirical analysis and the calibration of belief parameters use the Federal Reserve Bank of New York Survey of Consumer Expectations (SCE). The SCE Core Survey is administered monthly and elicits consumers' expectations about a range of household-level and aggregate economic and financial conditions. The analysis combines the Core Survey over 2013–2024 with the Public Policy Survey over 2015–2024. The latter is a special module fielded three times per year and elicits subjective beliefs about prospective changes in government policies, including assistance programs, social insurance, labor-market policies, taxes, and government fees. The Public Policy Survey is administered to respondents drawn from the SCE panel and can therefore be linked to their Core Survey responses.

The empirical evidence uses two questions from the Public Policy Survey. QP1 measures beliefs about the direction of future changes in the average income tax rate, while QP2 measures respondents' perceived personal exposure to the policy change described in the survey.

QP1 asks respondents to assign probabilities to alternative changes in the average income tax rate over the next 12 months:

What do you think is the percent chance that over the next 12 months there will be an increase, decrease, or no change in the average income tax rate?

Respondents report probabilities of an increase, a decrease, and no change, with an additional “no idea” option.

QP2 then asks about the perceived personal effect of the relevant tax change:

If the following policy changes were implemented in the next 12 months, how would you and your household be impacted?

For the average income tax rate, the response categories are “very negatively,” “somewhat negatively,” “no impact,” “somewhat positively,” and “very positively.” Respondents who assign a higher probability to a tax increase are asked about the personal impact of a tax increase; respondents who assign a higher probability to a tax decrease are asked about the personal impact of a tax decrease. When the two probabilities are equal, respondents are assigned approximately evenly across the two questions.

These two items provide the inputs for the measures of net tax-increase expectations and perceived tax exposure. Table 15 reports how the perceived-exposure sample is constructed. The main source of sample loss is that the QP2 tax-impact question is not available for every respondent-wave. The number of missing response is 79 observations. The final perceived-exposure sample contains 22,919 respondent-wave observations, or 74.5 percent of the merged sample. Of these observations, 70.1 percent are asked about the effect of a tax increase and 29.9 percent are asked about the effect of a tax decrease. Equal-probability observations account for 25.4 percent of the final perceived-exposure sample; they are a subset of, rather than a third category alongside, the tax-increase and tax-decrease question groups.

Table 15. Construction of the Perceived-Exposure Sample

Observations	N	% of merged sample	% of final sample
Merged Core–Public Policy sample	30,769	100.00	
QP2 tax-impact question not recorded	7,771	25.26	
QP2 response missing	79	0.26	
Final perceived-exposure sample	22,919	74.49	100.00
Asked about the personal impact of a tax increase	16,075	52.24	70.14
Asked about the personal impact of a tax decrease	6,844	22.24	29.86
Equal increase and decrease probabilities	5,830	18.95	25.44

Notes: The final perceived-exposure sample includes respondents for whom the QP2 tax-impact question was available and the response was nonmissing. The equal-probability observations overlap the tax-increase and tax-decrease question groups and are therefore not an additional mutually exclusive category.

The sample in Table 15 is the sample used to construct the perceived-exposure measure. Figures that additionally use QP1 probabilities are estimated on complete observations for both QP1 and QP2 variables.

Winsorization For continuous point forecasts used elsewhere in the analysis, including expected income growth, expected consumption growth, inflation expectations, and expected debt growth, values below -100 percent and identified nonresponse or data-entry values (999, 1000, 1200, and 2025) are first set to missing. The remaining observations are winsorized separately within each survey wave at the survey-weighted first and ninety-ninth percentiles. This wave-specific procedure prevents a small number of typed extreme values from disproportionately affecting the moments of the continuous expectation variables.

No winsorization is applied to the tax probabilities, the net tax-increase expectation measure defined below, or the five-point perceived-exposure measure because these variables are bounded by construction.

A.2 Variable Construction

Net tax-increase expectations Respondents report subjective probabilities that the average income tax rate will increase, remain unchanged, or decrease. Let p_{it}^{inc} and p_{it}^{dec} denote respondent i 's reported probabilities, in percentage points, of an increase and a decrease, respectively.

Using p_{it}^{inc} alone would not distinguish between respondents who assign the remaining probability primarily to no change and those who assign it primarily to a tax decrease. For example, two respondents may both assign a 40 percent probability to an increase while assigning very different probabilities to a decrease. The difference $p_{it}^{\text{inc}} - p_{it}^{\text{dec}}$ captures the directional tilt of beliefs toward an increase rather than a decrease. It is therefore interpreted as a net expectation measure, not as the probability of a tax increase itself.

Instead, I define net tax-increase expectations as

$$B_{it} = p_{it}^{\text{inc}} - p_{it}^{\text{dec}}, \quad B_{it} \in [-100, 100]. \quad (36)$$

A positive value indicates that the respondent assigns a higher probability to an income-tax-rate increase than to a decrease, whereas a negative value indicates that a decrease is viewed as more likely. A value of zero indicates that the respondent assigns equal probabilities to the two directions.

Perceived tax exposure The Public Policy Survey asks respondents how the anticipated change in the average income tax rate would affect them personally. The response is recorded on the following five-point scale:

Value	Response category
1	Very negatively
2	Somewhat negatively
3	No impact
4	Somewhat positively
5	Very positively

The direction referenced in the personal-impact question depends on the respondent's reported probabilities. Respondents who assign a higher probability to an increase than to a decrease are asked about the personal effect of a tax-rate increase. Those who assign a higher probability to a decrease are asked about the personal effect of a tax-rate decrease. When the two probabilities are equal, respondents are assigned approximately evenly between the two question wordings.

To place the responses on a common directional scale, I retain the original response under the tax-increase wording and reverse the response under the tax-decrease wording (**reverse coding**). If R_{it} denotes the original response, the aligned measure is

$$A_{it} = \begin{cases} R_{it}, & \text{under the tax-increase wording,} \\ 6 - R_{it}, & \text{under the tax-decrease wording.} \end{cases} \quad (37)$$

The transformation maps a positive perceived effect of a tax decrease into a negative perceived effect of the corresponding tax increase, and vice versa. Lower values of A_{it} therefore indicate that a tax-rate increase is perceived as personally harmful, whereas higher values indicate that it is perceived as personally beneficial.

For the descriptive figures, I aggregate the aligned measure into three categories:

$$\begin{aligned} \text{Negative exposure : } & A_{it} \in \{1, 2\}, \\ \text{Neutral exposure : } & A_{it} = 3, \\ \text{Positive exposure : } & A_{it} \in \{4, 5\}. \end{aligned} \quad (38)$$

Reverse coding aligns the direction of responses across the tax-increase and tax-decrease questions, but does not imply that the two question versions generate identical response distributions. For this reason, I conduct reverse-coding diagnostics in the following part.

Table 16. Descriptive Statistics for Net Tax-Increase Expectations

		Observations	Share	Mean net expectation	Std.
<i>Panel A: By perceived exposure</i>					
	Negative	16,220	74.18	27.77	50.65
	Neutral	3,110	17.42	18.14	47.67
	Positive	1,598	8.40	13.86	51.66
<i>Panel B: By income group and perceived exposure</i>					
Low income	Negative	3,932	61.45	30.45	50.91
	Neutral	1,471	28.11	19.60	49.06
	Positive	556	10.44	16.84	52.56
Middle income	Negative	5,916	79.47	28.22	51.58
	Neutral	874	12.05	17.47	45.77
	Positive	580	8.48	15.50	51.52
High income	Negative	6,245	83.29	25.00	49.39
	Neutral	735	10.65	14.45	45.89
	Positive	451	6.06	5.12	48.89

Notes: Weighted shares in Panel A are calculated over the complete-case sample. Weighted shares in Panel B are calculated separately within each income group. Observations are unweighted counts; means and standard deviations use SCE survey weights. Panel A uses 20,928 observations. Panel B uses 20,760 observations after dropping 168 observations with missing income-group information.

A.2.1 Reverse-Coding Diagnostics

The reverse coding in equation (37) is used to align the direction of the personal-impact responses, rather than to assume that the tax-increase and tax-decrease questions constitute identical measures. The diagnostic evidence supports using the aligned measure for qualitative comparisons across income groups: the relative income gradient is similar across the two questions. At the same time, the evidence indicates that a residual framing effect remains in the level and distribution of responses. Accordingly, the aligned measure is used as a descriptive classification of perceived exposure, not as a framing-invariant measure or a source of causal identification.

I examine these issues using two diagnostics. First, I test whether the relationship between income group and aligned exposure differs between respondents asked about a tax increase and those asked about a tax decrease. Second, I use the equal-probability subsample to test whether the two questions produce different response distributions when the direction of the question is not determined by respondents' relative increase and decrease probabilities. Table 17 reports the results.

Columns (1) and (2) show the same qualitative income ordering under both questions. Relative to low-income respondents, middle- and high-income respondents report lower values of the aligned exposure measure, indicating a more negative perceived effect of a tax increase. The

Table 17. Diagnostics for the Directionally Aligned Exposure Measure

	Separate samples		Pooled	Equal-probability
	Increase question (1)	Decrease question (2)	sample (3)	sample (4)
Middle-income group	−0.300*** (0.026)	−0.326*** (0.036)	−0.335*** (0.036)	
High-income group	−0.432*** (0.026)	−0.438*** (0.035)	−0.449*** (0.036)	
Increase-question indicator			−0.344*** (0.031)	−0.182*** (0.029)
Middle income × increase question			0.038 (0.043)	
High income × increase question			0.022 (0.042)	
Observations	15,930	6,803	22,733	5,830
Joint p -value: income × question			0.664	
Probability-level controls	No	No	No	Yes
Survey-year controls	No	No	No	Yes

Notes: The omitted income category is the low-income group. Standard errors clustered by respondent are reported in parentheses. Ordered-probit cut points are omitted. *** $p < 0.001$.

estimated coefficients are similar across the two samples. More formally, column (3) shows that the interactions between income group and the increase-question indicator are small and jointly insignificant ($p = 0.664$). Thus, there is no evidence that the relative income gradient in perceived exposure differs between the two questions. This result supports using the aligned measure for qualitative comparisons across income groups.

Column (4) examines the equal-probability subsample. Within this subsample, respondents asked about a tax increase and those asked about a tax decrease have similar income-group composition ($p = 0.659$) and survey-year composition ($p = 0.213$). Differences between the two groups are therefore unlikely to reflect these observable compositional differences.

Despite this balance, the distribution of the aligned exposure measure differs across the two questions ($\chi^2(4) = 64.85$, $p < 0.001$). After controlling for the common probability level and survey year, the increase-question indicator has a coefficient of -0.182 , with a standard error of 0.029 . Because lower values indicate a more negative perceived effect, this estimate shows that stating the question directly in terms of a tax increase shifts responses toward more negative categories.

Taken together, the diagnostics support directional alignment but not full measurement equivalence. The similar income gradients justify using the reverse-coded measure to describe qualitative differences across income groups. The residual framing effect, however, prevents treating the two questions as perfectly interchangeable or interpreting the association between perceived exposure and tax expectations causally.

Appendix B Equilibrium in OLG Models with Fiscal Beliefs

B.1 One-Type OLG

Output and Debt.

$$\begin{aligned}
 y_t &= (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k}) \right] \right) \\
 &= (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k \{ (1 - \tau_y) y_{t+k} - \tau_d d_{t+k} - (\tau_d - 1) \varepsilon_{t+k} \} \right] \right) \\
 &= (1 - \beta\omega) \left(d_t + \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k \{ (1 - \tau_y) y_{t+k} - \tau_d d_{t+k} \} \right] - (\tau_d - 1) \bar{E}_t \left[\sum_{k=0}^{\infty} (\beta\omega)^k \varepsilon_{t+k} \right] \right)
 \end{aligned}$$

- $S_\varepsilon = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t \varepsilon_{t+k}$

$$S_\varepsilon = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t \varepsilon_{t+k} = \sum_{k=0}^{\infty} (\beta\omega)^k \lambda^p \varepsilon_{t+k} = \sum_{k=0}^{\infty} (\beta\omega \rho_\varepsilon)^k \lambda^p \varepsilon_t = \underbrace{\frac{\lambda^p}{1 - \beta\omega \rho_\varepsilon}}_H \varepsilon_t$$

- $S_d = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t d_{t+k}$

$$\bar{E} d_{t+k+1} = \phi_d \bar{E}_t d_{t+k} + \phi_\varepsilon \bar{E}_t \hat{\varepsilon}_{t+k}$$

$$\begin{aligned}
 S_d &= \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t d_{t+k} = d_t + \sum_{k=0}^{\infty} (\beta\omega)^{k+1} \bar{E}_t d_{t+k+1} \\
 &= d_t + \sum_{k=0}^{\infty} (\beta\omega)^{k+1} (\phi_d \bar{E}_t d_{t+k} + \phi_\varepsilon \bar{E}_t \hat{\varepsilon}_{t+k}) = d_t + \beta\omega \phi_d S_d + \beta\omega \phi_\varepsilon \underbrace{\sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t \hat{\varepsilon}_{t+k}}_{\lambda^g S_\varepsilon = \lambda^g H \varepsilon_t} \\
 \therefore S_d &= \underbrace{\frac{1}{1 - \beta\omega \phi_d}}_{A_d} d_t + \underbrace{\frac{\beta\omega \phi_\varepsilon \lambda^g}{1 - \beta\omega \phi_d}}_{A_\varepsilon} H \varepsilon_t
 \end{aligned}$$

- $S_y = \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t y_{t+k}$

$$y_t = \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t$$

$$\begin{aligned}
 S_y &= \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t y_{t+k} = \psi_d \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t d_{t+k} + \psi_\varepsilon \sum_{k=0}^{\infty} (\beta\omega)^k \bar{E}_t \hat{\varepsilon}_{t+k} \\
 &= \psi_d S_d + \psi_\varepsilon \lambda^g S_\varepsilon \\
 &= \psi_d (A_d d_t + A_\varepsilon \varepsilon_t) + \psi_\varepsilon \lambda^g H \varepsilon_t \\
 &= \underbrace{\psi_d A_d d_t}_{B_d} + \underbrace{(\psi_d A_\varepsilon + \psi_\varepsilon \lambda^g H) \varepsilon_t}_{B_\varepsilon}
 \end{aligned}$$

Combined,

$$\begin{aligned}
y_t &= (1 - \beta\omega) (d_t + (1 - \tau_y)S_y - \tau_d S_d - (\tau_d - 1)S_\varepsilon) \\
&= (1 - \beta\omega) (d_t + (1 - \tau_y)(B_d d_t + B_\varepsilon \varepsilon_t) - \tau_d(A_d d_t + A_\varepsilon \varepsilon_t) - (\tau_d - 1)H \varepsilon_t) \\
&= (1 - \beta\omega) \left(\underbrace{[1 + (1 - \tau_y)B_d - \tau_d A_d]}_{\text{Coefficient on } d_t} d_t + \underbrace{[(1 - \tau_y)B_\varepsilon - \tau_d A_\varepsilon - (\tau_d - 1)H]}_{\text{Coefficient on } \varepsilon_t} \varepsilon_t \right) \\
&= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t \\
&= \psi_d d_t + \psi_\varepsilon \lambda^p \varepsilon_t
\end{aligned}$$

Note

$$\begin{aligned}
d_{t+1} &= \frac{1}{\beta}(d_t - t_t) + \frac{D}{Y} \underbrace{r_t}_{=0} - \frac{D}{Y} \underbrace{(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}])}_{=0} \\
&= \frac{1}{\beta} ((1 - \tau_d)d_t - \tau_y y_t - (\tau_d - 1)\varepsilon_t) \\
&= \frac{1}{\beta} \left((1 - \tau_d)d_t - \tau_y (\psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t) - (\tau_d - 1) \frac{\hat{\varepsilon}_t}{\lambda^p} \right) \\
&= \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t \\
&= \phi_d d_t + \phi_\varepsilon \lambda^p \varepsilon_t
\end{aligned}$$

Therefore,

$$\begin{aligned}
\phi_d &= \frac{1}{\beta} (1 - \tau_d - \tau_y \psi_d) \\
\phi_\varepsilon &= \frac{1}{\beta} \left(-\tau_y \psi_\varepsilon - \frac{\tau_d - 1}{\lambda^p} \right)
\end{aligned}$$

Inflation

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t.$$

Following a one-time unanticipated shock, there is no future uncertainty. Thus, under rational expectations, $\mathbb{E}_t \pi_{t+1} = \pi_{t+1}$. Therefore,

$$\pi_0 = \kappa \sum_{t=0}^{\infty} \beta^t y_t = \kappa (\psi_d \cdot PV(d) + \psi_\varepsilon \cdot PV(\hat{\varepsilon})).$$

- $PV(\hat{\varepsilon})$.

The perceived shock entering the realized paths satisfies $\hat{\varepsilon}_t = \lambda^p \varepsilon_t$ for all $t \geq 0$, with $\varepsilon_t = \rho^t \varepsilon_0$. Then

$$PV(\hat{\varepsilon}) \equiv \sum_{t=0}^{\infty} \beta^t \hat{\varepsilon}_t = \lambda^p \sum_{t=0}^{\infty} \beta^t \varepsilon_t = \lambda^p \frac{\varepsilon_0}{1 - \beta\rho}.$$

- $PV(d)$.

$$PV(d) \equiv \sum_{t=0}^{\infty} \beta^t d_t.$$

Given the debt law of motion $d_{t+1} = \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t$, we have

$$d_{t+1} = \phi_d^{t+1} d_0 + \phi_\varepsilon \sum_{j=0}^t \phi_d^{t-j} \hat{\varepsilon}_j = \phi_d^{t+1} d_0 + \phi_\varepsilon \lambda^p \varepsilon_0 \sum_{j=0}^t \phi_d^{t-j} \rho^j.$$

Equivalently, for $t \geq 1$,

$$d_t = \phi_d^t d_0 + \phi_\varepsilon \lambda^p \varepsilon_0 \sum_{j=0}^{t-1} \phi_d^{t-1-j} \rho^j.$$

Therefore,

$$\therefore PV(d) = d_0 + \sum_{t=1}^{\infty} \beta^t d_t = \frac{1}{1 - \beta \phi_d} \left[d_0 + \frac{\beta}{1 - \beta \rho} \phi_\varepsilon \lambda^p \varepsilon_0 \right].$$

Substituting in,

$$\begin{aligned} \pi_0 &= \kappa (\psi_d PV(d) + \psi_\varepsilon PV(\hat{\varepsilon})) \\ &= \kappa \left(\psi_d \frac{1}{1 - \beta \phi_d} \left[d_0 + \frac{\beta}{1 - \beta \rho} \phi_\varepsilon \lambda^p \varepsilon_0 \right] + \psi_\varepsilon \frac{\lambda^p \varepsilon_0}{1 - \beta \rho} \right). \end{aligned}$$

Since $d_0 = -\frac{D}{Y} \pi_0$, substituting and solving for π_0 gives:

$$\therefore \pi_0 = \frac{\kappa \varepsilon_0 \frac{\lambda^p}{1 - \beta \rho} \left[\psi_\varepsilon + \psi_d \frac{\beta \phi_\varepsilon}{1 - \beta \phi_d} \right]}{1 + \kappa \psi_d \frac{D/Y}{1 - \beta \phi_d}}.$$

Given π_0 , we can determine the initial debt, $d_0 = -\frac{D}{Y} \pi_0$.

B.2 Two-Type OLG

Output and Debt.

Guess-and-verify:

$$\begin{aligned} y_t &= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t \\ d_{t+1} &= \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t \end{aligned}$$

Note in two-type agent economy,

$$\hat{\varepsilon}_t = \bar{E}_t \varepsilon_t = \sum_{i=1}^2 \mu_i \lambda_i^p \varepsilon_t$$

Then

$$y_t^i = (1 - \beta \omega_i) \left(d_t^i + \bar{E}_t^i \left[\sum_{k=0}^{\infty} (\beta \omega_i)^k \{ (1 - \tau_y) y_{t+k} - \tau_d d_{t+k} \} \right] - (\tau_d - 1) \bar{E}_t^i \left[\sum_{k=0}^{\infty} (\beta \omega_i)^k \varepsilon_{t+k} \right] \right)$$

Note S_y, S_d, S_t in section 3.1 have i superscripts where

$$\begin{aligned}
S_\varepsilon^i &= \sum_{k=0}^{\infty} (\beta\omega_i)^k E_t \varepsilon_{t+k} = \frac{\lambda_i^p}{1 - \beta\omega_i \rho_\varepsilon} \varepsilon_t = H_i \varepsilon_t \\
S_d^i &= \underbrace{\frac{1}{1 - \beta\omega_i \phi_d}}_{A_d^i} d_t + \underbrace{\frac{\beta\omega_i \phi_\varepsilon \lambda_i^g}{1 - \beta\omega_i \phi_d}}_{A_\varepsilon^i} H_i \varepsilon_t \\
S_y^i &= \psi_d S_d^i + \psi_\varepsilon \lambda_i^g S_\varepsilon^i \\
&= \psi_d (A_d^i d_t + A_\varepsilon^i H_i \varepsilon_t) + \psi_\varepsilon \lambda_i^g H_i \varepsilon_t \\
&= \psi_d A_d^i d_t + (\psi_d A_\varepsilon^i + \psi_\varepsilon \lambda_i^g) H_i \varepsilon_t
\end{aligned}$$

Substitute in $S_y^i, S_d^i, S_\varepsilon^i$:

$$\begin{aligned}
y_t &= \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) \left[s_i d_t + (1 - \tau_y) (\psi_d A_d^i d_t + (\psi_d A_\varepsilon^i + \psi_\varepsilon \lambda_i^g) H_i \varepsilon_t) - \tau_d (A_d^i d_t + A_\varepsilon^i H_i \varepsilon_t) - (\tau_d - 1) H_i \varepsilon_t \right] \\
&= \left\{ \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) \left[s_i + ((1 - \tau_y) \psi_d - \tau_d) A_d^i \right] \right\} d_t \\
&+ \left\{ \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i \left[((1 - \tau_y) \psi_d - \tau_d) A_\varepsilon^i + (1 - \tau_y) \psi_\varepsilon \lambda_i^g - (\tau_d - 1) \right] \right\} \varepsilon_t \\
&= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t.
\end{aligned}$$

Coefficient on d_t :

$$\begin{aligned}
\psi_d &= \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) \left[s_i + (1 - \tau_y) \psi_d A_d^i - \tau_d A_d^i \right] \\
&= \underbrace{\sum_{i=1}^2 \mu_i (1 - \beta\omega_i) s_i}_{\equiv \Xi_s} + ((1 - \tau_y) \psi_d - \tau_d) \underbrace{\sum_{i=1}^2 \mu_i (1 - \beta\omega_i) A_d^i}_{\equiv \Xi_d}.
\end{aligned}$$

Equivalently,

$$\psi_d = \frac{\Xi_s - \tau_d \Xi_d}{1 - (1 - \tau_y) \Xi_d}.$$

Coefficient on ε_t :

$$\begin{aligned}
\psi_\varepsilon \sum_i \mu_i \lambda_i^p &= \sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i \left[((1 - \tau_y) \psi_d - \tau_d) A_\varepsilon^i + (1 - \tau_y) \psi_\varepsilon \lambda_i^g - (\tau_d - 1) \right] \\
&= ((1 - \tau_y) \psi_d - \tau_d) \underbrace{\sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i A_\varepsilon^i}_{\equiv \Xi_A} + (1 - \tau_y) \psi_\varepsilon \underbrace{\sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i \lambda_i^g}_{\equiv \Xi_G} \\
&- (\tau_d - 1) \underbrace{\sum_{i=1}^2 \mu_i (1 - \beta\omega_i) H_i}_{\equiv \Xi_H} \\
&= ((1 - \tau_y) \psi_d - \tau_d) \Xi_A + (1 - \tau_y) \psi_\varepsilon \Xi_G - (\tau_d - 1) \Xi_H.
\end{aligned}$$

Therefore,

$$\psi_\varepsilon = \frac{((1 - \tau_y)\psi_d - \tau_d)\Xi_A - (\tau_d - 1)\Xi_H}{\sum_i \mu_i \lambda_i^p - (1 - \tau_y)\Xi_G}.$$

Finally, from the government budget constraint,

$$\begin{aligned} d_{t+1} &= \frac{1}{\beta} \left((1 - \tau_d)d_t - \tau_y y_t - (\tau_d - 1)\varepsilon_t \right) = \frac{1}{\beta} \left((1 - \tau_d)d_t - \tau_y(\psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t) - (\tau_d - 1)\varepsilon_t \right) \\ &= \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t, \\ \phi_d &= \frac{1 - \tau_d - \tau_y \psi_d}{\beta}, \quad \phi_\varepsilon = \frac{-(\tau_d - 1) - \tau_y \psi_\varepsilon \sum_i \mu_i \lambda_i^p}{\beta \sum_i \mu_i \lambda_i^p}. \end{aligned}$$

Note

$$\begin{aligned} \Xi_s &\equiv \sum_{i=1}^2 \mu_i (1 - \beta \omega_i) s_i, \quad \Xi_d \equiv \sum_{i=1}^2 \mu_i (1 - \beta \omega_i) A_d^i, \quad \Xi_A \equiv \sum_{i=1}^2 \mu_i (1 - \beta \omega_i) H_i A_\varepsilon^i, \\ \Xi_H &\equiv \sum_{i=1}^2 \mu_i (1 - \beta \omega_i) H_i, \quad \Xi_G \equiv \sum_{i=1}^2 \mu_i (1 - \beta \omega_i) H_i \lambda_i^g. \end{aligned}$$

Inflation

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t.$$

Following a one-time unanticipated shock, there is no future uncertainty. Thus, under rational expectations, $\mathbb{E}_t \pi_{t+1} = \pi_{t+1}$. Iterating forward and imposing the no-bubbles condition yields

$$\pi_0 = \kappa \sum_{t=0}^{\infty} \beta^t y_t = \kappa (\psi_d \cdot PV(d) + \psi_\varepsilon \cdot PV(\hat{\varepsilon})).$$

- $PV(\hat{\varepsilon})$

The aggregate perceived shock entering the laws of motion is

$$\hat{\varepsilon}_t = P \varepsilon_t, \quad P \equiv \sum_i \mu_i \lambda_i^p, \quad \varepsilon_t = \rho^t \varepsilon_0.$$

Then

$$PV(\hat{\varepsilon}) \equiv \sum_{t=0}^{\infty} \beta^t \hat{\varepsilon}_t = P \sum_{t=0}^{\infty} \beta^t \varepsilon_t = P \frac{\varepsilon_0}{1 - \beta \rho}.$$

- $PV(d)$

$$PV(d) \equiv \sum_{t=0}^{\infty} \beta^t d_t.$$

Given the debt law of motion

$$d_{t+1} = \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t,$$

we have, for $t \geq 1$,

$$d_t = \phi_d^t d_0 + \phi_\varepsilon \sum_{j=0}^{t-1} \phi_d^{t-1-j} \hat{\varepsilon}_j = \phi_d^t d_0 + \phi_\varepsilon P \varepsilon_0 \sum_{j=0}^{t-1} \phi_d^{t-1-j} \rho^j \quad (t \geq 1).$$

Therefore,

$$PV(d) = d_0 + \sum_{t=1}^{\infty} \beta^t d_t = \frac{1}{1 - \beta \phi_d} \left[d_0 + \frac{\beta}{1 - \beta \rho} \phi_\varepsilon P \varepsilon_0 \right].$$

Substituting into $\pi_0 = \kappa(\psi_d PV(d) + \psi_\varepsilon PV(\hat{\varepsilon}))$,

$$\pi_0 = \kappa \left(\psi_d \frac{1}{1 - \beta \phi_d} \left[d_0 + \frac{\beta}{1 - \beta \rho} \phi_\varepsilon P \varepsilon_0 \right] + \psi_\varepsilon \frac{P \varepsilon_0}{1 - \beta \rho} \right).$$

Since $d_0 = -\frac{D}{Y} \pi_0$, substituting and solving for π_0 yields

$$\therefore \pi_0 = \frac{\kappa \varepsilon_0 \frac{P}{1 - \beta \rho} \left[\psi_\varepsilon + \psi_d \frac{\beta \phi_\varepsilon}{1 - \beta \phi_d} \right]}{1 + \kappa \psi_d \frac{D/Y}{1 - \beta \phi_d}}.$$

Given π_0 , we can determine the initial debt, $d_0 = -\frac{D}{Y} \pi_0$.

Appendix C TANK with Fiscal Beliefs

C.1 Model

The economy is populated by a share α of unconstrained households and $1 - \alpha$ share of hand-to-mouth households.

C.1.1 Households

Unconstrained households solve

$$\begin{aligned} \max_{C_{u,t}, N_{u,t}, b_{u,t}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_{u,t}^{1-\frac{1}{\sigma}}}{1 - \frac{1}{\sigma}} - \iota \frac{N_{u,t}^{1+\frac{1}{\psi}}}{1 + \frac{1}{\psi}} \right] \\ \text{s.t.} \quad & C_{u,t} + b_{u,t} = w_t N_{u,t} + Div_{u,t} - T_{u,t} + \frac{R_{t-1}^{\text{nominal}}}{\Pi_t} b_{u,t-1} \end{aligned}$$

where $b_{u,t} = \frac{B_{u,t}}{P_t}$ is the real value of bond holdings, $R_{t-1}^{\text{nom}} = R_{t-1} \mathbb{E}_{t-1} \Pi_t$ is the ex-ante nominal interest rate on bonds, and $Div_{u,t}$ denotes dividends.

Constrained (Hand-to-mouth) households solve

$$C_{h,t} = w_t N_{h,t} + Div_{h,t} - T_{h,t}$$

Let's discuss how household income is determined. First, following Angeletos, Lian, Wolf (2024), we assume that labor supply is intermediated by labor unions and abstract from heterogeneity

in labor supply. This is because this allows us to isolate the channel that this paper identifies: how the aggregate demand changes disproportionately in response to a fiscal shock depending on the fiscal beliefs.

Thus, labor unions demand identical hours worked from all households ($L_{it} = L_t$) and bargain on behalf of those households, equalizing the post-tax real wage and the average marginal rate of substitution between consumption and labor supply.

$$(1 - \tau_y)W_t = \frac{\iota L_t^{\frac{1}{\psi}}}{\int_0^1 C_{i,t}^{-1/\sigma} di}$$

Therefore,

$$L_{u,t} = L_{h,t} = L_t$$

Second, we assume that firm profits are distributed equally across households.

$$Div_{u,t} = Div_{h,t} = Div_t$$

Although we retain the notation Div_t and refer to these payments as dividends for convenience, they should not be interpreted as returns to household equity ownership. Such an interpretation would imply that hand-to-mouth households own shares in firms, raising the question of why households that are genuinely hand-to-mouth hold financial assets. Instead, Div_t is more naturally interpreted as a uniform profit rebate, or equivalently, a lump-sum transfer of firm profits to households. Hand-to-mouth households cannot save this transfer and therefore consume it immediately. Their budget constraint is thus given by

$$C_{h,t} = W_t N_t + Div_t - T_t$$

C.1.2 Firms

The production side of the economy is the same as in the textbook New Keynesian model. There is a continuum of monopolistically competitive retailers, who set prices subject to the Calvo friction, hire labor on a spot market, produce according to a technology that is linear in labor, and pay out all their profits as dividends back to the households. Combined with our aforementioned assumptions on labor supply, this guarantees that the supply block of our economy reduces exactly to the standard New Keynesian Phillips curve (NKPC).

C.1.3 Policy

$$\begin{aligned} \frac{R_t}{\Pi} &= R \mathbb{E}_t \left[\frac{\Pi_{t+1}}{\Pi} \right] \left(\frac{Y_t}{Y} \right)^\phi \\ D_{t+1} &= R_t (D_t - T_t) \frac{\mathbb{E}_t \Pi_{t+1}}{\Pi_{t+1}} \\ T_{i,t} &= \tau_y Y_{i,t} + \bar{T} - \epsilon_t + \tau_d (D_t - D + \epsilon_t) \end{aligned}$$

The policy authorities have full-information rational expectations. Only households hold distorted beliefs about the fiscal shock.

C.1.4 Exogenous

Exogenous tax shock follows an AR(1) process

$$\varepsilon_{t+1} = \rho_\varepsilon \varepsilon_t + u_t$$

I assume a one-time unexpected tax shock. This allows us to easily solve the problem. Furthermore, the sign of the shock is negative, i.e., tax increase, implying a fiscal consolidation effort that lasts for several quarters.

C.1.5 Aggregate conditions

$$\begin{aligned} N_t &= \alpha N_{u,t} + (1 - \alpha) N_{h,t} \\ C_t &= \alpha C_{u,t} + (1 - \alpha) C_{h,t} \\ Y_t &= C_t \\ D_t &= \alpha \frac{R_{t-1} b_{u,t-1}}{\Pi_t} \end{aligned}$$

The last equation represents the asset market clearing condition. The government debt is owned by the unconstrained households. The real value of nominal debt issued and held in the previous period is revalued once today's price level is realized.

C.1.6 Beliefs

Households have non full-information imperfect information about the shock. Specifically, unconstrained and hand-to-mouth households have different first-order and second-order beliefs about the tax shock.

First-order beliefs:

$$E_t^u \varepsilon_t = \lambda_p^u \varepsilon_t, \quad , E_t^h \varepsilon_t = \lambda_p^h \varepsilon_t,$$

where u and h denote unconstrained and hand-to-mouth households, respectively. This is analogous to the two-type OLG model in Section 3, where two groups of households hold different fiscal beliefs.

Then average belief is

$$\hat{\varepsilon}_t = \bar{E}_t \varepsilon_t = \bar{\lambda}_p \varepsilon_t$$

where $\bar{\lambda}_p^u = \alpha \lambda_p^u + (1 - \alpha) \lambda_p^h$.

Second-order beliefs:

$$E_t^u \bar{E}_t \varepsilon_t = \lambda_g^u \varepsilon_t, \quad , E_t^h \bar{E}_t \varepsilon_t = \lambda_g^h \varepsilon_t,$$

C.1.7 Information and Timing

At the beginning of period t , the fiscal innovation u_t is realized. The policy authority and firms observe the fiscal state under full information. Households instead form type-specific beliefs. Conditional on these beliefs, households make their consumption decisions. Unconstrained households choose consumption using their Euler equation and intertemporal budget

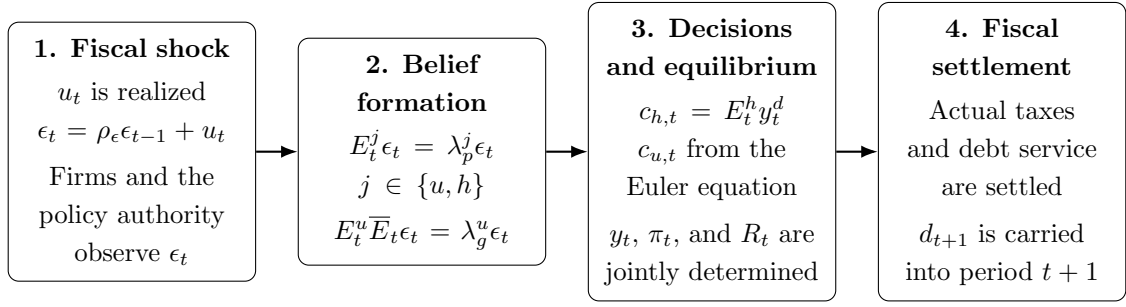


Figure 8. Information and within-period timing

constraint, whereas high-MPC households consume according to perceived current disposable income. Household decisions, firms' pricing decisions, and monetary policy jointly determine output, inflation, and the interest rate. At the end of the period, actual tax liabilities and government debt service are settled, and the resulting stock of government debt is carried into period $t + 1$.

C.2 Log-linearization of Households' Problem

The budget constraint for unconstrained agents in levels is given by:

$$C_{u,t} + b_{u,t} = Y_{u,t}^d + \frac{R_{t-1}^{\text{nominal}}}{\Pi_t} b_{u,t-1}$$

Log-linearizing the budget constraint around the steady state yields:

$$\begin{aligned} \hat{c}_{u,t} + \hat{b}_{u,t} &= \hat{y}_{u,t}^d + \frac{Rb}{\Pi Y} \hat{r}_{t-1} + \frac{R}{\Pi} \hat{b}_{u,t-1} \\ &= \hat{y}_{u,t}^d + \beta^{-1} \frac{b}{Y} \hat{r}_{t-1} + \beta^{-1} \hat{b}_{u,t-1}, \end{aligned}$$

where normalized deviations are defined relative to steady-state output Y as:

$$\hat{c}_{u,t} \equiv \frac{C_{u,t} - C_u}{Y}, \quad \hat{y}_{u,t}^d \equiv \frac{Y_{u,t}^d - Y_u^d}{Y}, \quad \hat{b}_{u,t} \equiv \frac{b_{u,t} - b_u}{Y},$$

and the ex-post real interest rate deviation is $\hat{r}_{t-1} \equiv \hat{R}_{t-1} - \hat{\pi}_t$, with $\beta = \frac{\Pi}{R}$.

Rearranging for initial assets $\hat{b}_{u,t-1}$ gives:

$$\hat{b}_{u,t-1} = \beta \left(\hat{c}_{u,t} - \hat{y}_{u,t}^d - \beta^{-1} \frac{b}{Y} \hat{r}_{t-1} \right) + \beta \hat{b}_{u,t}.$$

Iterating this equation forward K periods yields:

$$\hat{b}_{u,t-1} = E_t^u \left[\sum_{k=0}^K \beta^{k+1} \left(\hat{c}_{u,t+k} - \hat{y}_{u,t+k}^d - \beta^{-1} \frac{b}{Y} \hat{r}_{t+k-1} \right) + \beta^{K+1} \hat{b}_{u,t+K} \right].$$

Imposing the transversality condition,

$$\lim_{K \rightarrow \infty} E_t^u \left[\beta^{K+1} \hat{b}_{u,t+K} \right] = 0,$$

we obtain the intertemporal budget constraint:

$$\hat{b}_{u,t-1} = E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \left(\hat{c}_{u,t+k} - \hat{y}_{u,t+k}^d \right) - \frac{b}{Y} E_t^u \sum_{k=0}^{\infty} \beta^k \hat{r}_{t+k-1}.$$

Next, iterating the log-linearized consumption Euler equation forward k periods gives:

$$E_t^u \hat{c}_{u,t+k} = \hat{c}_{u,t} + \tilde{\sigma}_u \sum_{j=0}^{k-1} E_t^u \hat{r}_{t+j},$$

where $\tilde{\sigma}_u \equiv \sigma \frac{C_u}{Y}$ and σ denotes the intertemporal elasticity of substitution.

Discounting and summing the consumption stream over all future periods yields:

$$E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{c}_{u,t+k} = \frac{\beta}{1-\beta} \hat{c}_{u,t} + \frac{\tilde{\sigma}_u \beta}{1-\beta} E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{r}_{t+k}.$$

Combining the forward-iterated budget constraint with the Euler equation, we have:

$$\hat{b}_{u,t-1} + E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{y}_{u,t+k}^d = \frac{\beta}{1-\beta} \hat{c}_{u,t} + \frac{\tilde{\sigma}_u \beta}{1-\beta} E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{r}_{t+k} - \frac{b}{Y} E_t^u \sum_{k=0}^{\infty} \beta^k \hat{r}_{t+k-1}.$$

Solving for $\hat{c}_{u,t}$, the log-linearized consumption function for unconstrained households is:

$$\hat{c}_{u,t} = \frac{1-\beta}{\beta} \left(\hat{b}_{u,t-1} + E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{y}_{u,t+k}^d \right) - \sigma \frac{C_u}{Y} E_t^u \sum_{k=0}^{\infty} \beta^{k+1} \hat{r}_{t+k} + \frac{1-\beta}{\beta} \frac{b}{Y} E_t^u \sum_{k=0}^{\infty} \beta^k \hat{r}_{t+k-1}. \quad (39)$$

For hand-to-mouth households, consumption is simply determined by their current disposable income:

$$\hat{c}_{h,t} = E_t^h \hat{y}_{h,t}^d. \quad (40)$$

C.3 Solution

We conjecture the following linear laws of motion for $t \geq 1$:

$$\begin{aligned} y_t &= \psi_d d_t + \psi_\varepsilon \hat{\varepsilon}_t \\ d_{t+1} &= \phi_d d_t + \phi_\varepsilon \hat{\varepsilon}_t \\ \pi_t &= \omega_d d_t + \omega_\varepsilon \hat{\varepsilon}_t \end{aligned}$$

C.3.1 Output

The log-linearized consumption functions are

$$\begin{aligned} c_{u,t} &= \frac{1-\beta}{\beta} \left(b_{u,t-1} + E_t^u \sum_{k=0}^{\infty} \beta^{k+1} y_{t+k}^d \right) - \sigma_u^e E_t^u \sum_{k=0}^{\infty} \beta^{k+1} r_{t+k} + \frac{1-\beta}{\beta} \frac{b}{Y} E_t^u \sum_{k=0}^{\infty} \beta^k r_{t+k-1}, \\ c_{h,t} &= E_t^h y_t^d \end{aligned}$$

where

$$\sigma_u^e \equiv \sigma \frac{C_u}{Y}$$

and σ denotes the intertemporal elasticity of substitution.

Disposable income is

$$y_t^d = (1 - \tau_y)y_t - \tau_d d_t - (\tau_d - 1)\epsilon_t$$

Log-linearize asset market clearing condition, $D_t = \alpha \frac{R_{t-1} b_{u,t-1}}{\Pi_t}$,

$$d_t = \alpha \beta^{-1} \hat{b}_{u,t-1} + \frac{D}{Y} r_{t-1}$$

and at steady-state,

$$D = \alpha \frac{Rb}{\Pi}$$

This implies

$$\frac{D}{Y} = \alpha \beta^{-1} \frac{b}{Y}$$

By market clearing condition,

$$\begin{aligned} y_t &= \alpha c_{u,t} + (1 - \alpha) c_{h,t} \\ &= (1 - \beta) d_t + \alpha (1 - \beta) E_t^u \sum_{k=0}^{\infty} \beta^k y_{t+k}^d - \alpha \sigma_u^e E_t^u \sum_{k=0}^{\infty} \beta^{k+1} r_{t+k} \\ &\quad + (1 - \beta) \beta \frac{D}{Y} E_t^u \sum_{k=0}^{\infty} \beta^k r_{t+k} + (1 - \alpha) E_t^h y_t^d \end{aligned} \quad (41)$$

Define

$$S_\epsilon \equiv E_t^u \sum_{k=0}^{\infty} \beta^k \epsilon_{t+k}, \quad S_{\hat{\epsilon}} \equiv E_t^u \sum_{k=0}^{\infty} \beta^k \hat{\epsilon}_{t+k}, \quad S_d \equiv E_t^u \sum_{k=0}^{\infty} \beta^k d_{t+k}, \quad S_y \equiv E_t^u \sum_{k=0}^{\infty} \beta^k y_{t+k}$$

Since $r_t = \phi y_t$,

$$E_t^u \sum_{k=0}^{\infty} \beta^{k+1} r_{t+k} = \phi \beta S_y$$

Since

$$\epsilon_{t+1} = \rho_\epsilon \epsilon_t, \quad \hat{\epsilon}_t = \bar{\lambda}_p \epsilon_t, \quad E_t^u \epsilon_{t+k} = \lambda_p^u \rho_\epsilon^k \epsilon_t, \quad E_t^u \hat{\epsilon}_{t+k} = \lambda_g^u \rho_\epsilon^k \epsilon_t$$

It follows that

$$S_\epsilon = \frac{\lambda_p^u}{1 - \beta \rho_\epsilon} \epsilon_t \equiv H_p \epsilon_t, \quad S_{\hat{\epsilon}} = \frac{\lambda_g^u}{1 - \beta \rho_\epsilon} \epsilon_t \equiv H_g \epsilon_t$$

where

$$H_p \equiv \frac{\lambda_p^u}{1 - \beta \rho_\epsilon}, \quad H_g \equiv \frac{\lambda_g^u}{1 - \beta \rho_\epsilon}$$

Using the conjectured debt law

$$d_{t+1} = \phi_d d_t + \phi_\epsilon \widehat{\epsilon}_t$$

the discounted debt sum satisfies

$$\begin{aligned} S_d &= \sum_{k=0}^{\infty} \beta^k d_{t+k} = d_t + \beta \sum_{k=0}^{\infty} \beta^k d_{t+1+k} = d_t + \beta \sum_{k=0}^{\infty} \beta^k (\phi_d d_{t+k} + \phi_\epsilon \widehat{\epsilon}_{t+k}) \\ &= d_t + \beta \phi_d S_d + \beta \phi_\epsilon S_{\widehat{\epsilon}} \end{aligned}$$

Therefore,

$$S_d = \frac{1}{1 - \beta \phi_d} d_t + \frac{\beta \phi_\epsilon H_g}{1 - \beta \phi_d} \epsilon_t \equiv A_d d_t + A_\epsilon \epsilon_t$$

where

$$A_d \equiv \frac{1}{1 - \beta \phi_d}, \quad A_\epsilon \equiv \frac{\beta \phi_\epsilon H_g}{1 - \beta \phi_d}$$

Using the conjectured output rule

$$y_t = \psi_d d_t + \psi_\epsilon \widehat{\epsilon}_t$$

we obtain

$$S_y = \psi_d S_d + \psi_\epsilon S_{\widehat{\epsilon}} = \psi_d A_d d_t + (\psi_d A_\epsilon + \psi_\epsilon H_g) \epsilon_t$$

The discounted disposable-income term satisfies

$$E_t^u \sum_{k=0}^{\infty} \beta^k y_{t+k}^d = (1 - \tau_y) S_y - \tau_d S_d - (\tau_d - 1) S_\epsilon = (1 - \tau_y) S_y - \tau_d S_d - (\tau_d - 1) H_p \epsilon_t$$

Substituting these expressions and $y_t^d = (1 - \tau_y) y_t - \tau_d d_t - (\tau_d - 1) \epsilon_t$ into equation (41):

$$\begin{aligned} y_t &= (1 - \beta) d_t + \alpha(1 - \beta) [(1 - \tau_y) S_y - \tau_d S_d - (\tau_d - 1) H_p \epsilon_t] \\ &\quad - \alpha \sigma_u^e \phi \beta S_y + (1 - \beta) \frac{D}{Y} \beta \phi S_y \\ &\quad + (1 - \alpha) E_t^h [(1 - \tau_y) y_t - \tau_d d_t - (\tau_d - 1) \epsilon_t] \end{aligned}$$

Define

$$\begin{aligned} L &\equiv 1 - (1 - \alpha)(1 - \tau_y) \\ K &\equiv \alpha(1 - \beta)(1 - \tau_y) - \alpha \sigma_u^e \phi \beta + \alpha(1 - \beta) \frac{b\phi}{Y} \\ M &\equiv \alpha(1 - \beta) \tau_d \\ Q_d &\equiv (1 - \beta) - (1 - \alpha) \tau_d \\ Q_\epsilon &\equiv (1 - \alpha)(\tau_d - 1) \lambda_p^h + \alpha(1 - \beta)(\tau_d - 1) H_p = (\tau_d - 1) \left[(1 - \alpha) \lambda_p^h + \alpha(1 - \beta) H_p \right] \end{aligned}$$

The goods-market equilibrium can then be written as

$$L y_t = Q_d d_t + K S_y - M S_d - Q_\epsilon \epsilon_t$$

Substituting $y_t = \psi_d d_t + \psi_\epsilon \bar{\lambda}_p \epsilon_t$:

$$L (\psi_d d_t + \psi_\epsilon \bar{\lambda}_p \epsilon_t) = Q_d d_t + K [\psi_d A_d d_t + (\psi_d A_\epsilon + \psi_\epsilon H_g) \epsilon_t] - M (A_d d_t + A_\epsilon \epsilon_t) - Q_\epsilon \epsilon_t$$

- The coefficient on d_t :

$$L\psi_d = Q_d + K\psi_d A_d - MA_d$$

Hence,

$$\psi_d = \frac{Q_d - MA_d}{L - KA_d} = \frac{Q_d - \frac{M}{1 - \beta\phi_d}}{L - \frac{K}{1 - \beta\phi_d}}$$

- The coefficient on ϵ_t :

$$L\bar{\lambda}_p\psi_\epsilon = K(\psi_d A_\epsilon + \psi_\epsilon H_g) - MA_\epsilon - Q_\epsilon$$

Therefore,

$$\psi_\epsilon = \frac{(K\psi_d - M)A_\epsilon - Q_\epsilon}{L\bar{\lambda}_p - KH_g} = \frac{(K\psi_d - M)\frac{\beta\phi_\epsilon H_g}{1 - \beta\phi_d} - Q_\epsilon}{L\bar{\lambda}_p - KH_g}$$

The derivation requires

$$|\beta\rho_\epsilon| < 1, \quad |\beta\phi_d| < 1, \quad \phi_d \neq 0, \quad \rho_\epsilon \neq 0$$

together with nonzero denominators in the expressions for ψ_d and ψ_ϵ .

C.3.2 Inflation

Inflation is determined by the New Keynesian Phillips curve:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t.$$

Following a one-time unanticipated shock, there is no future uncertainty along the realized equilibrium path for $t \geq 1$. Thus,

$$E_t \pi_{t+1} = \pi_{t+1}.$$

Forward iteration of the Phillips curve gives

$$\pi_t = \kappa \sum_{j=0}^{\infty} \beta^j y_{t+j},$$

provided that $\lim_{J \rightarrow \infty} \beta^{J+1} \pi_{t+J+1} = 0$.

Using the conjectured output rule $y_t = \psi_d d_t + \psi_\epsilon \hat{\epsilon}_t$, inflation satisfies

$$\pi_t = \kappa \psi_d \sum_{j=0}^{\infty} \beta^j d_{t+j} + \kappa \psi_\epsilon \sum_{j=0}^{\infty} \beta^j \hat{\epsilon}_{t+j}.$$

Define

$$PV_t(d) \equiv \sum_{j=0}^{\infty} \beta^j d_{t+j}, \quad PV_t(\hat{\epsilon}) \equiv \sum_{j=0}^{\infty} \beta^j \hat{\epsilon}_{t+j}$$

Then

$$\pi_t = \kappa [\psi_d PV_t(d) + \psi_\epsilon PV_t(\hat{\epsilon})]. \quad (42)$$

- $PV_t(\hat{\varepsilon}_t)$

$$PV_t(\hat{\varepsilon}) = \sum_{j=0}^{\infty} \beta^j \hat{\varepsilon}_{t+j} = \sum_{j=0}^{\infty} (\beta \rho_{\varepsilon})^j \hat{\varepsilon}_t = \frac{1}{1 - \beta \rho_{\varepsilon}} \hat{\varepsilon}_t = \frac{\bar{\lambda}_p}{1 - \beta \rho_{\varepsilon}} \varepsilon_t.$$

- $PV_t(d_t)$

Debt evolves according to $d_{t+1} = \phi_d d_t + \phi_{\varepsilon} \hat{\varepsilon}_t$.

The discounted present value of debt satisfies

$$\begin{aligned} PV_t(d) &= d_t + \sum_{j=0}^{\infty} \beta^{j+1} d_{t+j+1} = d_t + \beta \sum_{j=0}^{\infty} \beta^j (\phi_d d_{t+j} + \phi_{\varepsilon} \hat{\varepsilon}_{t+j}) \\ &= d_t + \beta \phi_d PV_t(d) + \beta \phi_{\varepsilon} PV_t(\hat{\varepsilon}) \end{aligned}$$

Solving for $PV_t(d)$ gives

$$PV_t(d) = \frac{1}{1 - \beta \phi_d} d_t + \frac{\bar{\lambda}_p \beta \phi_{\varepsilon}}{(1 - \beta \phi_d)(1 - \beta \rho_{\varepsilon})} \varepsilon_t$$

Substituting into (42):

$$\pi_t = \kappa \psi_d \left[\frac{1}{1 - \beta \phi_d} d_t + \frac{\beta \phi_{\varepsilon}}{(1 - \beta \phi_d)(1 - \beta \rho_{\varepsilon})} \hat{\varepsilon}_t \right] + \kappa \psi_{\varepsilon} \frac{1}{1 - \beta \rho_{\varepsilon}} \hat{\varepsilon}_t.$$

Matching the coefficients on d_t and $\hat{\varepsilon}_t$:

$$\pi_t = \omega_d d_t + \omega_{\varepsilon} \hat{\varepsilon}_t,$$

where

$$\omega_d = \frac{\kappa \psi_d}{1 - \beta \phi_d}, \quad \omega_{\varepsilon} = \frac{\kappa}{1 - \beta \rho_{\varepsilon}} \left[\psi_{\varepsilon} + \frac{\beta \phi_{\varepsilon} \psi_d}{1 - \beta \phi_d} \right]$$

C.3.3 Debt

For $t \geq 1$, the inflation-surprise term vanishes along the realized continuation path:

$$\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}] = 0.$$

Hence,

$$\begin{aligned} d_{t+1} &= \frac{1}{\beta} (d_t - t_t) + \frac{D}{Y} r_t - \frac{D}{Y} \underbrace{(\pi_{t+1} - \mathbb{E}_t[\pi_{t+1}])}_{=0, t \geq 1} \\ &= \frac{1}{\beta} [(1 - \tau_d) d_t - \tau_y y_t - (\tau_d - 1) \varepsilon_t] + \frac{D}{Y} \phi y_t \end{aligned}$$

Substituting $y_t = \psi_d d_t + \psi_{\varepsilon} \hat{\varepsilon}_t$:

$$\begin{aligned} d_{t+1} &= \frac{1}{\beta} \left[(1 - \tau_d) d_t - \tau_y (\psi_d d_t + \psi_{\varepsilon} \hat{\varepsilon}_t) - (\tau_d - 1) \frac{\hat{\varepsilon}_t}{\bar{\lambda}_p} \right] + \frac{D}{Y} \phi (\psi_d d_t + \psi_{\varepsilon} \hat{\varepsilon}_t) \\ &= \left[\frac{1}{\beta} (1 - \tau_d - \tau_y \psi_d) + \frac{D}{Y} \phi \psi_d \right] d_t + \left[\frac{1}{\beta} \left(-\tau_y \psi_{\varepsilon} - \frac{\tau_d - 1}{\bar{\lambda}_p} \right) + \frac{D}{Y} \phi \psi_{\varepsilon} \right] \hat{\varepsilon}_t \\ &= \phi_d d_t + \phi_{\varepsilon} \hat{\varepsilon}_t \end{aligned}$$

Therefore,

$$\begin{aligned}\phi_d &= \frac{1}{\beta} (1 - \tau_d - \tau_y \psi_d) + \frac{D}{Y} \phi \psi_d \\ \phi_\varepsilon &= \frac{1}{\beta} \left(-\tau_y \psi_\varepsilon - \frac{\tau_d - 1}{\bar{\lambda}_p} \right) + \frac{D}{Y} \phi \psi_\varepsilon\end{aligned}$$

C.3.4 Inflation and debt in initial period, $t = 0$

The initial period must be solved separately because the period-0 price level revalues nominal debt inherited from $t = -1$. Note

$$D_0 = \alpha \frac{R_{-1} b_{u,-1}}{\Pi_0} \quad (\text{level})$$

Since inherited debt is at steady state and the shock is unanticipated, R_{-1} and $b_{u,-1}$ are predetermined, and the only first-order movement in D_0 comes from surprise inflation:

$$d_0 = -\frac{D}{Y} \pi_0 \quad (\text{log-linearized})$$

At $t = 0$,

$$\pi_0 = \frac{\kappa \psi_d}{1 - \beta \phi_d} d_0 + \frac{\kappa \bar{\lambda}_p}{1 - \beta \rho_\varepsilon} \left[\psi_\varepsilon + \frac{\beta \phi_\varepsilon \psi_d}{1 - \beta \phi_d} \right] \varepsilon_0$$

Substituting the initial debt revaluation condition $d_0 = -\frac{D}{Y} \pi_0$ into the inflation equation gives:

$$\pi_0 = -\frac{\kappa \psi_d}{1 - \beta \phi_d} \frac{D}{Y} \pi_0 + \frac{\kappa \bar{\lambda}_p}{1 - \beta \rho_\varepsilon} \left[\psi_\varepsilon + \frac{\beta \phi_\varepsilon \psi_d}{1 - \beta \phi_d} \right] \varepsilon_0$$

Therefore,

$$\pi_0 = \frac{\frac{\kappa \bar{\lambda}_p \varepsilon_0}{1 - \beta \rho_\varepsilon} \left[\psi_\varepsilon + \frac{\beta \phi_\varepsilon \psi_d}{1 - \beta \phi_d} \right]}{1 + \frac{D}{Y} \frac{\kappa \psi_d}{1 - \beta \phi_d}}$$

Given π_0 , then

$$\begin{aligned}d_0 &= -\frac{D}{Y} \pi_0 \\ y_0 &= \psi_d d_0 + \psi_\varepsilon \bar{\lambda}_p \varepsilon_0 \\ r_0 &= \phi y_0 \\ d_1 &= \phi_d d_0 + \phi_\varepsilon \bar{\lambda}_p \varepsilon_0\end{aligned}$$

C.4 Results

In this model, I conduct the following counterfactual analysis (Table 18). Holding the population-weighted average belief fixed at 0.5, the figure plots the differences in the responses of output and government debt relative to the benchmark following a one-time innovation that generates

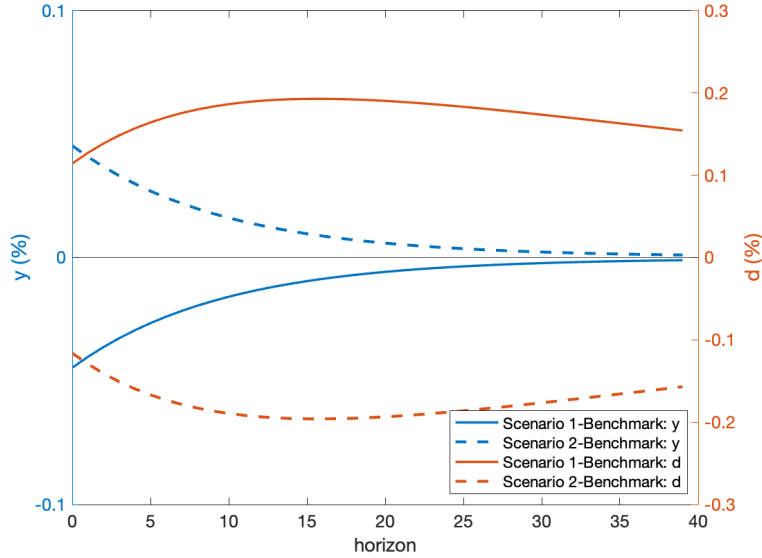
Table 18. Counterfactual Analysis

Scenarios	λ^H	λ^U	average
Benchmark	0.5	0.5	0.5
Scenario 1: High-MPC close to FIRE	0.99	0.4614	0.5
Scenario 2: High-MPC strongly non-FIRE	0.01	0.5386	0.5

a persistent tax increase. In Scenario 1, hand-to-mouth (HtM), or high-MPC, households hold beliefs that are close to FIRE: $\lambda_p^h = \lambda_g^u = 0.99$. In Scenario 2, HtM households hold beliefs that are strongly non-FIRE: $\lambda_p^h = \lambda_g^u = 0.01$. In each scenario, the beliefs of unconstrained, or low-MPC, households are chosen so that the population-weighted average belief remains equal to 0.5.

All parameters are calibrated the same as in the main analysis except for the population share and the wealth share. The share of population is set to be 0.927 for unconstrained households and 0.073 for hand-to-mouth households. Share of wealth is set to be 1 for unconstrained and 0 for constrained.

Figure 9. Differences in Output and Debt Responses Relative to the Benchmark



Notes: Impulse responses to a one-time tax increase of 1% of GDP. t is reported in percent of GDP, so a value of 1 corresponds to 1% of GDP.

In Scenario 1, output is lower and government debt is higher than in the benchmark. Because high-MPC households perceive a larger fraction of the tax increase, they reduce their consumption more sharply. The resulting contraction in aggregate demand lowers output and, consequently, the endogenous tax base. This decline in the tax base offsets part of the direct revenue gain from the tax increase, leaving government debt higher than in the benchmark.

In Scenario 2, the opposite occurs. Output is higher and government debt is lower than in the benchmark. Because high-MPC households perceive only a small fraction of the tax increase, they reduce their consumption by less. The contraction in aggregate demand and output is therefore attenuated, which preserves a larger tax base. As a result, the fiscal consolidation

produces a larger improvement in government debt relative to the benchmark.

These results are consistent with the main analysis; the effectiveness of fiscal consolidation depends not only on the average belief in the economy, but also on how that belief is distributed across households. Beliefs held by high-MPC households have a disproportionately large effect on contemporaneous aggregate demand and therefore on the endogenous fiscal response.

However, a limitation of the current specification is that second-order beliefs affect the equilibrium only through unconstrained households. Because HtM households base their consumption solely on perceived current disposable income, their second-order beliefs do not independently enter their consumption decisions.

Appendix D Measuring Group-Specific Tax Changes Using the CEX

SCE asks households about the change in their taxes over the next twelve months while holding income fixed, essentially the effective tax rate, so the objective benchmark must be measure a change in the effective tax rate at a monthly frequency and separately for each group. No existing data source provides both. Group-specific effective tax rates, $r_{g,t} = T_{g,t}/Y_{g,t}$, are available only at an annual frequency. The CEX is fielded quarterly, but its income and tax questions refer to the preceding twelve months, and the BLS imputes income-tax liabilities on an annual basis. The CBO series are also released with a lag of several years. The NIPA, in contrast, record tax payments quarterly, but only for the household sector as a whole.

I therefore proceed in two steps. First, I measure the quarterly realized change in the *aggregate* effective tax rate over the survey horizon using the NIPA. Second, I map this aggregate change into group-specific changes using a group-specific loading, κ_g , estimated from the annual CEX cross-section. The second step requires an assumption about how an aggregate change in the tax rate is distributed across groups, which I state and discuss below.

Step 1: Aggregate realized tax change. Let $r_t = W055_t/PI_t$ denote the aggregate effective personal tax rate in quarter t , where W055 is personal current taxes and PI is personal income (BEA, NIPA). For a survey response in quarter t , the realized change over the survey's twelve-month horizon is

$$z_t = 100 \times \left[\log \left(\frac{1}{4} \sum_{j=0}^3 r_{t+j} \right) - \log r_{t-1} \right],$$

which compares the average effective rate over the four quarters beginning with the survey quarter to the rate in the last quarter before the survey. Because z_t is a log difference, it approximates the percent change in the effective rate.

Step 2: From aggregate to group-specific changes. The aggregate rate is an income-weighted average of group rates,

$$r_t = \sum_g s_{g,t} r_{g,t}, \quad s_{g,t} = Y_{g,t}/Y_t,$$

so a quarterly observation of Δr_t gives one restriction on the G unobserved group-specific changes $\{\Delta r_{g,t}\}_{g=1}^G$. The restriction alone cannot pin down how the aggregate change is divided across groups. Identifying the group-specific changes therefore requires additional structure, and I impose the following two assumptions.

(A1) *Equal percentage-point changes:* all groups experience the same change in their effective tax rate, $\Delta r_{g,t} = \Delta r_t$ for all g .

(A2) *Stable composition:* income shares s_g are constant over the sample period, and κ_g is evaluated at sample-average effective tax rates, $\kappa_g = \bar{r}/\bar{r}_g$.

(A2) ensures that changes in the aggregate rate reflect changes in group rates rather than shifts in the income distribution, since then $\Delta r_t = \sum_g s_g \Delta r_{g,t}$. Given (A2), (A1) is consistent with this aggregation identity because $\sum_g s_g = 1$.

To translate (A1) into a statement about proportional changes, which is the object measured by z_t , divide both sides of $\Delta r_{g,t} = \Delta r_t$ by r_g . To a first order,

$$d \log r_g = \frac{dr_g}{r_g} = \frac{dr}{r_g} = \frac{r}{r_g} \cdot \frac{dr}{r} = \kappa_g d \log r, \quad \kappa_g \equiv \frac{r}{r_g}.$$

Applying this to the twelve-month change in Step 1 gives the group-specific realized tax change

$$z_{g,t} = \kappa_g z_t. \tag{43}$$

Equation (43) is a first-order approximation. The exact counterpart under (A1) is $z_{g,t} = 100 \log[1 + \kappa_g (e^{z_t/100} - 1)]$; the two differ negligibly for the changes observed in the sample.

The loading κ_g can be written in terms of observable shares:

$$\kappa_g = \frac{r}{r_g} = \frac{T/Y}{T_g/Y_g} = \frac{Y_g/Y}{T_g/T} = \frac{\text{income share of group } g}{\text{tax share of group } g}. \tag{44}$$

By construction, $\sum_g (T_g/T) \kappa_g = 1$. Assumption (A1) implies that a group with a below-average effective tax rate ($\kappa_g > 1$) experiences a larger *proportional* change in its tax rate for a given aggregate change. The alternative assumption of equal proportional changes across groups corresponds to $\kappa_g = 1$ for all g .

Measurement. I estimate κ_g from the BLS Consumer Expenditure Survey (CEX) Interview Survey for 2015–2023, which overlaps with the SCE sample period. Groups are defined as in the main text: group H consists of low-income / hand-to-mouth households whose income is below \$50,000 and group L consists of all remaining households. Tax liabilities include (i) federal and state income taxes, taken from the BLS tax-imputation files (`ntaxi`), which are computed with NBER TAXSIM; (ii) property taxes on owner-occupied dwellings; and (iii) general sales taxes, imputed with the IRS Optional State Sales Tax Tables using each household’s state of residence, income, and family size. Income is CEX total income before taxes. All shares use CEX sample weights. I compute κ_g separately for each year and average across years. The `ntaxi` files are not yet available for 2024 and 2025, so I apply the 2015–2023 average to those years, consistent with (A2).

Estimates. Table 19 reports the estimates. Group H accounts for 13.8 percent of income but only 8.0 percent of broad-base tax liabilities, reflecting the progressivity of the tax system: its effective tax rate is roughly 60 percent of the aggregate rate. Its loading therefore exceeds one ($\kappa_H = 1.736$), while $\kappa_L = 0.936$ is slightly below one. Under (A1), a given aggregate change in the effective tax rate thus translates into a proportional change for group H that is about 1.85 times as large as for group L. Because κ_g is the average of annual ratios, it need not equal the ratio of the averaged shares shown in the table.

Table 19. Group tax loadings, CEX 2015–2023 average

	H	L
Income share, Y_g/Y	0.138	0.862
Broad-base tax share, T_g/T	0.080	0.920
Tax loading, κ_g	1.736	0.936

For group H, whose effective tax rate is below average, (A1) therefore implies a larger proportional response than (A1') equal proportional changes ($\kappa_g = 1$).

Caveats. First, the tax bases differ: NIPA personal current taxes cover federal, state, and local income taxes and personal property taxes, but classify general sales taxes and real-estate taxes as taxes on production and imports, whereas κ_g uses the broad base. [ROBUSTNESS: κ_g on the income-tax base, or z_t with Table 3.5 sales and property taxes added.] Second, CEX income taxes are statutory liabilities imputed by TAXSIM, while NIPA records payments, including taxes on realized capital gains and final settlements. Third, the CEX under-represents top incomes and suppresses state identifiers for some households, which affects the sales-tax imputation. Fourth, payroll taxes are excluded from both measures, consistent with the survey question.

Appendix E Robust Inference on the Belief Parameter

Conventional inference for a response ratio can be misleading when its denominator is imprecisely estimated. I therefore construct Fieller-type confidence sets by directly inverting the ratio restriction. For each candidate value λ_0 , I test

$$H_0 : \beta_g^{\text{survey}} - \lambda_0 \beta_g^{\text{objective}} = 0. \quad (45)$$

Operationally, I estimate

$$\text{survey}_{t+12|t,g} - \lambda_0 \text{objective}_{t+12|t,g} = a_g(\lambda_0) + \rho_g(\lambda_0) \varepsilon_t^{\text{FIRE}} + \Pi_g(\lambda_0)' W_{t-1} + v_{g,t}(\lambda_0), \quad (46)$$

and retain λ_0 whenever $H_0 : \rho_g(\lambda_0) = 0$ is not rejected. Newey–West standard errors are used throughout.

The resulting Anderson–Rubin (AR) 90 percent confidence set is obtained by grid search over $\lambda_0 \in [-5, 5]$ (step 0.01), collecting all values for which the AR test of $H_0 : \lambda_g^p = \lambda_0$ fails to reject at the 10% level. In every specification the accepted set is disconnected and both endpoints coincide with the boundary of the search grid (−5 and/or 5), so no specification yields a bounded

90% confidence interval for λ_g^p . Thus, the data do not provide a bounded confidence interval for the level of λ_g^p . This result motivates treating the estimated response ratios as benchmark calibration targets and evaluating the model over alternative values in $[0, 1]$.

Table 20. Anderson–Rubin 90% Confidence Sets for λ_g^p

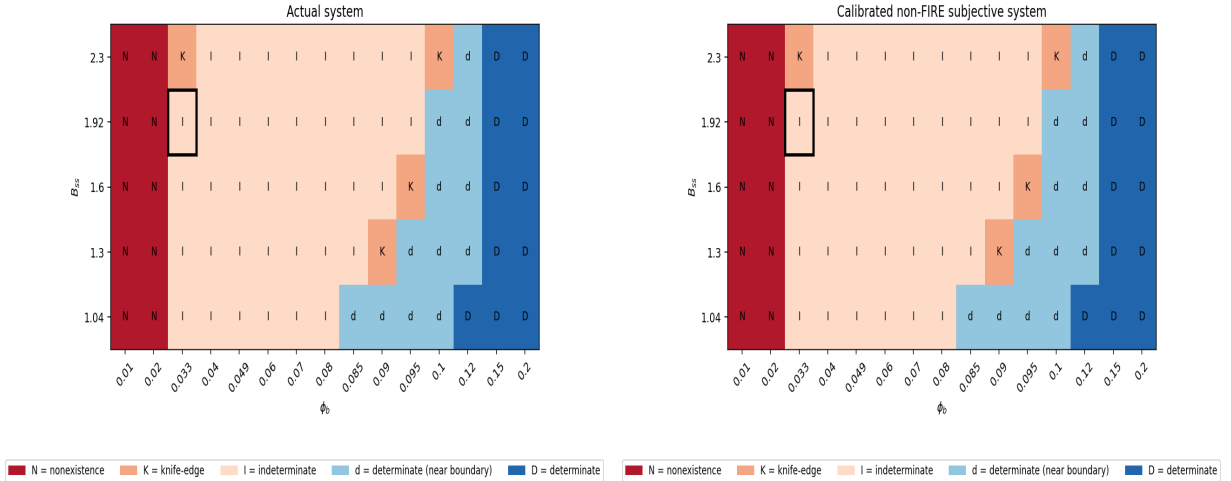
Specification	Sample	Group	F -stat	$\hat{\lambda}_{2SLs}$	90% AR confidence set
total_cps	Baseline	H (income < \$50k)	1.95	0.112	$[-5.00, -0.41] \cup [0.02, 5.00]$
		L (income $\geq$ \$50k)	1.80	0.180	$[-5.00, -1.10] \cup [-0.01, 5.00]$
	Excl. COVID	H (income < \$50k)	0.73	0.303	$[-5.00, -0.08] \cup [0.12, 5.00]$
		L (income $\geq$ \$50k)	0.65	0.651	$[-5.00, -0.58] \cup [0.21, 5.00]$

Notes: F -stat is the first-stage F -statistic; $F < 10$ throughout indicates a weak instrument.

Appendix F Debt Feedback and Equilibrium Determinacy

This appendix characterizes the determinacy region over combinations of the steady-state debt level, B_{ss} , and ϕ_b , for a target MPC of 0.25 in both the actual and subjective economies. We determine equilibrium existence and determinacy using the winding-number method. In the figures, N denotes nonexistence of an equilibrium, K the boundary between nonexistence and indeterminacy, I indeterminacy, d determinacy with a root close to one, and D determinacy. In both economies, when $B_{ss} = 1.04$, equilibrium is determinate only if $\phi_b \geq 0.12$. We therefore set $\phi_b = 0.12$.

Figure 10. Determinacy Regions in the Actual and Subjective Economies



This calibration implies a half-life of government debt of approximately five to six quarters, which is faster than typical empirical estimates. The relatively strong fiscal feedback is required by a reinforcing debt–inflation mechanism in the model. When government debt rises, the tax rule raises taxes. Higher taxes increase the tax wedge entering the wage Phillips curve, putting upward pressure on inflation. The Taylor rule then raises the nominal interest rate, increasing debt-service costs and thereby further raising government debt. Without sufficiently strong feedback from debt to taxes, this feedback loop can cause debt to diverge. For the target MPC

of 0.25 and $B_{ss} = 1.04$, a value of $\phi_b \geq 0.12$ is therefore necessary to ensure a determinate equilibrium.

Appendix G HANK with Belief–MPC Correlation

This appendix derives the two-stage equilibrium used in Section 4.1. The logic has two steps. Informed households must first forecast the aggregate path, which requires taking a stand on how many other households are informed; solving that forecasting problem is the *subjective equilibrium*. Only then can the *actual* equilibrium be computed, by aggregating informed and uninformed behavior with the true informed shares.

Second-order beliefs. Collect the second-order beliefs in a matrix Q ,

$$Q = \begin{bmatrix} q_{HH} & q_{HL} \\ q_{LH} & q_{LL} \end{bmatrix},$$

where q_{ij} is the share of group j that group i believes to be informed. Assumption 1 makes this belief a statement about a population-wide fraction rather than about a particular group, so $q_{ij} = \lambda_i^g$ for both j , and

$$Q = \begin{bmatrix} \lambda_H^g & \lambda_H^g \\ \lambda_L^g & \lambda_L^g \end{bmatrix}.$$

Household responses. Let $z = (y, r, b, \pi, i)^\top$, let J_h^X denote the FIRE Jacobian, $\mathcal{E}_h^X$ the myopic Jacobian that retains only contemporaneous responses, and define the forward-looking kernels $K_h^X = J_h^X - \mathcal{E}_h^X$ and $K_h^{X\varepsilon} = J_h^{X\varepsilon} - \mathcal{E}_h^{X\varepsilon}$ for $X \in \{C, A\}$. An informed household in group h acts on its forecast $\hat{z}_h$; an uninformed household responds only to realized inputs:

$$\begin{aligned} X_h^I &= \mathcal{E}_h^X \cdot z + K_h^X \cdot \hat{z}_h + J_h^{X\varepsilon} \cdot \varepsilon \\ X_h^U &= \mathcal{E}_h^X \cdot z + \mathcal{E}_h^{X\varepsilon} \cdot \varepsilon \end{aligned}$$

Averaging within the group with the true informed share λ_h^p ,

$$\begin{aligned} X_h &= \lambda_h^p \cdot X_h^I + (1 - \lambda_h^p) \cdot X_h^U \\ &= \mathcal{E}_h^X z + \lambda_h^p K_h^X \hat{z}_h + [\mathcal{E}_h^{X\varepsilon} + \lambda_h^p K_h^{X\varepsilon}] \varepsilon. \end{aligned}$$

Fixed-point expectations of the informed. An informed household in group h must forecast the behavior of *both* groups. It knows the information structure and the Jacobians, so it can compute what an informed household in group g would do; the only thing it does not know is *who* is informed. Formally this is the assumption $E_h^I[\hat{z}_g] = \hat{z}_g$: the belief structure is common knowledge, so group g 's forecast is not itself uncertain from h 's point of view. Under it,

$$\begin{aligned} E_h^I[X_g^U] &= \mathcal{E}_g^X \hat{z}_h + \mathcal{E}_g^{X\varepsilon} \varepsilon \\ E_h^I[X_g^I] &= \mathcal{E}_g^X \hat{z}_h + K_g^X \hat{z}_g + J_g^{X\varepsilon} \varepsilon \end{aligned}$$

Group h 's informed households expect a fraction q_{hg} of group g to be informed, so

$$\begin{aligned} E_h^I[X_g] &= q_{hg}E_h^I[X_g^I] + (1 - q_{hg})E_h^I[X_g^U] \\ &= \mathcal{E}_g^X \hat{z}_h + q_{hg}K_g^X \hat{z}_g + [\mathcal{E}_g^{X\varepsilon} + q_{hg}K_g^{X\varepsilon}]\varepsilon, \end{aligned}$$

and, aggregating across groups,

$$\hat{X}_h = \sum_{g \in \{H, L\}} \mu_g \{ \mathcal{E}_g^X \hat{z}_h + q_{hg}K_g^X \hat{z}_g + [\mathcal{E}_g^{X\varepsilon} + q_{hg}K_g^{X\varepsilon}]\varepsilon \}.$$

Because $\hat{X}_h$ depends on both $\hat{z}_h$ and $\hat{z}_g$, the two groups' forecasts must be solved jointly:

$$\hat{X}_h = \hat{X}_h(\hat{z}_H, \hat{z}_L), \quad \hat{X}_L = \hat{X}_L(\hat{z}_H, \hat{z}_L).$$

Subjective general equilibrium. Observer h closes its perceived economy with the same equilibrium conditions as the true economy. Let L and F denote the lag and lead operators, let $\bar{\tau}$ be the steady-state aggregate tax rate, and let $\varpi \equiv \sum_g \mu_g \varsigma_g \omega_g$ be the direct-revenue coefficient. Then $\hat{z}_h$ solves

$$\begin{aligned} \hat{X}_h &= \sum_{g \in \{H, L\}} \mu_g \{ \mathcal{E}_g^X \hat{z}_h + q_{hg}K_g^X \hat{z}_g + [\mathcal{E}_g^{X\varepsilon} + q_{hg}K_g^{X\varepsilon}]\varepsilon \} & X \in \{c, a\} \\ \hat{a}_h &= \hat{b}_h & \text{asset market} \\ \hat{\pi}_h &= \beta_{nk}F\hat{\pi}_h + \kappa_w \left[\frac{1}{\text{EIS}}\hat{c}_h + \frac{1}{\text{Frisch}}\hat{y}_h + \tilde{\tau}_h^{\text{eff}} \right] & \text{wage Phillips curve} \\ \hat{i}_h &= \rho_i L \hat{i}_h + (1 - \rho_i)(\phi_\pi \hat{\pi}_h + \phi_y \hat{y}_h) & \text{Taylor rule} \\ \hat{r}_h &= L \hat{i}_h - \hat{\pi}_h & \text{Fisher equation} \\ [I - (1 + R^{ss} - \phi_b)L]\hat{b}_h + \bar{\tau}\hat{y}_h - \frac{B^{ss}}{Y^{ss}}\hat{r}_h &= -\varpi\varepsilon & \text{government budget} \\ x &= (1 - \rho_x L)^{-1}\varepsilon & \text{tax shock} \end{aligned}$$

Goods-market clearing $\hat{y}_h = \hat{c}_h$ then holds by Walras's law and is not imposed separately. Stacking the two observers, the system is linear in $(\hat{z}_H, \hat{z}_L)$ and depends on the second-order beliefs only through Q :

$$A(Q) \begin{bmatrix} \hat{z}_H \\ \hat{z}_L \end{bmatrix} = R(Q)\varepsilon \quad \implies \quad \begin{bmatrix} \hat{z}_H \\ \hat{z}_L \end{bmatrix} = A(Q)^{-1}R(Q)\varepsilon.$$

Setting Q to a matrix of ones recovers the FIRE path, which is the nesting check reported in the text.

Actual equilibrium. With the perceived paths in hand, the actual equilibrium aggregates informed and uninformed behavior using the *true* informed shares λ_g^p rather than the believed shares Q :

$$X = \sum_{g \in \{H, L\}} \mu_g \cdot X_g = \sum_g \mu_g \{ \mathcal{E}_g^X z + \lambda_g^p K_g^X \hat{z}_g + [\mathcal{E}_g^{X\varepsilon} + \lambda_g^p K_g^{X\varepsilon}]\varepsilon \},$$

together with the same wage Phillips curve, asset-market clearing, Taylor rule, Fisher equation, and government budget constraint, now evaluated at the realized path z . Writing the resulting system as

$$Az = R\varepsilon + B_p \begin{bmatrix} \hat{z}_H \\ \hat{z}_L \end{bmatrix} \implies z = A^{-1} \{R + B_p A(Q)^{-1} R(Q)\} \varepsilon,$$

which makes the two-stage structure explicit: $A(Q)^{-1} R(Q)$ is the forecasting stage governed by second-order beliefs, and A^{-1} is the equilibrium stage, with the true informed shares entering through B_p .

Appendix H Government Debt Maturity Structure

This section incorporates a maturity structure for government debt. The main findings remain unchanged: average fiscal beliefs increase the output cost of fiscal consolidation, whereas fiscal belief heterogeneity offsets this cost by one-third. These results are also robust to alternative tax timing specifications and prior predictive analysis. Section H.1 outlines the model environment, Section H.2 presents the quantitative results, and Section H.3 provides robustness checks.

H.1 Environment: Fiscal Policy

To model the maturity structure of government debt, I assume that debt consists of nominal bonds whose coupons decay geometrically at rate ρ . Let B_t denote the number of bonds outstanding at the end of period t and Q_t their nominal price, so that the real market value of debt is $\mathcal{B}_t \equiv Q_t B_t / P_t$. Setting $\rho = 0$ recovers one-period debt.

Bond price. No arbitrage between the long-term bond and a one-period nominal bond with rate i_t requires

$$Q_t(1 + i_t) = 1 + \rho Q_{t+1}, \quad (47)$$

so that $Q_t = \sum_{k \geq 0} \rho^k \prod_{j=0}^k \frac{1}{1+i_{t+j}}$. In the steady state, $Q = 1/(1 + i^{ss} - \rho)$, or equivalently $(1 + i^{ss})Q = 1 + \rho Q$. Let q_t denote the log deviation of Q_t , let i_t and π_t denote deviations from the steady state, and define

$$\omega \equiv \frac{\rho Q}{1 + \rho Q} = \frac{\rho}{1 + i^{ss}}.$$

Log-linearizing (47) yields $q_t = -i_t + \omega q_{t+1}$, and solving forward,

$$q_t = - \sum_{k \geq 0} \omega^k i_{t+k}. \quad (48)$$

The log bond price moves inversely with the entire expected path of short-term rates, with weights that decline geometrically at rate ω .

Budget constraint. The government pays the coupon on its outstanding bonds and finances the deficit by issuing new bonds:

$$Q_t B_t = (1 + \rho Q_t) B_{t-1} + P_t (G - T_t).$$

Dividing by P_t and multiplying and dividing the first term by $Q_{t-1}P_{t-1}$ expresses the constraint in terms of the real market value of debt:

$$\mathcal{B}_t = (1 + R_t^b) \mathcal{B}_{t-1} + G - T_t, \quad 1 + R_t^b \equiv \frac{1 + \rho Q_t}{Q_{t-1}(1 + \Pi_t)}, \quad (49)$$

where R_t^b is the ex-post real holding return on government debt between $t - 1$ and t : a bond bought at Q_{t-1} pays a coupon of one and leaves ρ bonds worth ρQ_t . Let $b_t \equiv (\mathcal{B}_t - \mathcal{B}^{ss})/Y^{ss}$ and $\Delta T_t \equiv (T_t - T^{ss})/Y^{ss}$. To first order, (49) becomes

$$b_t = (1 + r^{ss}) b_{t-1} + \frac{\mathcal{B}^{ss}}{Y^{ss}} r_t^b - \Delta T_t, \quad (50)$$

which has the same form as with one-period debt, except that the return on debt now reflects changes in the bond price:

$$r_t^b = \omega q_t - q_{t-1} - \pi_t. \quad (51)$$

Realized returns. After the shock in period 0, the path is deterministic. For $t \geq 1$, the price q_{t-1} already reflects the realized path, so (48) implies $q_{t-1} = -i_{t-1} + \omega q_t$. Substituting into (51),

$$r_t^b = i_{t-1} - \pi_t, \quad t \geq 1,$$

which is the same return as on one-period nominal debt. In period 0, the pre-shock price is at its steady state, $q_{-1} = 0$, so

$$r_0^b = \omega q_0 - \pi_0 = -\pi_0 - \sum_{k \geq 0} \omega^{k+1} i_k.$$

Long-term debt thus introduces an additional revaluation channel on impact. In addition to the revaluation through inflation, $-\pi_0$, the bond price term ωq_0 revalues outstanding debt in response to the change in the entire expected path of nominal rates. When the shock lowers that path, the market value of outstanding debt rises, offsetting part of the debt reduction.

Calibration of the maturity structure. I calibrate ρ to the maturity of marketable U.S. Treasury debt. Using security-level data from the Monthly Statement of the Public Debt (MSPD) from January 2001 to March 2026, I compute for each month the weighted average maturity (WAM) of marketable securities, defined as the average remaining time to maturity weighted by each security's outstanding face value. The sample-average WAM is 64.5 months, or 21.5 quarters. Under the geometric structure, the undiscounted average maturity of the payment stream is $1/(1 - \rho)$, which gives $\rho = 1 - 1/21.5 = 0.954$ at a quarterly frequency. The implied discounted decay is $\omega = \rho/(1 + i^{ss}) \approx 0.949$, so the Macaulay duration of the debt portfolio, $1/(1 - \omega)$, is about 19.5 quarters. Finally, $\mathcal{B}^{ss}/Y^{ss} = 1.04$ is the market value of liquid assets per capita in the SCF, consistent with the market-value definition of $\mathcal{B}_t$.

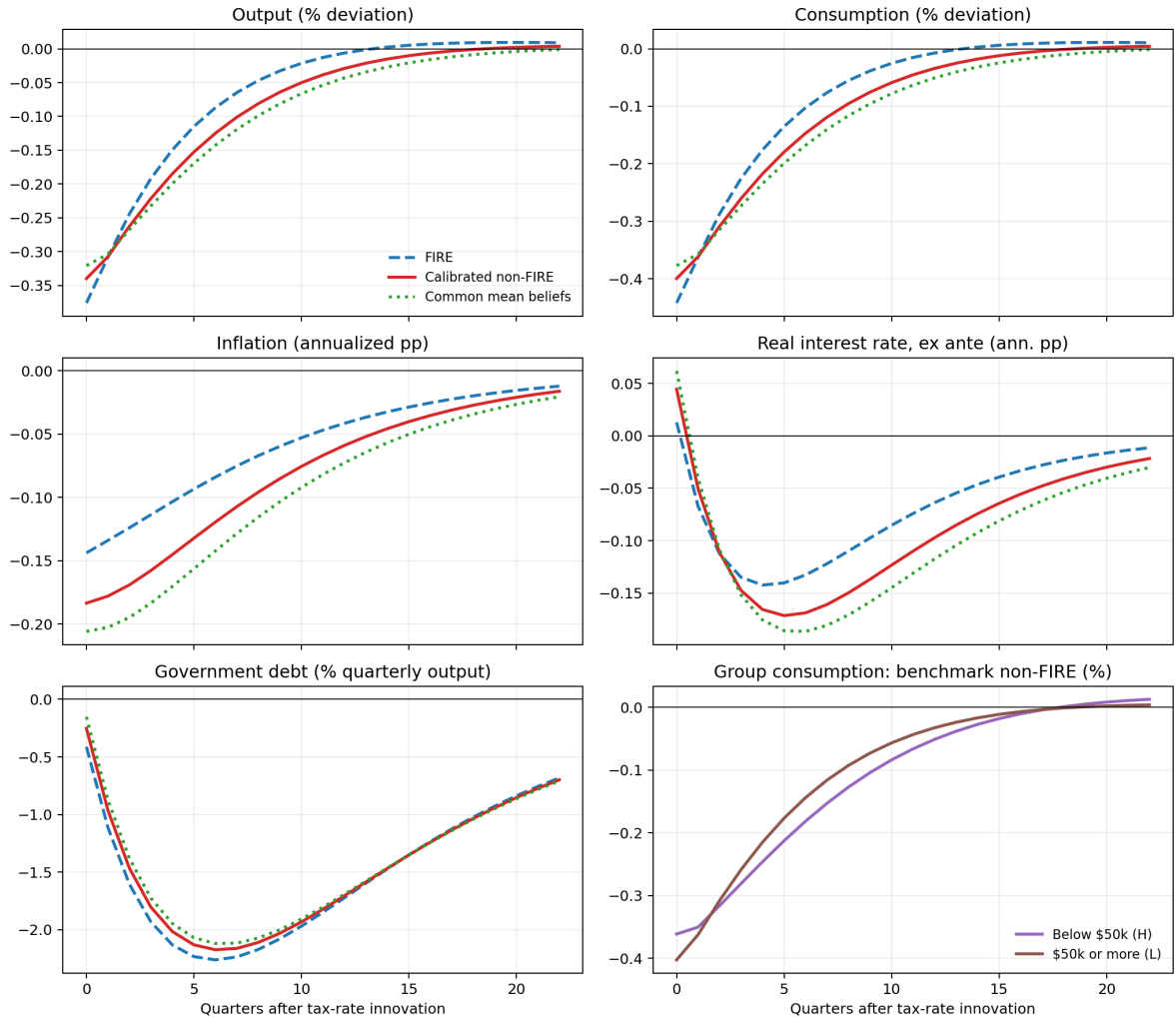
Calibration of the maturity structure. I calibrate ρ to match the empirical maturity profile of marketable U.S. Treasury debt. Using security-level data from the Monthly Statement of the Public Debt (MSPD) from January 2001 through March 2026, I calculate the monthly weighted average maturity (WAM) of marketable securities, weighting each issue by its outstanding face value. The sample-average WAM is 64.5 months (21.5 quarters). Under a geometric decay

structure, the undiscounted average maturity of the payment stream equals $1/(1 - \rho)$, yielding $\rho = 1 - 1/21.5 = 0.954$ at a quarterly frequency. Accounting for discounting yields the effective decay factor $\omega = \rho/(1 + i^{ss}) \approx 0.949$. This corresponds to a Macaulay duration, which measures the present-value-weighted maturity of the debt stream, of $1/(1 - \omega) \approx 19.5$ quarters. I use this Macaulay duration as the empirical target to calibrate ω , which directly governs the model's debt-revaluation dynamics.

H.2 Quantitative results

Figure 11 compares the responses to a one-percentage-point innovation in the labor-income tax rate under three belief specifications: full-information rational expectations (FIRE), common mean beliefs, and calibrated heterogeneous beliefs. Under the common-belief counterfactual, both groups receive the population-average first-order attention weight, $\lambda_H^{PE} = \lambda_L^{PE} = 0.540$, and the higher-order weight remains common across groups, as in the baseline. The counterfactual therefore preserves the population-average belief while removing its cross-group dispersion. I organize the discussion around two steps. Moving from FIRE to common beliefs isolates the *level* effect of fiscal beliefs. Moving from common to heterogeneous beliefs isolates its *dispersion* effect.

Figure 11. Impulse Responses Under Labor-income-tax Consolidation



Level effect of fiscal beliefs: from FIRE to common beliefs. Consumption falls by 0.442 percent on impact under FIRE but by only 0.337 percent under common beliefs, and output falls by 0.376 and 0.321 percent, respectively. Every household observes its current tax payment, so the attenuation does not come from the current tax. It reflects the underweighting of the expected continuation of the tax increase and of its general-equilibrium consequences.

The same underweighting makes the contraction more persistent. FIRE households anticipate the future relief as well as the future burden: the tax innovation decays, debt reduction lowers future taxes through the debt-feedback term of the fiscal rule, and real rates fall. Non-FIRE households underweight this relief as well, so their spending recovers more slowly. Output returns to steady state in quarter 13 under FIRE but only in quarter 22 under common beliefs.

Because the wage Phillips curve is forward looking and wage setters have full information, the more persistent contraction in demand translates into a deeper decline in wage and price inflation, even though the impact decline in output is smaller. Impact inflation is -0.144 annualized percentage point under FIRE and -0.206 under common beliefs. Because the policy rule is inertial, the nominal rate initially falls by less than expected inflation, and the impact real rate rises by more under common beliefs (0.013 versus 0.062). As the policy rate cuts accumulate, the real rate turns negative from the second quarter onward and falls well below its FIRE counterpart (-0.074 versus -0.131 at $h = 11$).

Dispersion effect of fiscal beliefs: from common to heterogeneous beliefs. The heterogeneous-belief economy lies between FIRE and common beliefs for every variable and at every horizon. Impact inflation is -0.183 , the impact real rate is 0.044 , output is 0.039 percent below steady state at $h = 11$, and output returns to steady state in quarter 19. Belief dispersion therefore moves the economy back toward FIRE.

The group-level responses clarify this mechanism. Group L perceives a tax shock more correctly ($\lambda_L^p = 0.685$) than the common benchmark, leading to a faster consumption recovery than under common beliefs (-0.259 versus -0.274 percent at $h = 3$). Group H , by contrast, perceives the tax shock less precisely ($\lambda_H^{PE} = 0.303$), resulting in a slower recovery (-0.281 versus -0.260 percent). While the two groups feature comparable marginal propensities to consume (0.265 and 0.241), their shares in aggregate spending are highly asymmetric: Group H accounts for just 7.3 percent of aggregate consumption and 6.4 percent of liquid assets. The aggregate response is therefore governed mainly by group L . Weighting these beliefs by consumption shares yields an effective aggregate belief parameter of about $0.073 \times 0.310 + 0.927 \times 0.685 \approx 0.658$, substantially above the population average of 0.550 . Belief heterogeneity thus places beliefs where most of the expectation-sensitive spending occurs.

This is the quantitative counterpart of the mechanism derived in Section 3.2. The aggregate path depends not only on the population mean of the informed share but also on its covariance with each group's weight in aggregate demand, including its marginal propensity to consume, its liquid-asset share, and its labor-income share. In this calibration the MPC gap is small, so the covariance operates mainly through the consumption, asset, and income shares. A representative-agent or common-belief specification sets this covariance to zero by construction and therefore understates effective fiscal beliefs.

Debt dynamics. With long-term debt, the response of debt differs markedly across belief specifications on impact, but the differences fade within about three years. On impact, the revaluation of outstanding debt through inflation and the bond price differs across specifications. Afterward, the debt feedback in the tax rule makes the inherited gap decay geometrically at rate $1 + r^{ss} - \phi_b = 0.885$ per quarter.

Using the government budget constraint (50) and $b_{-1} = 0$, impact debt equals the revaluation of outstanding debt minus the change in tax revenue:

$$b_0 = \underbrace{-\frac{\mathcal{B}^{ss}}{Y^{ss}} \pi_0}_{\text{inflation}} + \underbrace{\frac{\mathcal{B}^{ss}}{Y^{ss}} \omega q_0}_{\text{bond price}} - \left(\underbrace{\bar{\tau} y_0}_{\text{tax base}} + \underbrace{x_0}_{\text{direct}} \right),$$

where the debt-feedback term is zero on impact. Table 21 reports each term.

Table 21. Impact Debt Decomposition

	Debt	Revaluation		Tax revenue	
	b_0	Bond price	Inflation	Tax base	Direct
FIRE	-0.412	0.493	0.037	-0.058	1.000
Hetero.	-0.252	0.647	0.048	-0.053	1.000
Common	-0.152	0.745	0.054	-0.050	1.000

Notes: Entries are in percent of quarterly steady-state output. Debt equals the sum of the two revaluation terms minus the sum of the two tax-revenue terms.

On impact, the revaluation of long-term debt offsets more than half of the debt reduction from higher tax revenue: 56 percent under FIRE, and 73 and 84 percent under heterogeneous and common beliefs. The revaluation works through two channels, and belief distortions amplify both. First, the tax increase lowers the expected path of nominal rates, which raises the market value of outstanding debt by 0.49 percent of quarterly output under FIRE, 0.65 percent under heterogeneous beliefs, and 0.74 percent under common beliefs. This bond-price channel is larger under non-FIRE beliefs because the deeper and more persistent disinflation they generate lowers the rate path further through the Taylor rule. Second, the deeper disinflation raises the real value of debt directly, but this inflation channel is small (0.04 to 0.05 percent). Impact debt therefore falls by 0.41 percent of quarterly output under FIRE but only by 0.25 and 0.15 percent under heterogeneous and common beliefs.

These differences are not long-lived. Because the revaluation occurs only on impact and the debt feedback in the tax rule shrinks any inherited gap by about 11.5 percent per quarter, the debt paths converge: at the end of year 1, debt is 1.93, 1.80, and 1.73 percent of quarterly output below steady state under FIRE, heterogeneous beliefs, and common beliefs, and by the end of year 3 the gap is less than 0.05 percentage points.

H.3 Fiscal consolidation effectiveness and output costs

Table 22 presents the main quantitative result. Because debt reduction differs by at most 5 percent across specifications (Panel D), differences in the sacrifice ratio and in fiscal effectiveness

are driven almost entirely by differences in cumulative output loss (Panel C). Consequently, the ranking across specifications directly mirrors the cumulative output dynamics shown in Figure 11.

Table 22. Beliefs and the Measures of Fiscal Consolidation

Horizon	Scenario			Decomposition of Δ		
	FIRE	Common	Hetero.	Level (F→C)	Dispersion (C→H)	Disp./ Level
<i>Panel A: Sacrifice ratio $\mathcal{S}_s(h)$ (cumulative output loss per unit debt reduction)</i>						
Impact ($h = 0$)	0.913	2.111	1.347	1.197	-0.763	-0.64
End of year 1 ($h = 3$)	0.580	0.651	0.627	0.071	-0.024	-0.34
End of year 3 ($h = 11$)	0.892	1.139	1.058	0.247	-0.081	-0.33
End of year 5 ($h = 19$)	1.744	2.343	2.142	0.599	-0.201	-0.34
<i>Panel B: Fiscal effectiveness $\mathcal{E}_s(h)$ (debt reduction per unit of cumulative output loss)</i>						
Impact ($h = 0$)	1.095	0.474	0.742	-0.622	0.269	0.43
End of year 1 ($h = 3$)	1.724	1.536	1.595	-0.188	0.059	0.31
End of year 3 ($h = 11$)	1.121	0.878	0.945	-0.243	0.067	0.28
End of year 5 ($h = 19$)	0.573	0.427	0.467	-0.147	0.040	0.27
<i>Panel C: Cumulative output loss $\mathcal{L}_s(h)$ (pp-quarters)</i>						
Impact ($h = 0$)	0.376	0.321	0.340			
End of year 1 ($h = 3$)	1.120	1.124	1.131			
End of year 3 ($h = 11$)	1.650	2.055	1.928			
End of year 5 ($h = 19$)	1.616	2.222	2.014			
<i>Panel D: Debt reduction $\mathcal{D}_s(h)$</i>						
Impact ($h = 0$)	0.412	0.152	0.252			
End of year 1 ($h = 3$)	1.932	1.726	1.804			
End of year 3 ($h = 11$)	1.849	1.804	1.822			
End of year 5 ($h = 19$)	0.927	0.949	0.940			

Notes: Because $\mathcal{E}$ is a nonlinear transformation of $\mathcal{S}$, the Disp./Level ratios differ across Panels A and B.

Level effect. On impact, the common-belief distortion lowers the sacrifice ratio from 0.420 to 0.387 and raises fiscal effectiveness from 2.281 to 2.584, because non-FIRE fiscal beliefs attenuate the immediate contraction in spending. The ranking reverses within the first year, as the contraction becomes more persistent under common beliefs, corresponding to the crossing of the output IRF paths. By the end of year 5, cumulative output loss is 2.165 percentage-point quarters under common beliefs, compared with 1.514 under FIRE, an increase of about 45 percent. Debt reduction rises by only about 5 percent, from 0.975 to 1.026. The sacrifice ratio consequently increases from 1.552 to 2.111, and fiscal effectiveness falls from 0.644 to 0.474, a decline of about 27 percent. The average belief distortion thus shifts the output cost of consolidation from impact toward later quarters and lowers its medium-run effectiveness.

Dispersion effect. Belief dispersion moves both measures in the opposite direction from the level effect at every horizon. On impact it slightly raises the sacrifice ratio (+0.007) and lowers

effectiveness (-0.046). From the first year onward, as the output recovery under dispersion accelerates relative to common beliefs (Figure 7), it lowers the sacrifice ratio (-0.079 at $h = 11$ and -0.208 at $h = 19$) and raises effectiveness ($+0.084$ and $+0.052$). Over the medium run, the dispersion component offsets about one third of the level effect on the sacrifice ratio ($0.208/0.559 \simeq 0.37$) and about one third of the level effect on effectiveness ($0.052/0.171 \simeq 0.30$ at $h = 19$). The two offsets differ only because $\mathcal{E}$ is the inverse of $\mathcal{S}$. Both show that the heterogeneous-belief economy lies between FIRE and common beliefs. At the end of year 5, fiscal effectiveness under heterogeneous beliefs is 0.525, about 10 percent higher than under common beliefs. This offset reflects the mechanism in Section 4.3: heterogeneous beliefs place close to full-information rational-expectations on the households that account for most aggregate spending, which brings the economy closer to FIRE.

Implications. The results have two implications for tax-based consolidation under non-FIRE fiscal beliefs. First, misperception postpones the output cost of consolidation. Because households underweight the continuation of the tax increase and its general-equilibrium consequences, the impact contraction is attenuated, but the recovery is delayed. Over the medium horizon, a consolidation that retires essentially the same amount of debt therefore incurs a larger cumulative output cost. Evaluations based on short-run responses thus understate the cost of consolidation when beliefs depart from full information.

Second, this cost depends on how misperception is distributed across households, not only on its average. A common-belief specification with the same population-average fiscal beliefs understates medium-run fiscal effectiveness by about 10 percent (0.474 versus 0.525), because it ignores that beliefs closer to full-information rational-expectation is concentrated among the households that account for most aggregate spending. Belief dispersion offsets roughly one third of the level effect on the sacrifice ratio, so a representative-agent or common-belief specification overstates the cost of non-FIRE beliefs by a quantitatively meaningful margin.

H.4 Robustness

H.4.1 An alternative tax timing

Table 23 reports the results. The settlement timing deepens the impact contraction in both non-FIRE economies: impact output falls by 0.359 percent under heterogeneous beliefs and 0.342 percent under common beliefs, compared with 0.340 percent and 0.321 percent in the baseline.

From the end of the first year onward, settlement raises the sacrifice ratio in both non-FIRE economies, and the level component becomes larger: $+0.419$ at $h = 11$ and $+1.420$ at $h = 23$, compared with $+0.247$ and $+0.903$ in the baseline. The dispersion component grows in proportion, to -0.136 and -0.456 . As a result, the ratio of the two components is essentially unchanged: dispersion offsets 32 percent of the level component at the end of years 3 and 6, compared with 33 percent in the baseline. The tax timing therefore affects the size of the belief distortion but not the share of it that is offset by the cross-group allocation of beliefs.

Impact ratios are the exception and are not informative. On impact, the fall in the expected path of nominal rates raises the market value of long-term debt, which offsets most of the debt

Table 23. Robustness: Tax Timing

<i>Panel A: Sacrifice ratio $\mathcal{S}_s(h)$</i>					
Horizon	FIRE	Baseline timing		Settlement timing	
		Common	Hetero.	Common	Hetero.
Impact ($h = 0$)	0.913	2.111	1.347	16.012	2.113
End of year 1 ($h = 3$)	0.580	0.651	0.627	0.763	0.702
End of year 3 ($h = 11$)	0.892	1.139	1.058	1.312	1.176
End of year 6 ($h = 23$)	2.573	3.477	3.175	3.994	3.537

<i>Panel B: Decomposition into level ($F \rightarrow C$) and dispersion ($C \rightarrow H$)</i>						
Horizon	Level	Baseline timing		Level	Settlement timing	
		Dispersion	Disp./Level		Dispersion	Disp./Level
Impact ($h = 0$)	+1.197	−0.763	−0.64	+15.098	−13.899	−0.92
End of year 1 ($h = 3$)	+0.071	−0.024	−0.34	+0.183	−0.061	−0.33
End of year 3 ($h = 11$)	+0.247	−0.081	−0.33	+0.419	−0.136	−0.32
End of year 6 ($h = 23$)	+0.903	−0.302	−0.33	+1.420	−0.456	−0.32

Notes: $\mathcal{S}_s(h) = \mathcal{L}_s(h)/\mathcal{D}_s(h)$, normalized by the impact direct-tax revenue. The FIRE path is identical under both timings. Level = $\mathcal{S}_C - \mathcal{S}_F$ and Dispersion = $\mathcal{S}_H - \mathcal{S}_C$; a negative component lowers the sacrifice ratio. At impact under the settlement timing both components are close to zero, so their ratio is not informative.

reduction from the direct revenue gain. Impact debt reduction is therefore close to zero (0.02 percent of quarterly output under settlement with common beliefs), so the ratio reaches 16.0.

H.4.2 Prior predictive analysis.

The baseline-prior column of Table 24 reports the results. Three findings stand out. First, both components of the decomposition have robust signs. The level component raises the sacrifice ratio in 92 to 95 percent of draws, while the dispersion component lowers it in 94 to 97 percent. The two components have opposite signs in 87 to 92 percent of draws. Beyond impact, the median ratio of dispersion to level is -0.32 , -0.34 , and -0.34 at the end of years 1, 3, and 6, close to the one-third offset in the baseline calibration.

Second, the sign of the dispersion component is determined by the ordering of beliefs across groups. Among the 498 draws with $\lambda_L^p > \lambda_H^p$, the dispersion component is negative at $h = 23$ in every draw. Among the 16 draws that reverse the ordering, it is negative in none. The magnitude of the offset also increases with the belief gap (correlation -0.37). The mechanism is therefore governed by which group is better informed, as the analytical decomposition in Section 3.2 implies.

Third, λ^g matters little. Its correlation with the dispersion component at $h = 23$ is only 0.17, and the dispersion component is negative in 96 to 98 percent of draws within each tercile of the λ^g prior. This supports the identification argument of Section 4.2 and justifies restricting λ^g rather than estimating it.

One caveat should be stated plainly. The analysis retains only draws that yield a determinate equilibrium, which excludes 14 percent of the baseline draws. In every retained draw, debt

Table 24. Prior Predictive Analysis: Baseline and Extended Priors

	Baseline priors		Extended priors	
Horizon	Level	Dispersion	Level	Dispersion
<i>Share of draws in which the component lowers $\mathcal{S}$</i>				
Impact ($h = 0$)	0.08	0.94	0.11	0.97
End of year 1 ($h = 3$)	0.08	0.94	0.05	0.97
End of year 3 ($h = 11$)	0.06	0.97	0.06	0.97
End of year 6 ($h = 23$)	0.05	0.97	0.07	0.97
<i>Dispersion component at $h = 23$, by sign of $\lambda_L^p - \lambda_H^p$</i>				
$\lambda_L^p > \lambda_H^p$	1.00		1.00	
$\lambda_L^p \leq \lambda_H^p$	0.00		0.00	
<i>Correlation of the signed dispersion component at $h = 23$ with</i>				
$\lambda_L^p - \lambda_H^p$	−0.37		−0.12	
λ^g	+0.17		+0.11	
MPC target	—		+0.10	
B^{ss}	—		+0.13	
Draws used / solved / requested	515/600/600		131/166/200	

Notes: A valid sacrifice ratio requires positive cumulative debt reduction under all three specifications. Statistics use the determinate draws only. Debt reduction is positive under all three specifications in 412 of 515 baseline and 100 of 131 extended draws at $h = 0$; 514 of 515 baseline and 131 of 131 extended draws at $h = 23$, and the shares at those horizons use these draws. The conditional dispersion entries use $n = 498$ and $n = 16$ in the baseline exercise and $n = 127$ and $n = 4$ in the extended exercise. The 34 failed extended draws are parameter combinations for which the MPC target is unattainable. Correlations are between the drawn parameter and the signed dispersion component at $h = 23$; a negative correlation indicates a larger offsetting effect of dispersion. A dash indicates a parameter that is fixed under that prior.

reduction is positive at every reported horizon, so the sacrifice ratio is well defined and no further selection is required. The evidence should nonetheless be read as support for the signs of the two components within determinate, debt-reducing calibrations, not as a probability statement over all admissible economies.

Extending the priors to the household block. The baseline prior predictive exercise holds the aggregate MPC target and the liquid-asset supply fixed, because varying them requires re-solving the stationary household problem for each draw. As a more demanding check, I also draw these two parameters, the one-quarter aggregate rebate MPC over $[0.15, 0.30]$ and B^{ss} over $[1.04, 2.60]$, so that the degree of MPC heterogeneity and the level of household wealth are themselves randomized. Of 200 draws, 166 solve and 131 yield a determinate equilibrium. The 34 failures are parameter combinations for which no rule-of-thumb scale reproduces the drawn MPC target.

Beyond impact, the level component raises the sacrifice ratio in 93 to 95 percent of draws, and the dispersion component lowers it in 97 percent. The median ratio of dispersion to level is −0.37, −0.36, and −0.33 at the end of years 1, 3, and 6. The median ratio of dispersion to level

is -0.32 , -0.30 , and -0.31 at the end of years 1, 3, and 6. The sign of the dispersion component again tracks the ordering of attention without exception (127 of 127 draws with $\lambda_L^p > \lambda_H^p$, and none of the 4 draws with the reverse ordering), and it remains only weakly correlated with λ^g ($+0.11$). The dispersion component is also only weakly correlated with the drawn MPC target ($+0.10$) and with B^{ss} ($+0.13$), and neither changes its sign. The distributional mechanism therefore does not depend on the particular MPC calibration or asset supply of the baseline economy.